
Cardiovascular Physiology

Outline

- Functional anatomy of the heart
- Electrophysiology of cardiac muscle
- Innervation of the heart
- Coronary circulation
- Cardiac cycle
- Regulation of cardiac output
- Electrocardiogram (ECG)
- Arterial blood pressure and its regulation
- Hypertension → causes, types, complications, treatment
- Shock → stages and types

Cardiovascular System

- **Functions** → Tissue blood flow
 - Delivery of metabolic substrates (O_2 , glucose...) to cells
 - Washout of tissue metabolic by-products (CO_2 , H^+ , amines,...)
 - Maintenance of tissue hydration (cell volume & interstitial space)
 - Systemic heat exchange via cutaneous and pulmonary circulation
 - Systemic fluid & electrolyte homeostasis via the renal circulation
 - Distribute hormones and other agents that regulate cell function
- **Components:**
 - Pump + Distributive + Micro-vascular network

Cardiovascular System

1. Pump (Heart)

- Acts as two pumps in series
 - RA & RV → drive the pulmonary circulation
 - LA & LV → drive the systemic circulation

2. Distribution

- Beyond the elastic portion of arterial tree, muscular arteries and arterioles act to control distribution of blood
- These segments control blood flow and blood pressure by adjusting their luminal diameters as their walls actively contract or relax

Cardiovascular System

3. Micro - vascular network → Exchange

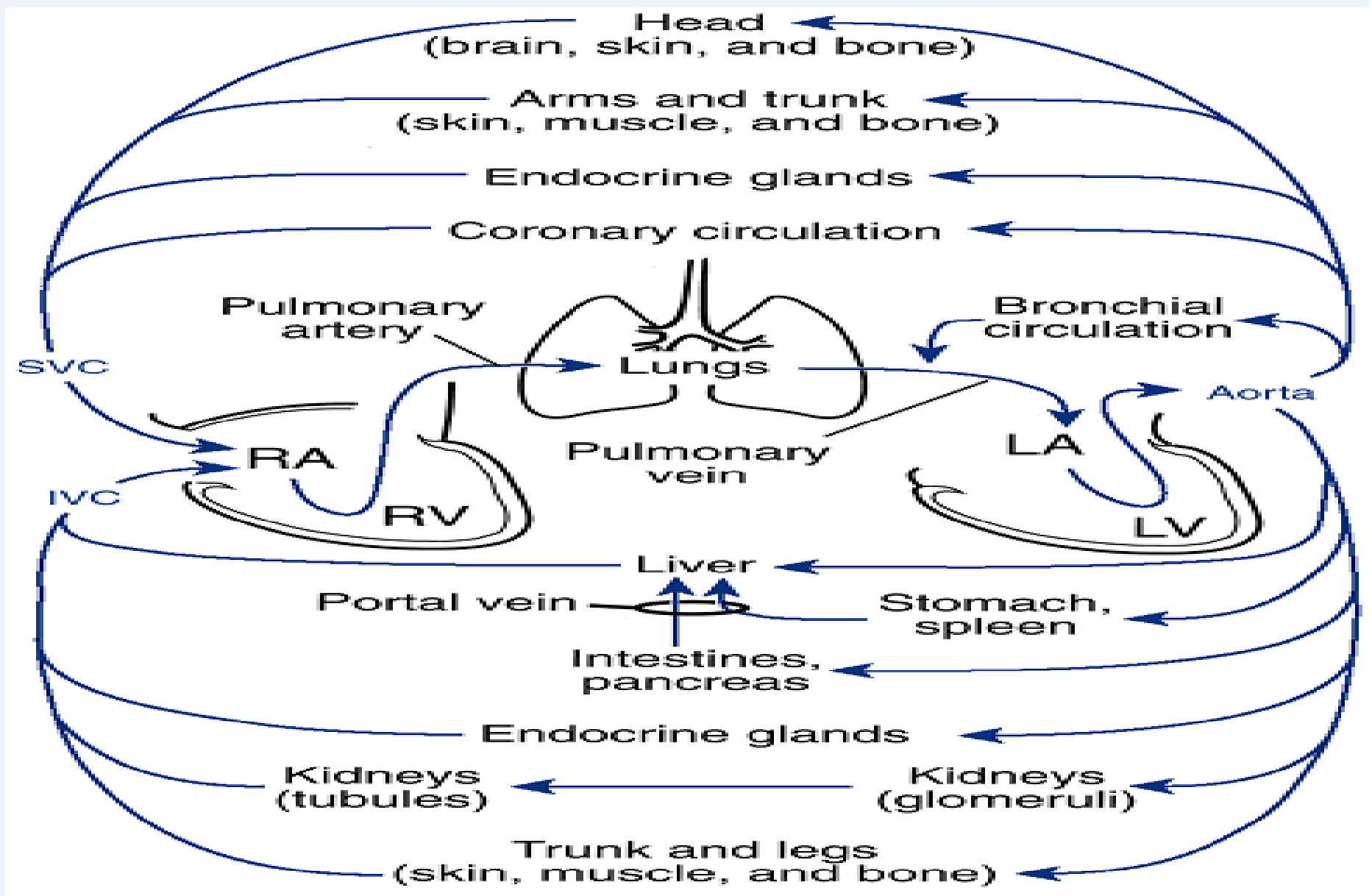
- From arterioles blood enters capillaries and post-capillary venules
- These vascular segments are under **low pressure**
- Their walls allow O_2 & CO_2 , nutrients & metabolic products, hormonal signals etc. to pass back and forth between tissues and vascular segment
- Blood return to RA carried by conductive systems

Heart

- 5 basic structures:
 1. Myocardium (Cardiac muscle)
 2. Valves
 3. Electrical conducting systems
 4. Nerves which regulate the heart
 5. Coronary blood vessels which supply the heart

Myocardium

- Has 4 chambers
 - Atria → thin walled ($\approx 2\text{mm}$)
 - Ventricles → thick walled ($\text{LV} \approx 8\text{-}9\text{ mm}$, $\text{RV} \approx 3\text{-}4\text{ mm}$)
 - RA → receive deoxygenated blood from vena cava
 - No valve between RA and vena cava
 - LA → receive oxygenated blood from the lung (pulmonary vein)
 - No valve between LA and pulmonary vein
 - RV → Pump blood to the lungs for oxygenation
 - LV → Pump oxygenated blood to the rest of the body



Heart Valves

- Valves **direct blood flow**
- Has fibrous connective tissue which prevent enlargement of valve opening and anchor valve flaps

1. **Inlet valves** → Atrioventricular valves

- Tricuspid valve (RA/RV) + Mitral valve (LA/LV)

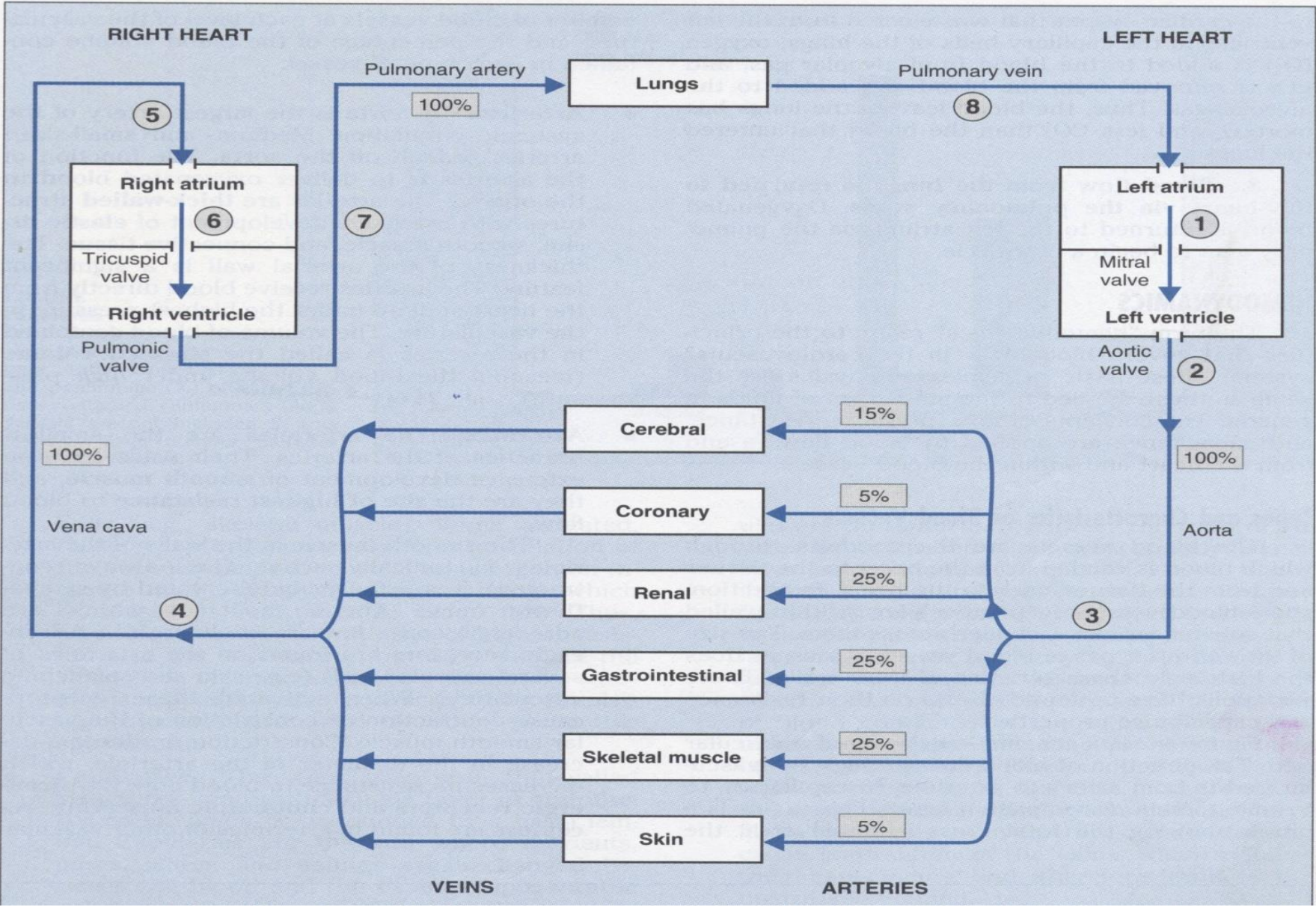
2. **Outlet valves** → Semilunar valves

- Aortic valve (LV/Aorta) + Pulmonic valve (RV/Pulmonary artery)

NB: Valves open/close by pressure on either side (passive)

Valve closure prevents backflow of blood during and after contraction

Closure of valves → **Phonocardiogram (heart sounds)**



Myocardial Fibers

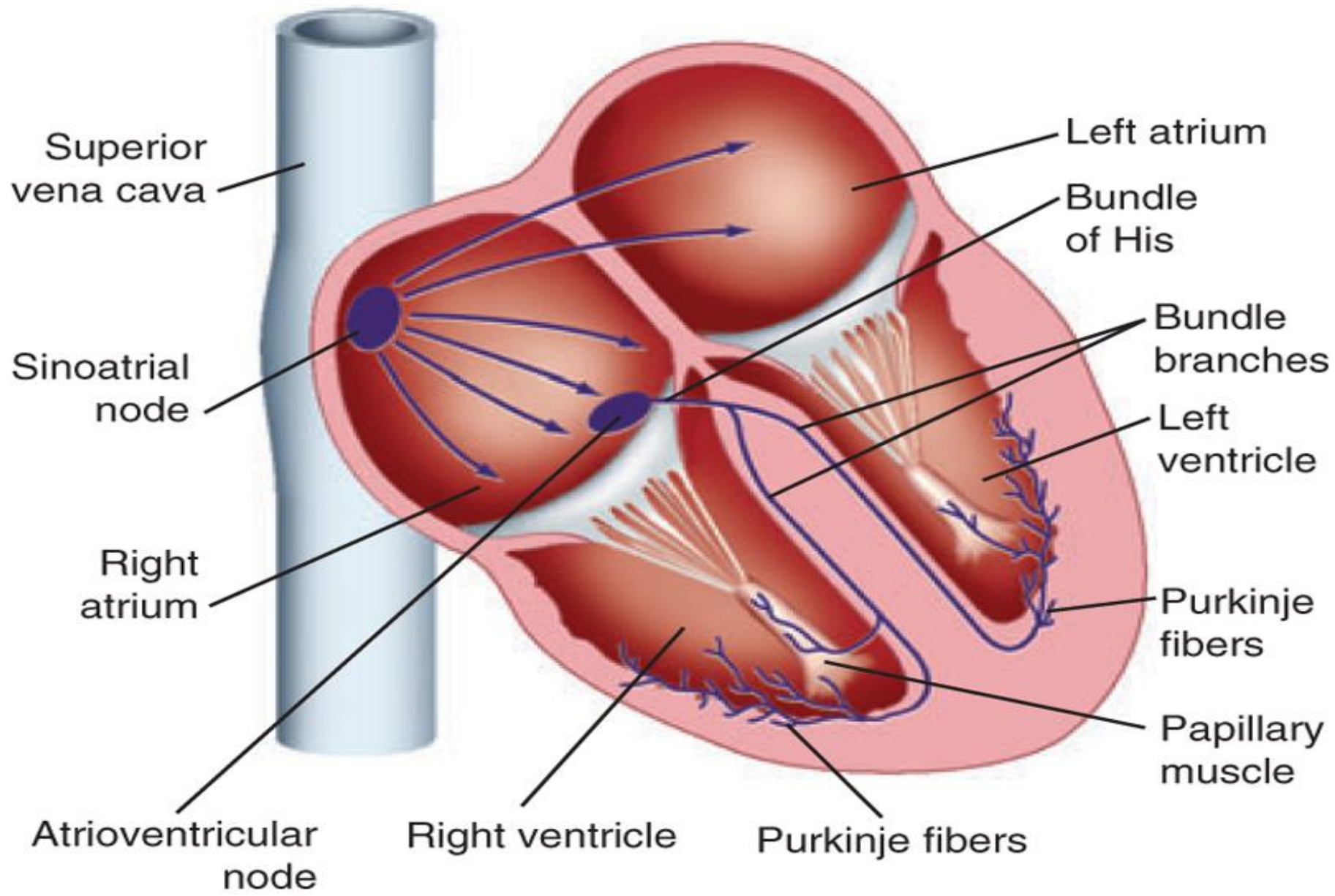
- Two types based on morphology & functional criteria
 1. Pacemaker & conducting systems:
 - Generate excitatory impulse and send to working myocardial cells
 2. Working myocardium of atria and ventricles
 - Do mechanical work of pumping
- Cardiac muscle fibers are connected with each other in series by intercalated discs (gap junction, desmosomes)
- Intercalated discs offer no obstacle to the conduction of excitation
- Action potential spreads cell to cell, past the intercalated disc

Myocardial Fibers

- Electrical activity:
 - The heart has 2 separate syncytium:
 - Atrial and ventricular
 - Atria and ventricle behave functionally as a syncytium (excitation anywhere in atria or ventricles spread all over unexcited fibers)
 - These are separated from each other by **fibrous tissue** (electric resistant, surrounds valves)
 - Action potential is conducted from the atria into the ventricle by way of the **A-V bundle only**

Electrical Activity

- Spread of cardiac excitation
 - Sinoatrial node (SA node)
 - Normal pacemaker, drive heart at a rate of 70 bpm (rest)
 - Excitation spreads over working myocardium of atria
 - Atrial excitation
 - Depolarization is complete in about 0.1 second
 - Impulse propagation spreads to the AV node

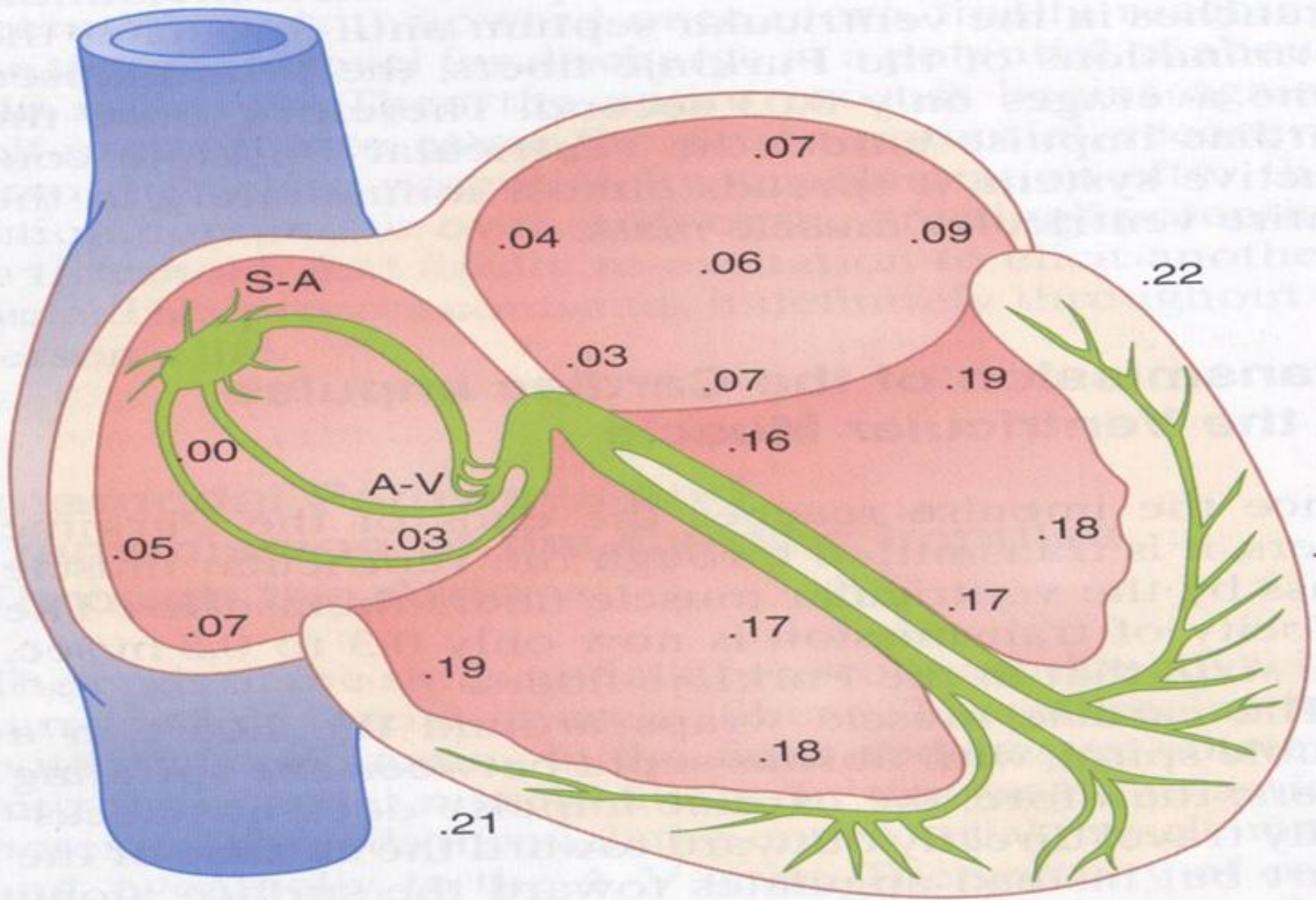


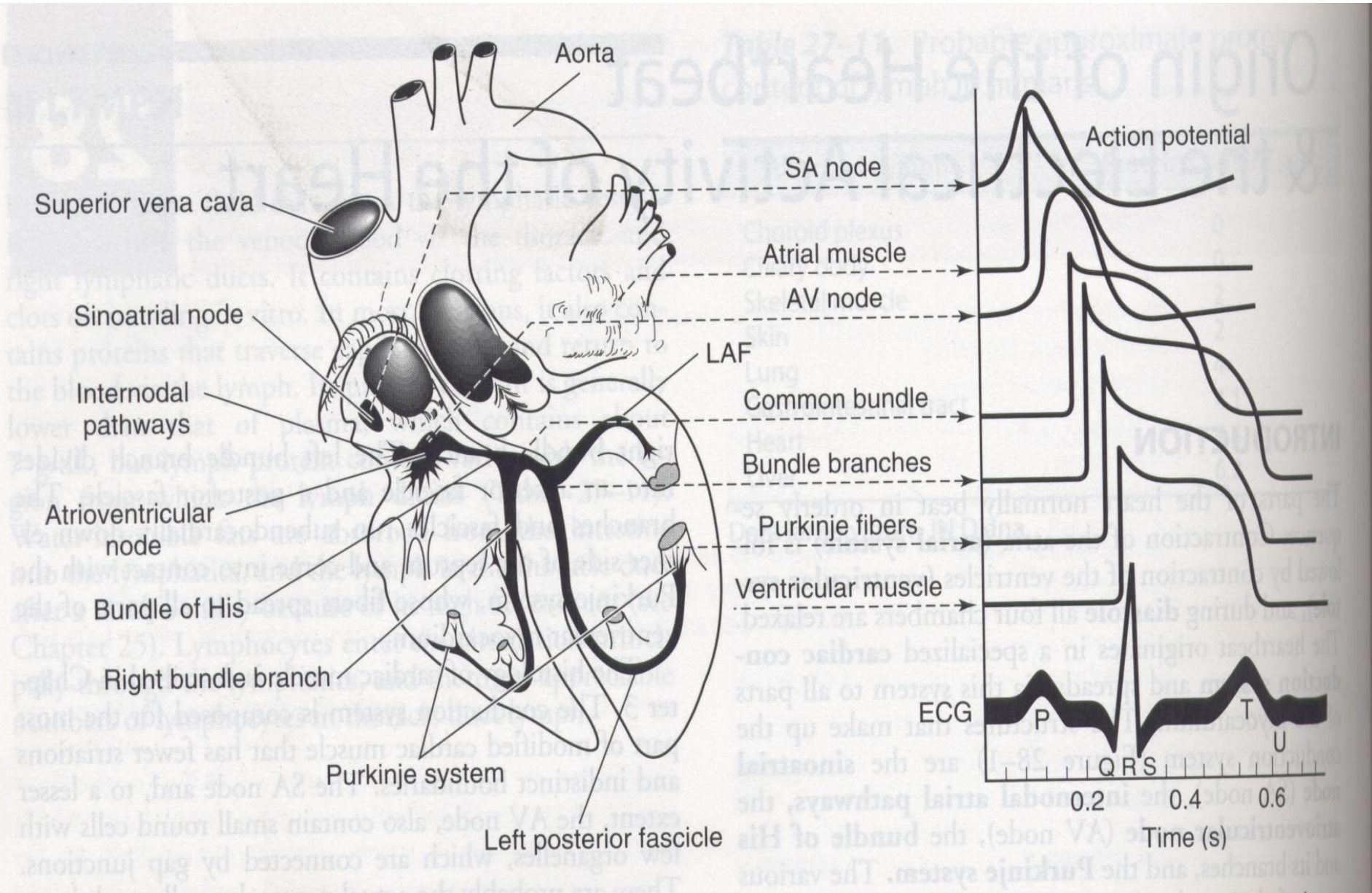
Electrical Activity

- Atrioventricular node (AV node)
 - The only pathway for impulse conduction to the ventricles
 - The rest of AV boundary consist of unexcitable connective tissue
 - Potential pacemaker (if SA node fails)
 - AV nodal delay → conduction in the AV node is slow ($\approx 0.05\text{m/s}$)
 - Allows atria to empty before ventricular contraction begins
- (Important for ventricular filling)

Electrical Activity

- Atrioventricular bundle (AV bundle)
 - Bundle of His, bundle branches and Purkinje fibers
 - Propagation from AV node to the ventricular myocardium
 - Propagation is fast for rapid successive ventricular excitation
 - Impulse propagation is fastest in the Purkinje fibers
- Ventricular excitation follows





ANS Innervation

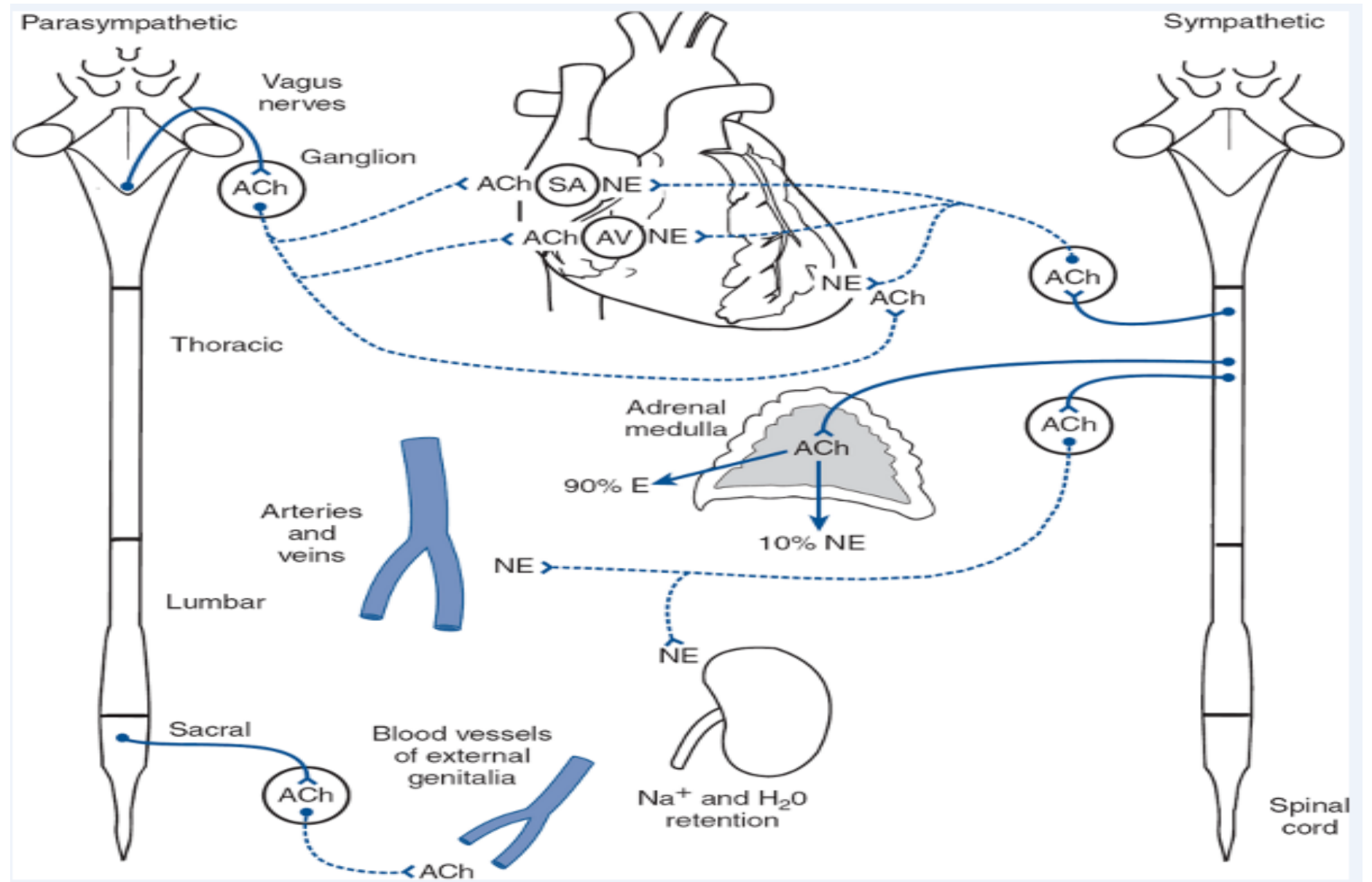
1. Sympathetic NS

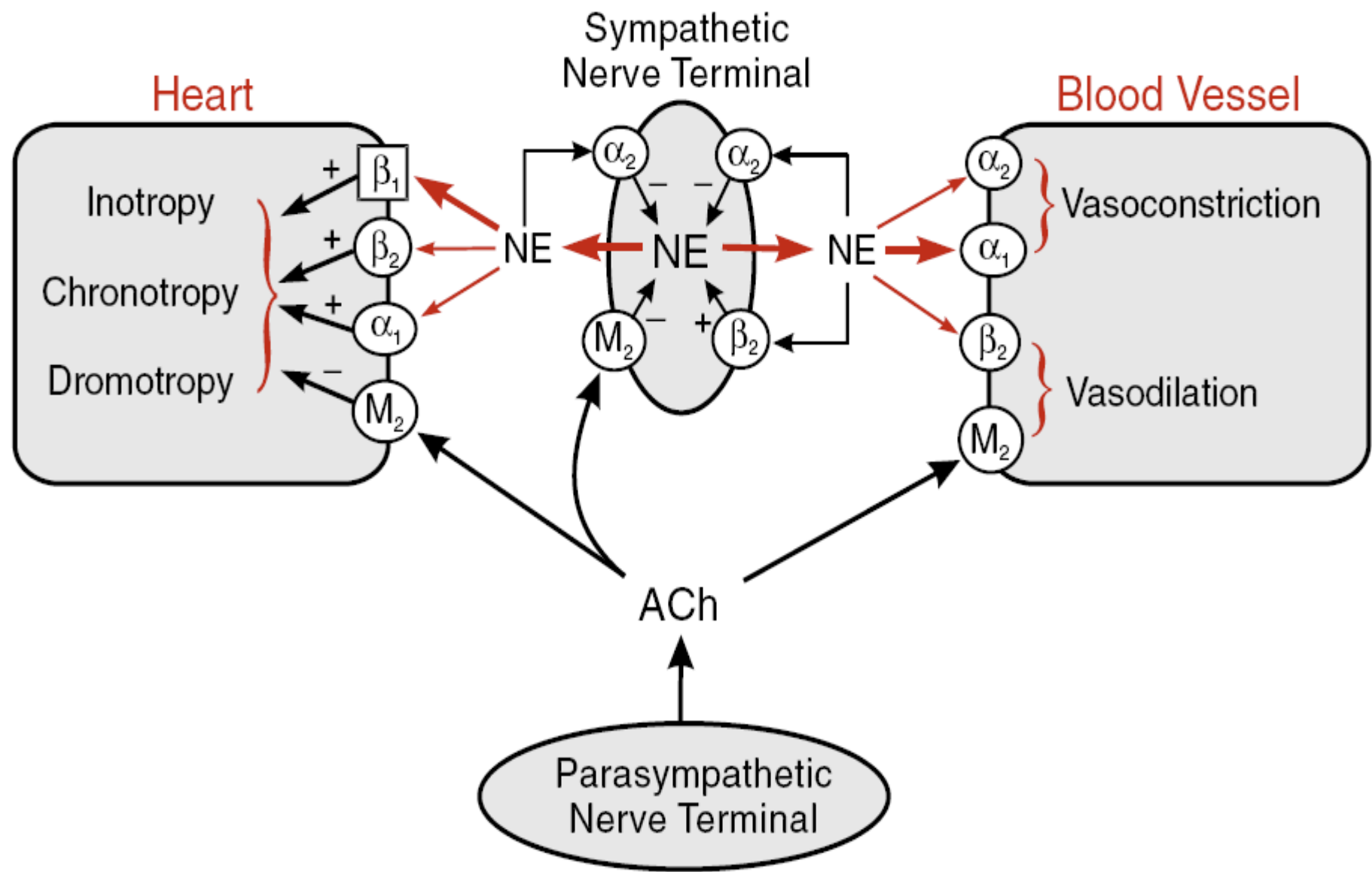
- Innervate all parts of the heart (*NE affinity: $\beta_1 > \beta_2$ & α_1*)
- Sympathetic stimulation of the heart produce:
 - **+ve Chronotropy** ($\uparrow$ cardiac frequency, HR)
 - **+ve Dromotropy** ($\uparrow$ conduction velocity of myocardial conduction)
 - **+ve Inotropy** ($\uparrow$ cardiac contractility)
- Sympathetic stimulation of blood vessels $\rightarrow$
 - **Vasoconstriction (α)** &
 - **Vasodilation (β_2)**

ANS Innervation

2. Parasympathetic NS

- Parasympathetic nerves (Vagus) innervate:
 - Mainly **SA node + AV node**
 - Atrial muscle (lesser extent)
 - Ventricular muscle (very little)
- Ach binds to postsynaptic muscarinic receptors (**M₂**) in the heart and produce **negative Inotropy + Chronotropy + Dromotropy**
- Ach also binds to **presynaptic M₂** receptors and **inhibit NE release** from sympathetic nerve terminals



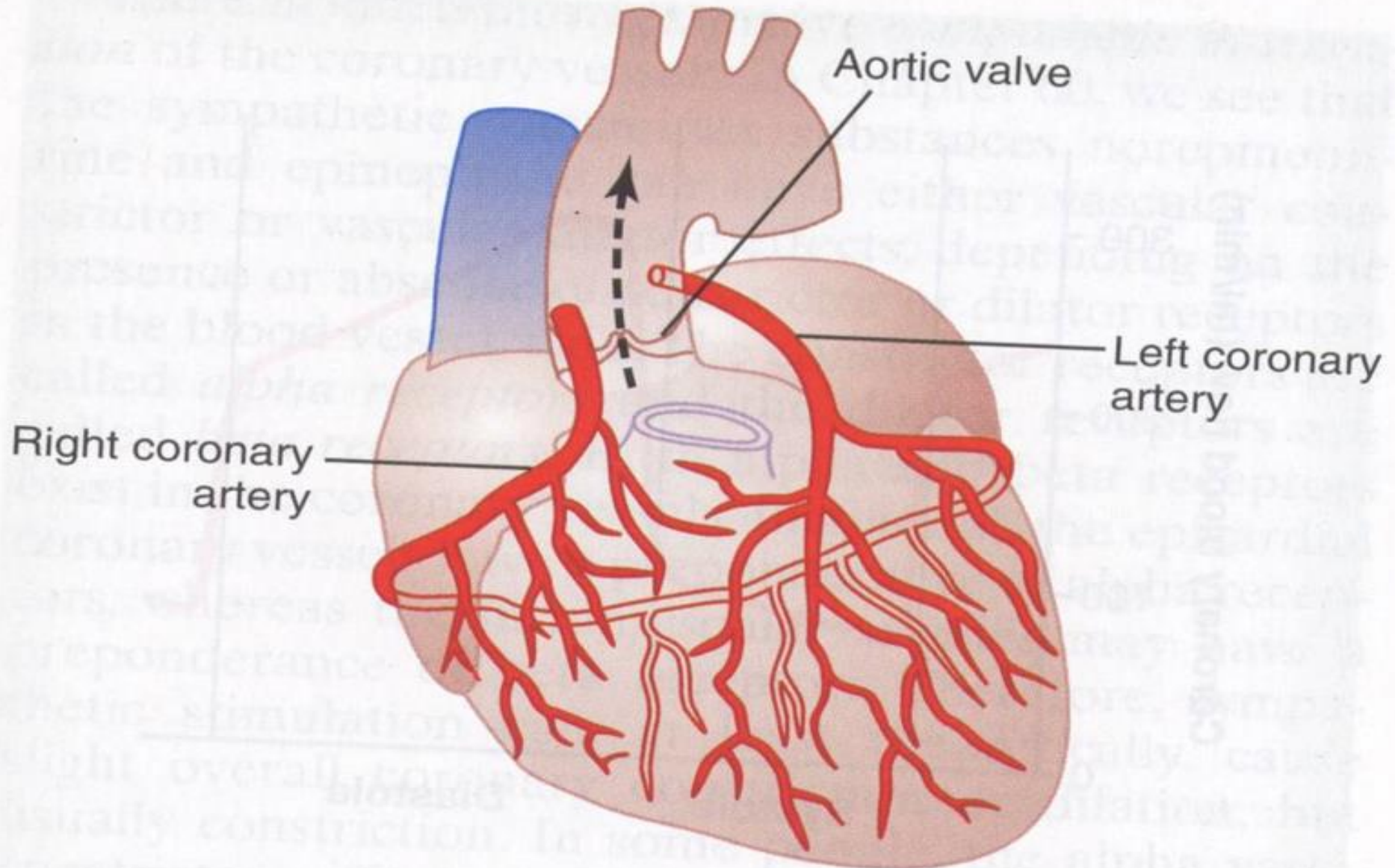


ANS Innervation

- Ventricular escape rhythm
 - LBBB, RBBB
 - Strong stimulation of vagus, completely blocks excitatory signal transmission to the ventricles
 - After a moment, however, purkinje fibers develops a rhythm of its own and causes ventricular contraction (*Ventricular escape*)
 - Ventricular rhythm rate: 20-40bpm
 - QRS complex: wider ($\geq 120\text{ms}$)

Coronary Circulation

- **Coronary vessels**
 - Carry blood to the heart
 - Heart depends strongly on aerobic metabolism
 - Brief period of low oxygen leads to → damage to myocardium
- 2 main **coronary arteries** arise from the aorta (just above aortic valve):
 - Left coronary artery → supply posterior part
 - Right coronary artery → supply right part
- **Coronary veins** return blood to coronary sinus located near the RA



Coronary Blood Flow

In the Coronary Artery:

Diastolic Flow > Systolic Flow



WHY?

- 1. More access to coronary arteries: when semilunar valves are open, they partially occlude holes in the coronary artery**
- 2. Contraction of the heart compresses coronary arteries (increase resistance, decrease flow)**

Cardiac cycle

Mechanics of Heart Function

Process	Action
Cardiac cycle	Sequence of events in 1 heartbeat. Blood is pumped through the entire cardiovascular system.
Systole	Contraction phase—usually refers to ventricular contraction.
Diastole	Relaxation phase—the atria and ventricles are filling. Lasts longer than systole.
Stroke volume (SV)	Amount of blood ejected from either ventricle in a single contraction. Starling's Law of the Heart states that degree of cardiac muscle stretch can increase force of ejected blood. More blood filling the ventricles $\uparrow$ SV.
Cardiac output (CO)	Amount of blood pumped through the cardiovascular system per min. $CO = SV \times \text{Heart rate (HR)}$

Cardiac cycle

PHASE 1. ATRIAL SYSTOLE: AV VALVES OPEN; AORTIC AND PULMONIC VALVES CLOSED

PHASE 2. ISOVOLUMETRIC CONTRACTION: ALL VALVES CLOSED

PHASE 3. RAPID EJECTION: AORTIC AND PULMONIC VALVES OPEN; AV VALVES REMAIN CLOSED

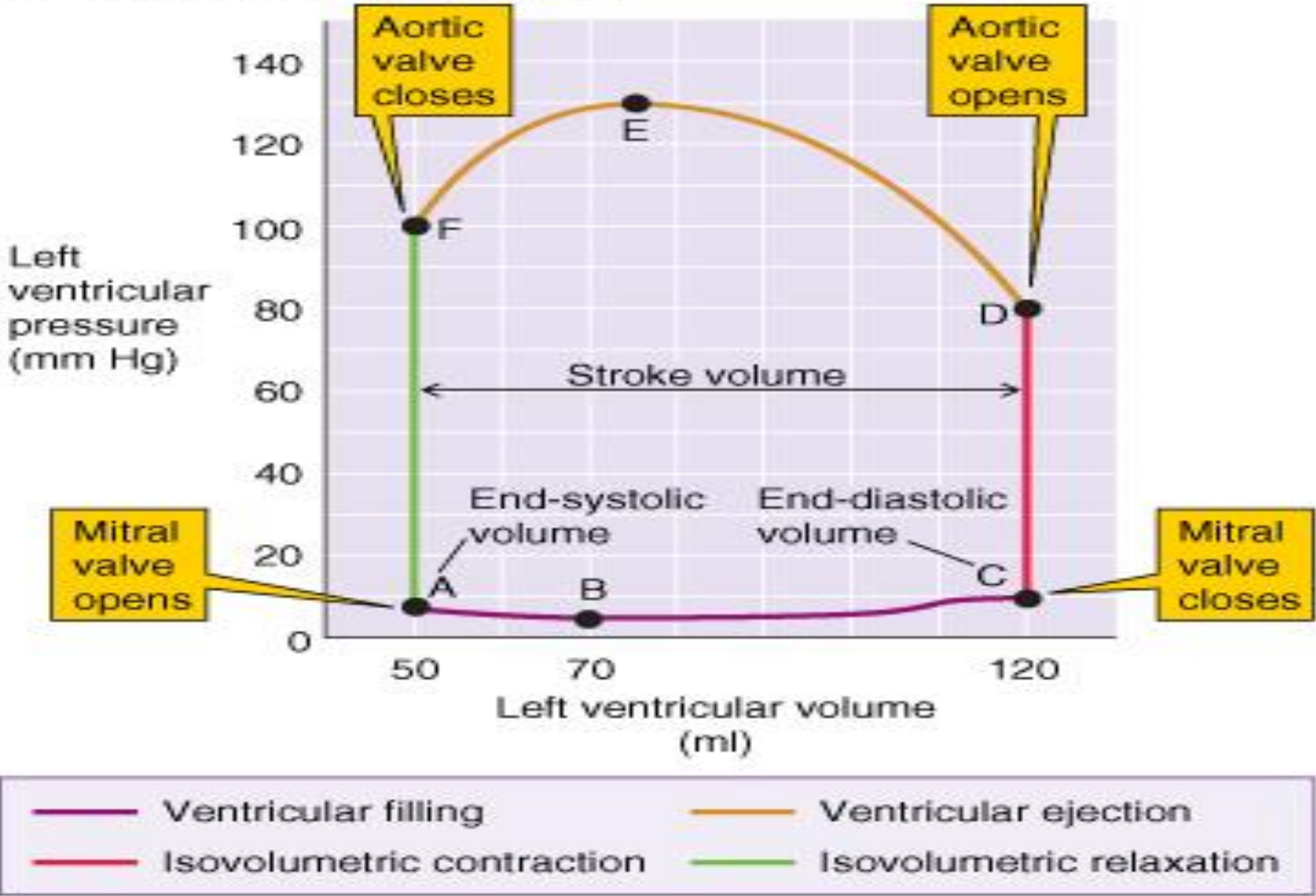
PHASE 4. REDUCED EJECTION: AORTIC AND PULMONIC VALVES OPEN; AV VALVES REMAIN CLOSED

PHASE 5. ISOVOLUMETRIC RELAXATION: ALL VALVES CLOSED

PHASE 6. RAPID FILLING: AV VALVES OPEN; AORTIC AND PULMONIC VALVES CLOSED

PHASE 7. REDUCED FILLING: AV VALVES OPEN; AORTIC AND PULMONIC VALVES CLOSED

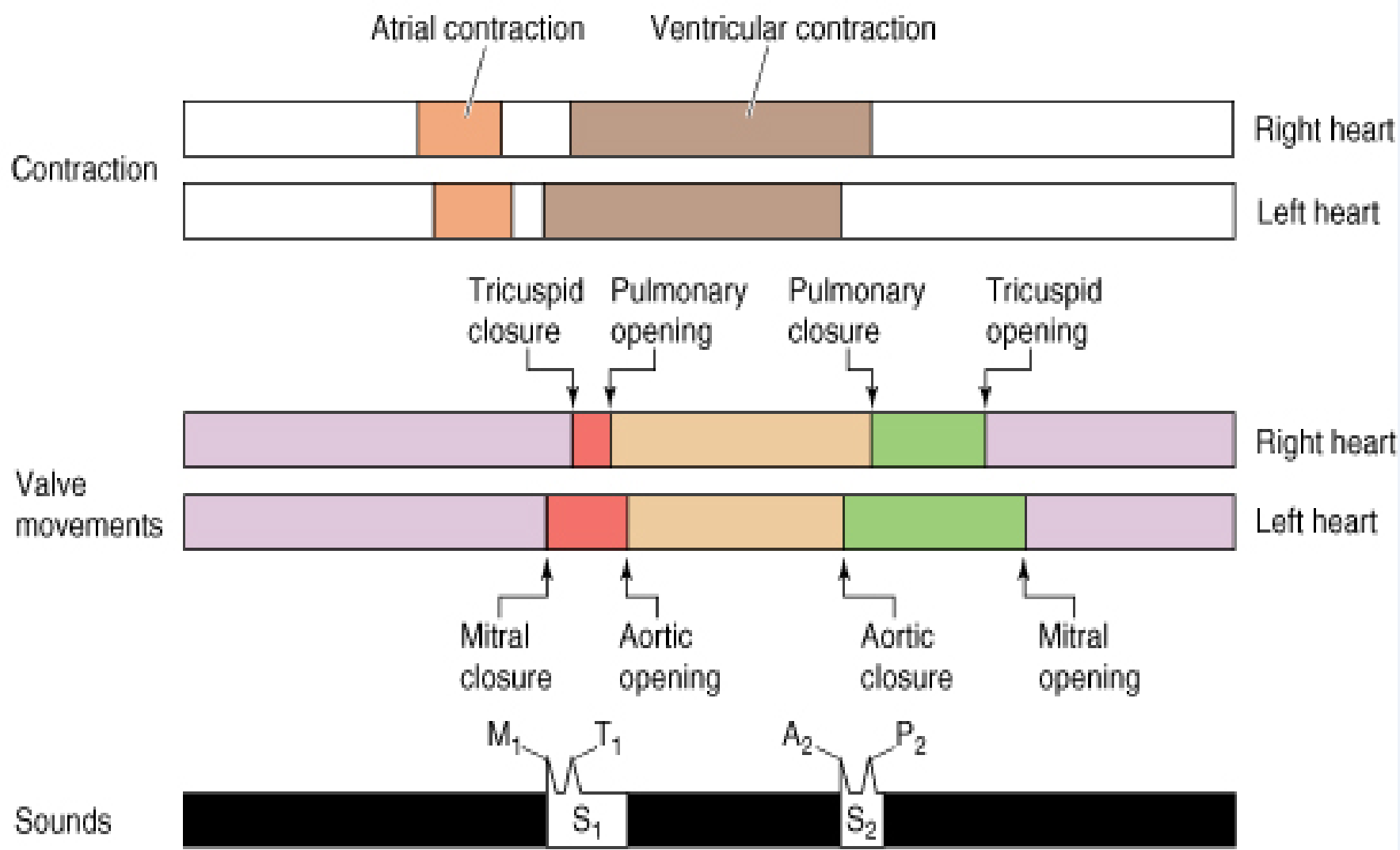
A PRESSURE-VOLUME LOOP



Heart Sounds

- Sounds associated (usually) with valve closure
 - **First heart sound** → Closure of AV valves (M & T)
 - Occurs at the beginning of isovolumic contraction
 - **Second heart sound** → Closure of semilunar valves (A & P)
 - Occur at the end of ejection (Isovolumic relaxation)
 - Sometimes splitted, since Aortic valve closes slightly before Pulmonic valve
 - **Third heart sound** (sometimes) → due to rapid ventricular filling
 - **Fourth heart sound** (occasionally) → during atrial contraction

C TIMING OF EVENTS IN CARDIAC CYCLE



Cardiac Output

- Stroke volume (SV)
 - Volume ejected during each beat ($SV \approx 0.07\text{L/min} \approx 70\text{ml}$)
 - Depends on venous return (can be equal to venous return)
- Heart rate (HR) ≈ 72 beats/min
- Cardiac output (CO)
 - All blood pumped around circuit once each minute
 - $CO = HR \text{ (bpm)} \times SV \text{ (L/beat)} \approx 5 \text{ L/min}$ (at rest)
 - CO increases when needed (eg. Exercise...)
- Cardiac Reserve = $CO_{\text{Max}} - CO_{\text{Min}}$

Ejection Fraction (EF)

- Measurement of ventricular performance
- Fraction of end diastolic volume (EDV) ejected from ventricles

$$EF = SV/EDV$$

- A healthy man has EF of 50% or more

Regulation of CO

- Intrinsic regulation
 - Arterial pressure (afterload)
 - Filling pressure (preload)
- Extrinsic regulation
 - Parasympathetic and sympathetic effects

Cardiac Output Regulation → 2 Categories :

$$C.O. = (H.R.) (S.V.)$$

1. INTRINSIC

a) Homeometric

- stretch atria
- ↑ irritability of SA Node
- ↑ Heart Rate

b) Heterometric

- stretch ventricles
- ↑ force of contraction
- ↑ Stroke Volume

What can stretch the atria or ventricles?

INCREASED PRELOAD
(End Diastolic Volume)



cont'd...

2. EXTRINSIC

a) Parasympathetic

L Vagus Nerve



AV Node

Atrial muscle fibers



R Vagus Nerve



SA Node

Atrial muscle fibers

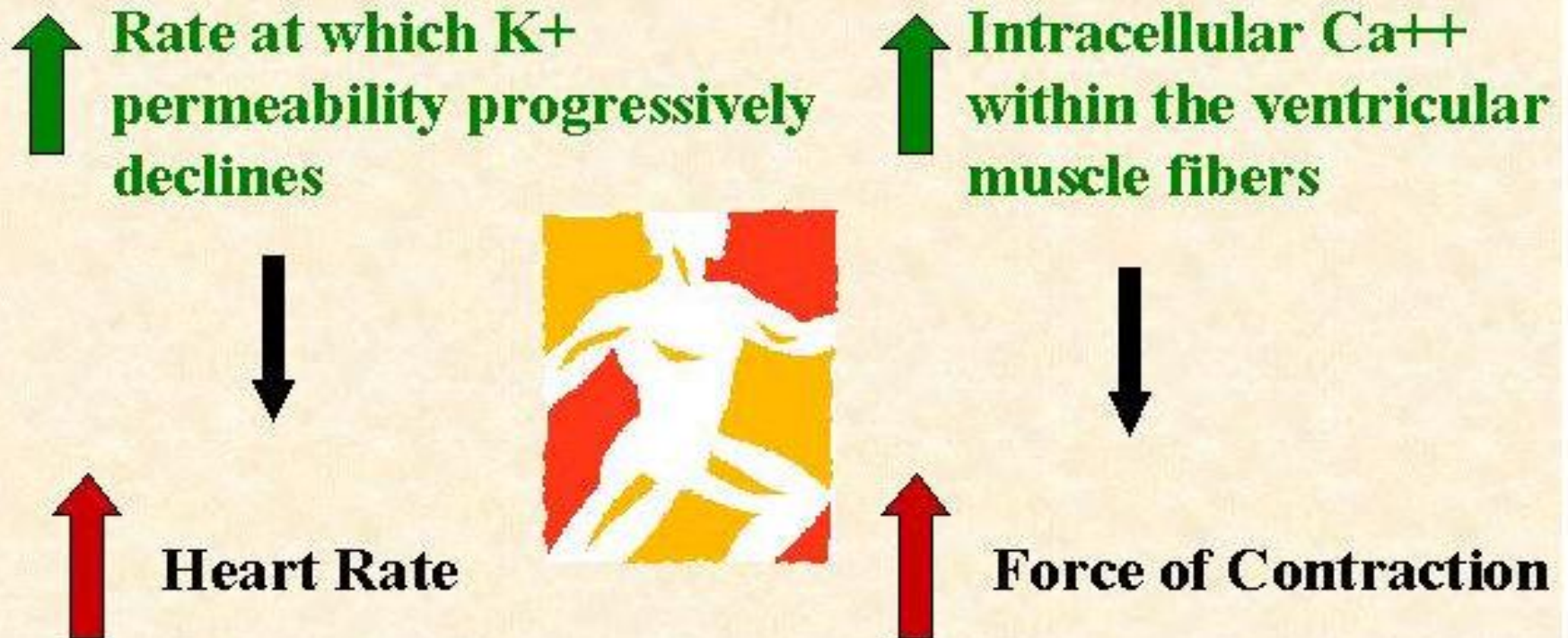


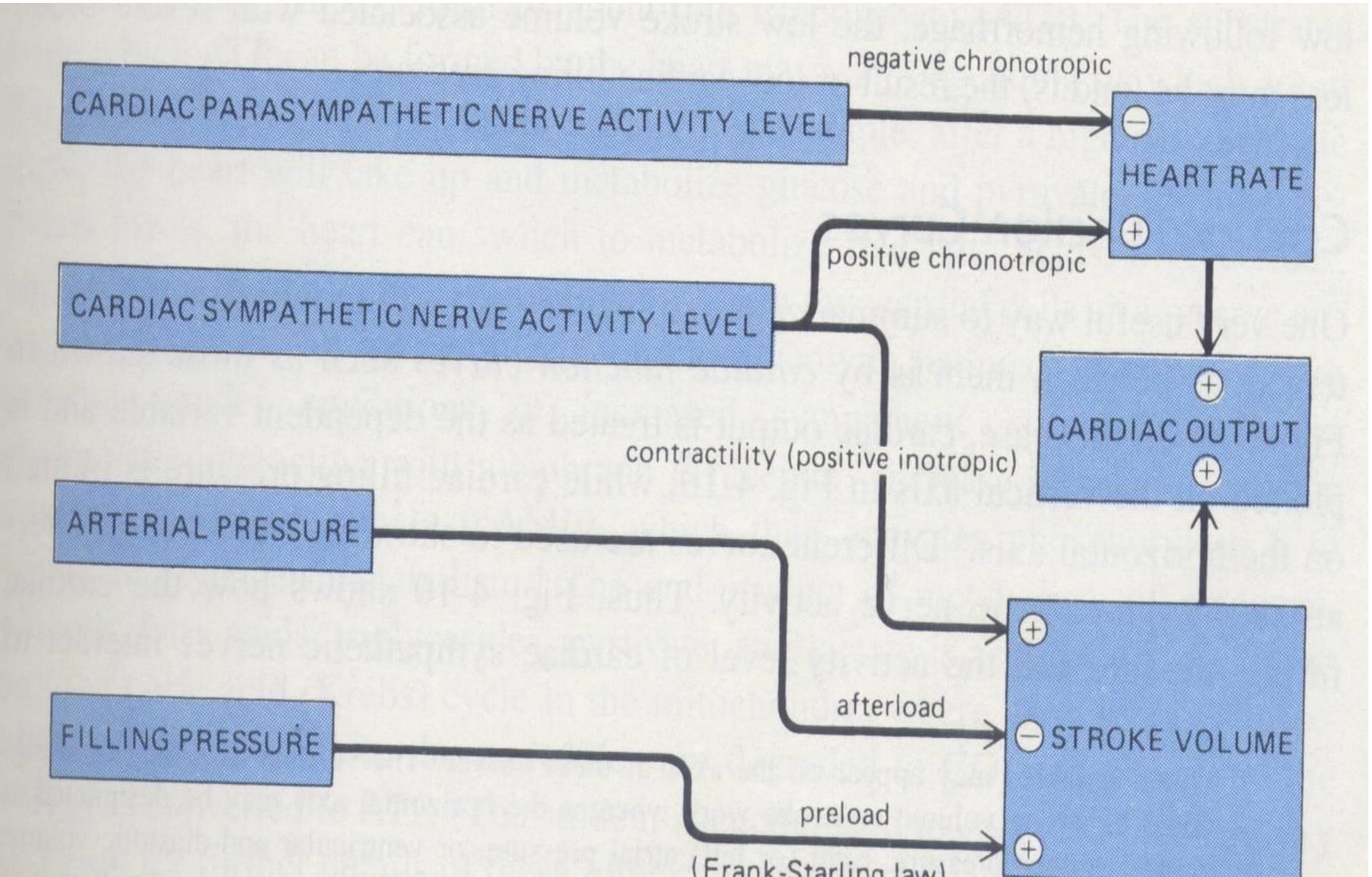
Heart Rate by



**Rate at which K⁺
progressively declines**

b) Sympathetic





Regulation of CO

- SV is determined by:
 - Preload + Afterload + Myocardial contractility
 - $\uparrow$ Preload $\rightarrow \uparrow$ SV
 - $\uparrow$ Afterload $\rightarrow \downarrow$ SV
 - $\uparrow$ Myocardial contractility $\rightarrow \uparrow$ SV

Clinical Correlates

Abnormal Heart Sounds

- **Murmur** – valvular disease
 - Turbulent blood flow (may be normal when flow increases)
- **Stenosis** → valves do not open properly (eg narrowing)
 - Swish sound
- **Regurgitations** → valvular insufficiency, incompetency
 - Valve does not close properly
 - Whistle sound = murmur
 - Timing of murmur indicates which valve is damaged

Examples Valvular Diseases:

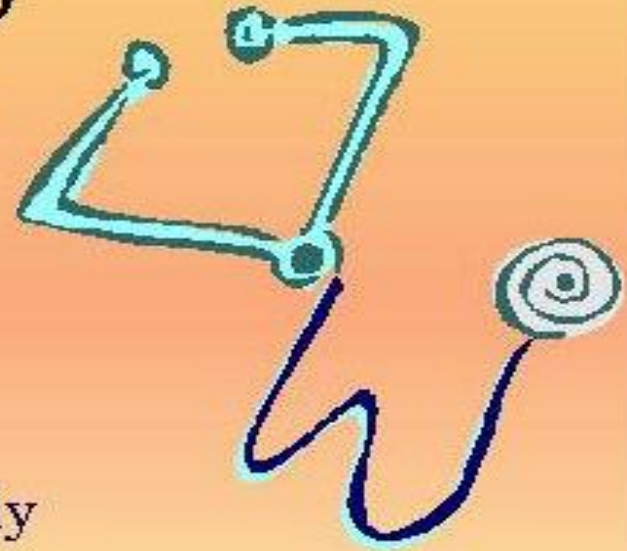
1. Regurgitations

(Insufficiency, Incompetency)

- valve does not close properly
 - Whistle sound*
- = MURMUR

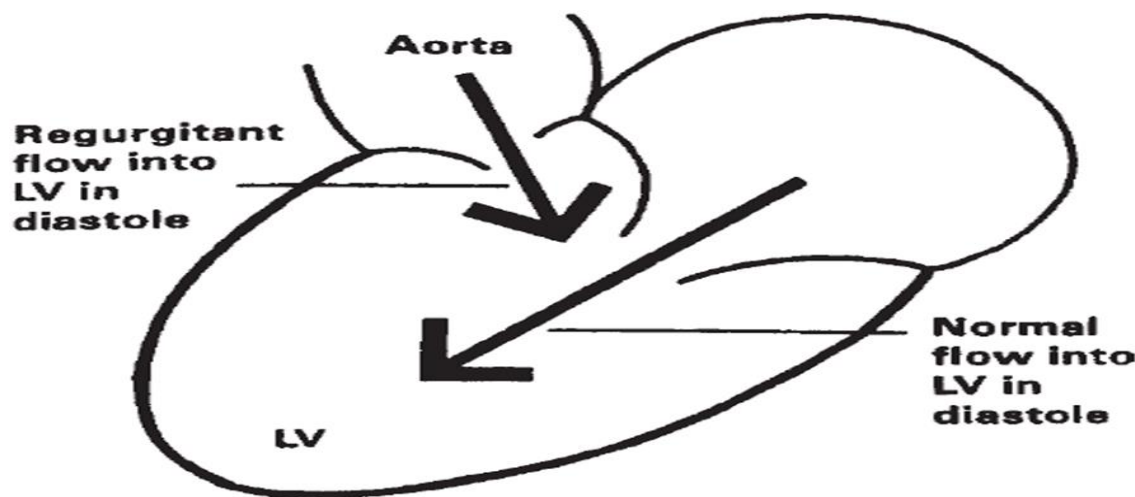
2. Stenosis

- valve does not open completely
- Swish sound*



Abnormal Heart Sounds

- **Mitral insufficiency**
 - Regurgitation of blood into left atrium during ventricular systole
 - Associated murmur between first and second heart sound
- **Murmur throughout systole** → Stenotic Pulmonic or Aortic valve
- **Murmur throughout diastole** → Stenotic tricuspid or mitral valve



Abnormal Heart Sounds

- Causes of valvular diseases

- Rheumatic fever

- Caused by repeated pharyngeal infection with group-A streptococcus
- Due to autoimmune reaction, patients develop fever, joint pain, mitral valve inflammation...
- >50% with acute rheumatic fever develop rheumatic heart disease, mainly in mitral and aortic valves (stenosis or regurgitation)

- Inflammation of endocardium...

Myocardial Hypertrophy

1. Physiological hypertrophy

- Physical exercise
- Heart wt. = 500gm. (350 gm. In sedentary)
- Length and thickness of myocardial cells increased
- Stroke volume is larger & HR is slower

Myocardial Hypertrophy

2. Pathological hypertrophy

- Aortic stenosis → unilateral hypertrophy (LV)
- Aortic insufficiency → cavity enlargement
- Degree of compensation limited → Thus myocardial fiber diameter increases → Distance between capillaries and cell interior increases (diffusion distance increases) → Thus inadequate O_2 , nutrition → heart failure (myocardial insufficiency)

Coronary Pathologies

1. Myocardial Ischemia

- Reduced blood flow to the heart → Heart Attack
- Causes:
 - **Vascular Spasm** → spastic contraction of arterioles
 - **Atherosclerosis**
 - Atheroma + plaque
 - Cells migrate to endothelium (retaining cholesterol and lipids)
 - **Thromboembolism**
 - Ultimate consequence of atherosclerosis
 - Blood clot breaks off, stops and impedes blood flow

Coronary Pathologies

*Myocardial Ischemia

- reduced blood flow to heart

→ Heart Attack

CAUSES: a) Vascular Spasm

spastic contraction of
arterioles

endothelial relaxing factor



cont'd ...

Coronary Pathologies... cont'd

b) Atherosclerosis

atheroma



cells migrate to endothelium (retaining cholesterol
and lipids)



plaque

★ ~ 75% of college men have fatty streaks
in their arteries

c) Thromboembolism

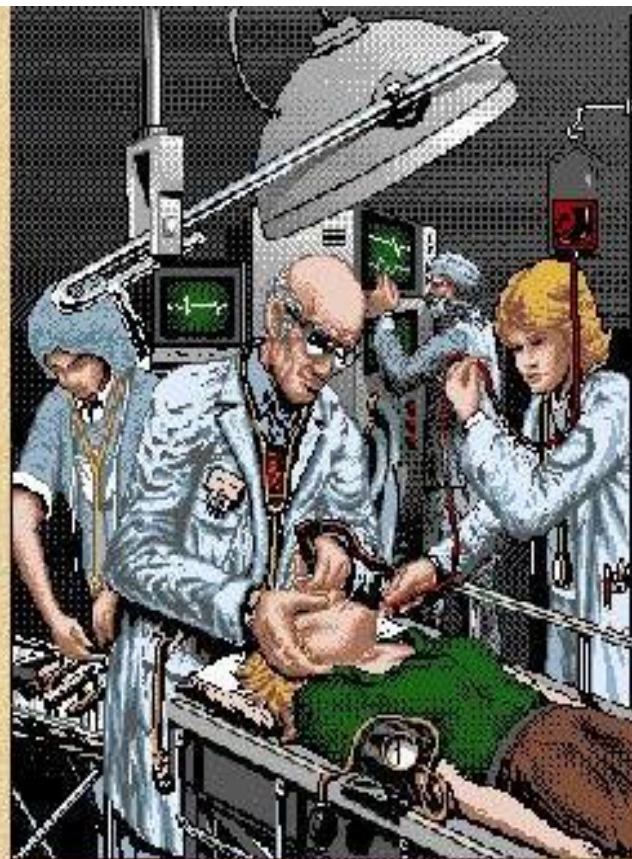
- ultimate consequence of atherosclerosis
- blood clot breaks off, stops and impedes blood flow





Angina

**= visceral pain caused by
myocardial ischemia**



**- heart's demand for Oxygen > capacity of circulation to
supply it**



PHARMACOLOGY

1. Nitrates (Nitroglycerin)

- a. Nitrostat
- b. Nitrodur
- c. Transdurm-Nitro

2. B- Blockers

3. Calcium Channel Blockers

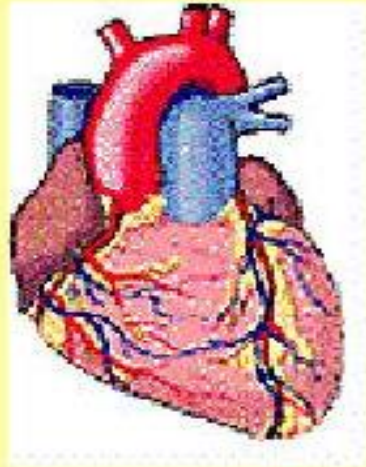
1. Slow Ca^{++} channels

2. ↓ Force of Contraction
Demand for Oxygen

3. Dilation of Coronary arteries

- a. Verapamil (Isoptin) (Calan)
- b. Nifedipine (Procardia)
- c. Diltiazem (Cardizem)
- d. Nicardipine (Cardene)

Congestive Heart Failure



- Deficiency in myocardium: ↑ End Systolic Volume
 - Heart does not sufficiently empty: ↑ Ventricular Pres.
- Results in (Pulmonary) Edema

PHARMACOLOGY

Treatment for Congestive Heart Failure

1. Diuretics

2. Cardiac glycosides

a. digitoxin (Lanoxin)

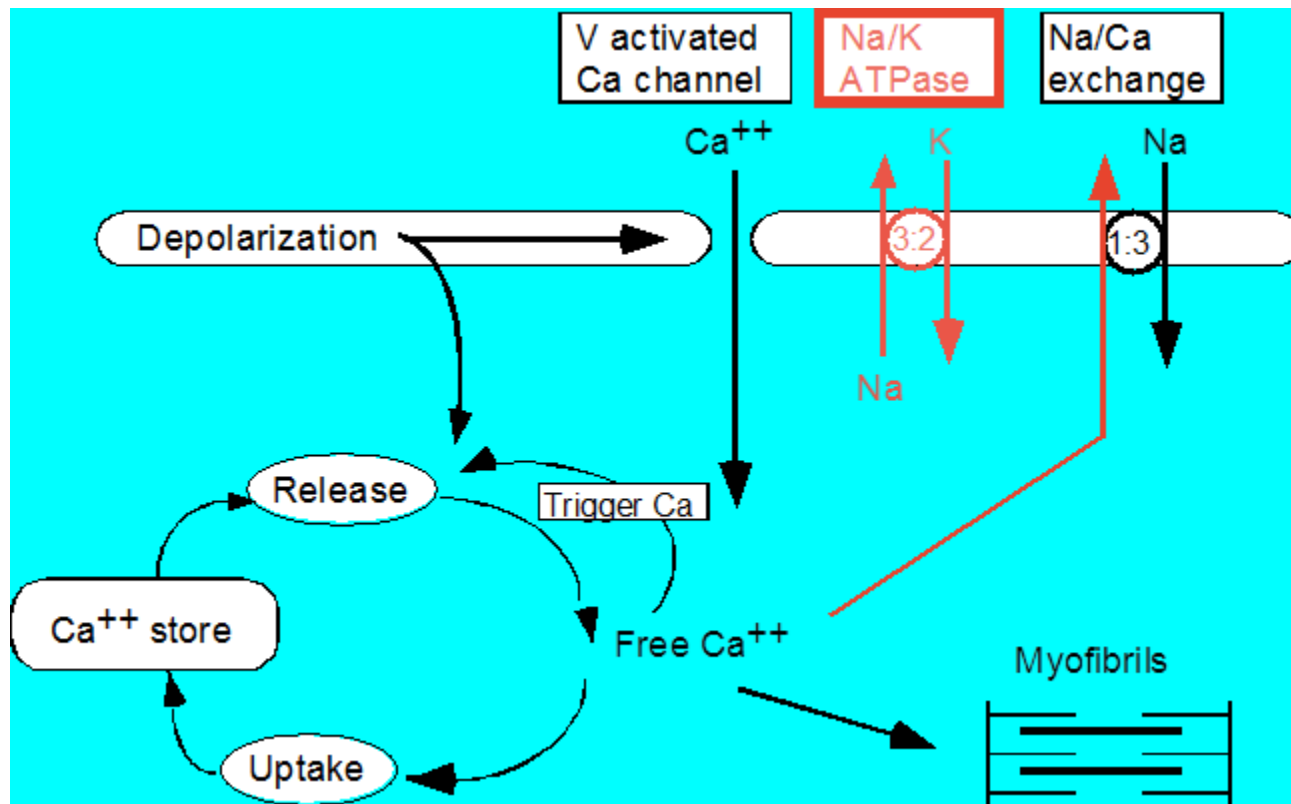
b. digitoxin (Crystodigin)



Cardiac Glycosides

- Inhibit $\text{Na}^+\text{-K}^+\text{-ATPase}$
- Leads to increase in intracellular Ca^{2+} through the $\text{Na}^+/\text{Ca}^{2+}$ exchange pump (by decreasing the availability of sodium to pump calcium out)
 - *(Inhibition of $\text{Na}^+\text{-K}^+\text{-ATPase} \rightarrow$ inhibition of $\text{Na}^+\text{-Ca}^{2+}$ exchanger)*
- Leads to enhanced contractility

Inotropic action of digoxin



Na/K ATPase ↓

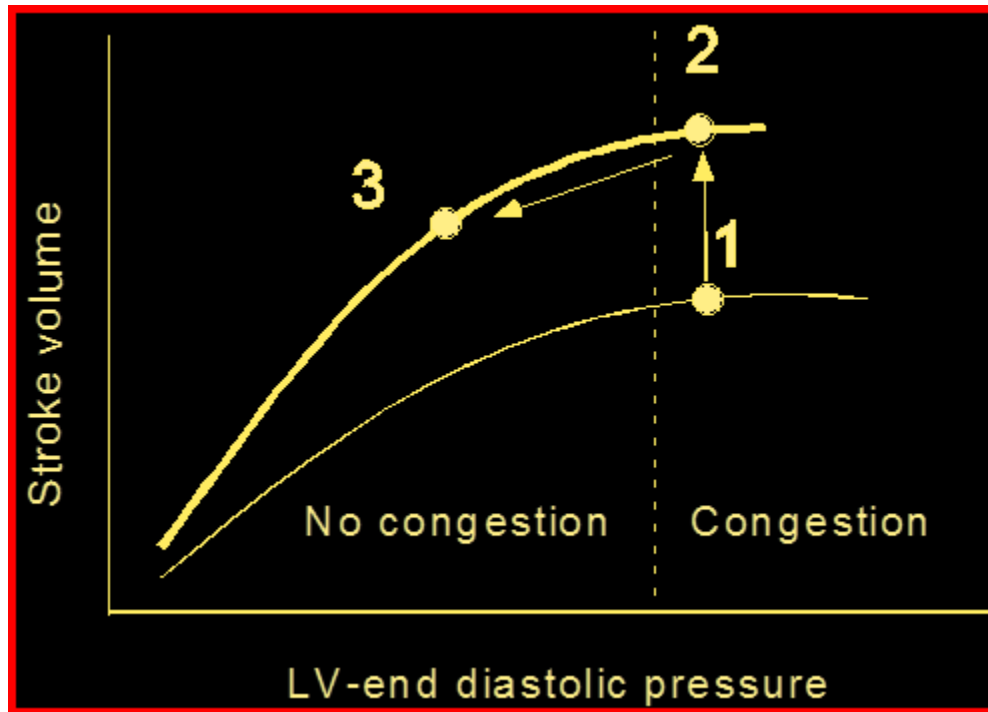
Na_i ↑

Na gradient ↓

Ca extrusion ↓

Contractility ↑

Overall effect of digoxin in HF

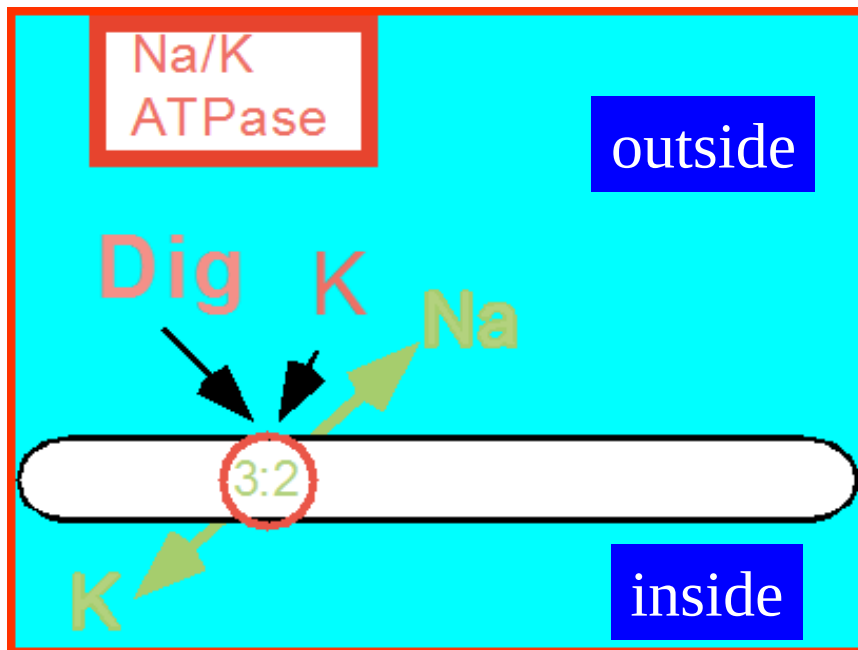


1 $\Rightarrow$ 2: immediate increase in inotropic function and SV

2 $\Rightarrow$ 3: delayed hemodynamic benefits due to:

- improved renal function
- lowered SNA and RAS

K interactions with digoxin



Dig and K compete for the same binding site therefore:

1. hyperK ↓ effect of dig
2. hypoK ↑ effect of dig

Corrollaries:

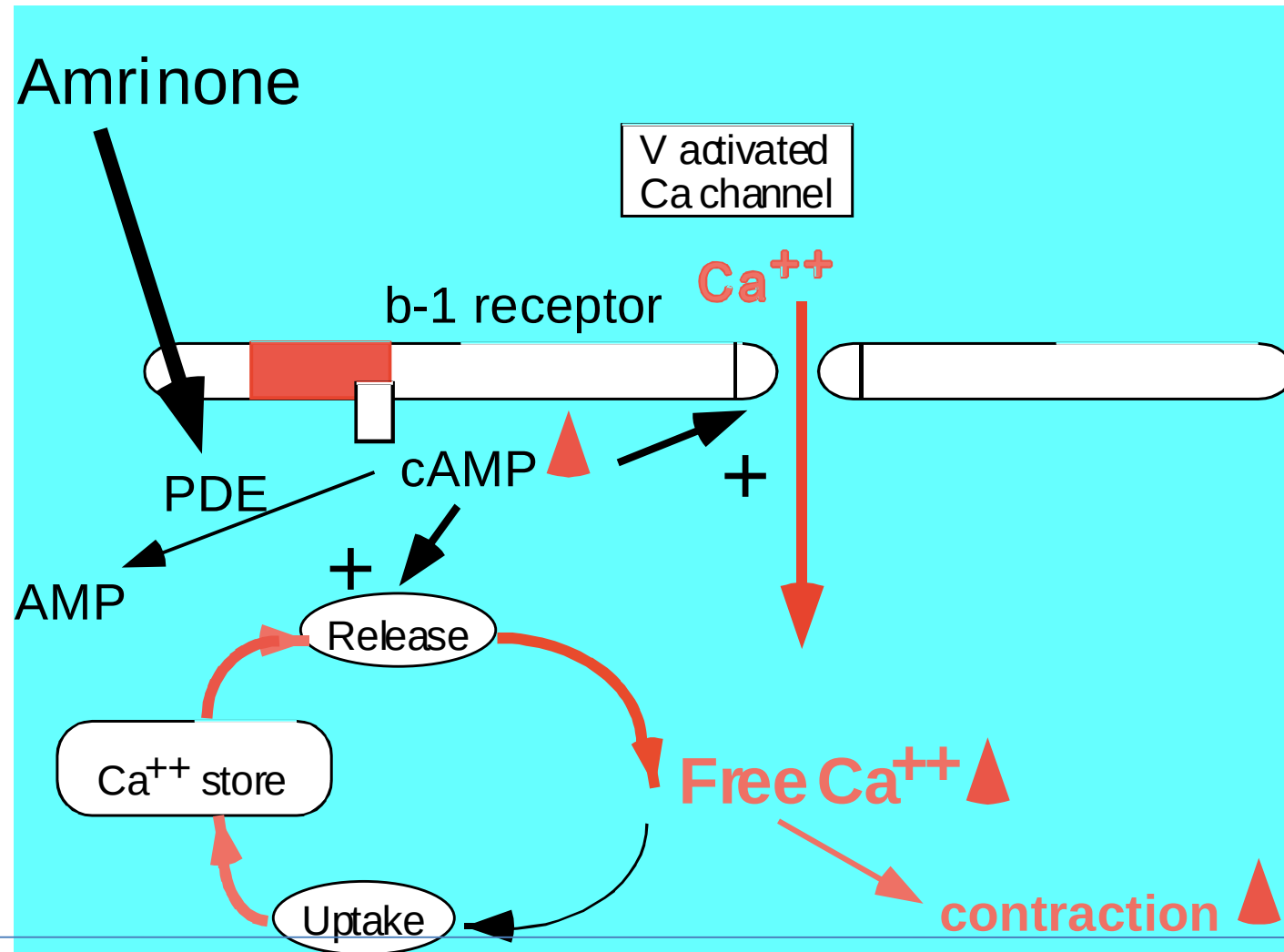
1. Administration of K is useful to treat dig overdose.
2. hypoK can produce dig toxicity (arrhythmias).
3. Kaliuretics can be a problem

Ca / Mg interactions with dig

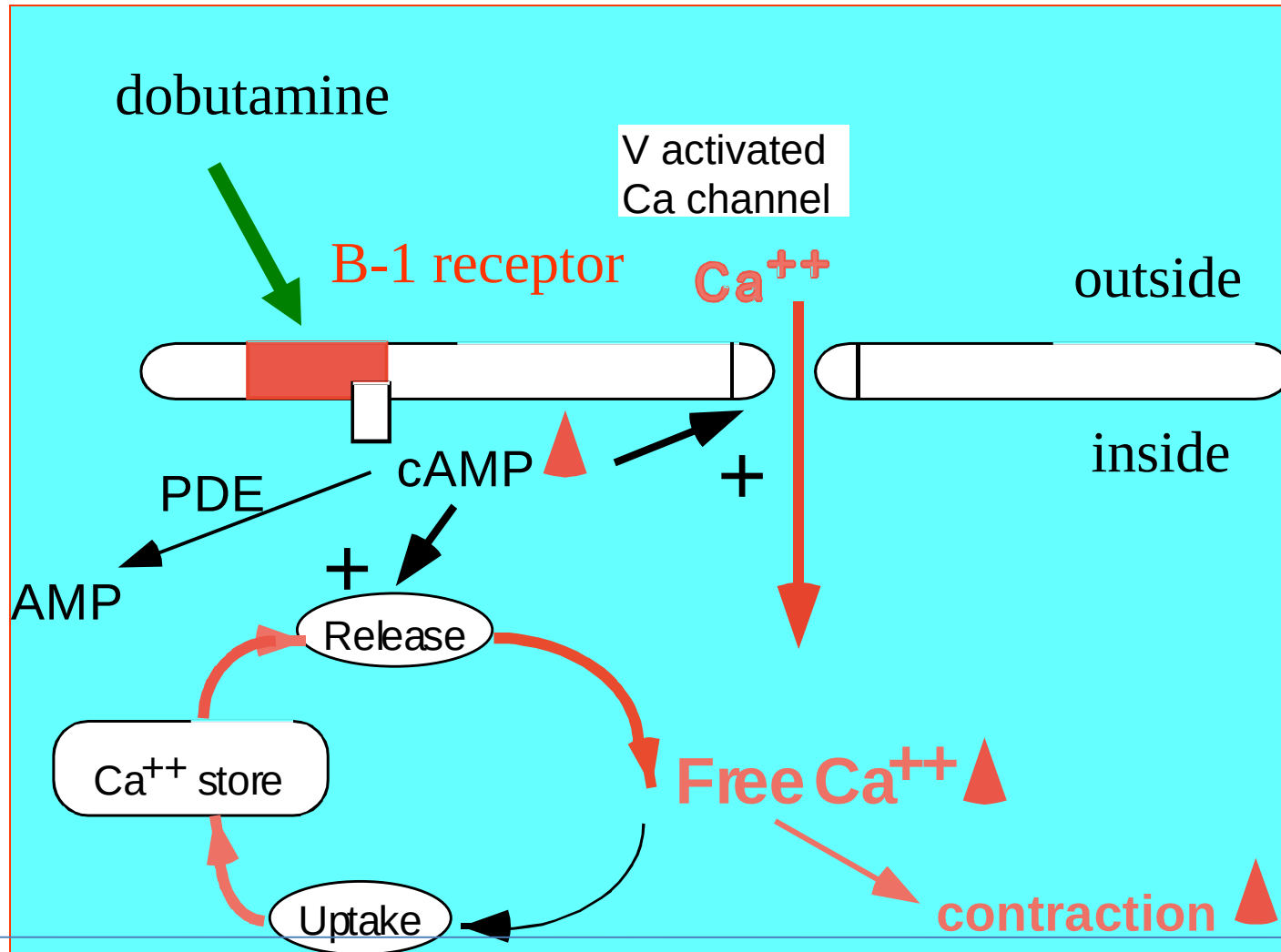
Hypercalcemia ↑ risk of dig toxicity (arrhythmias)
Additive inotropic effects

Hypomagnesemia has the same effect (↓ [K⁺]_i)

PDEIs increase heart contractility



Beta-1 agonists increase heart contractility

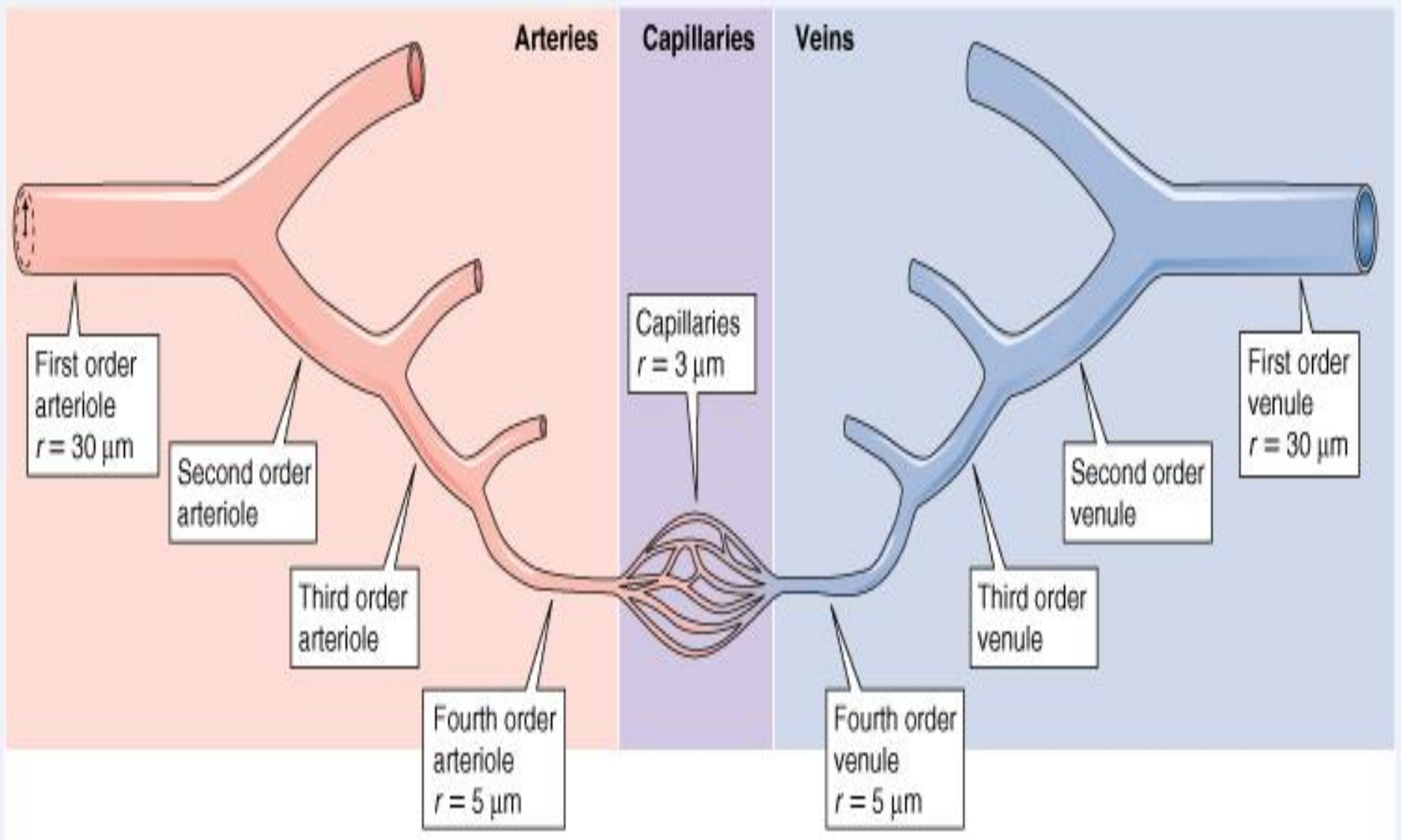


Physiology of Peripheral Circulation

Vascular System

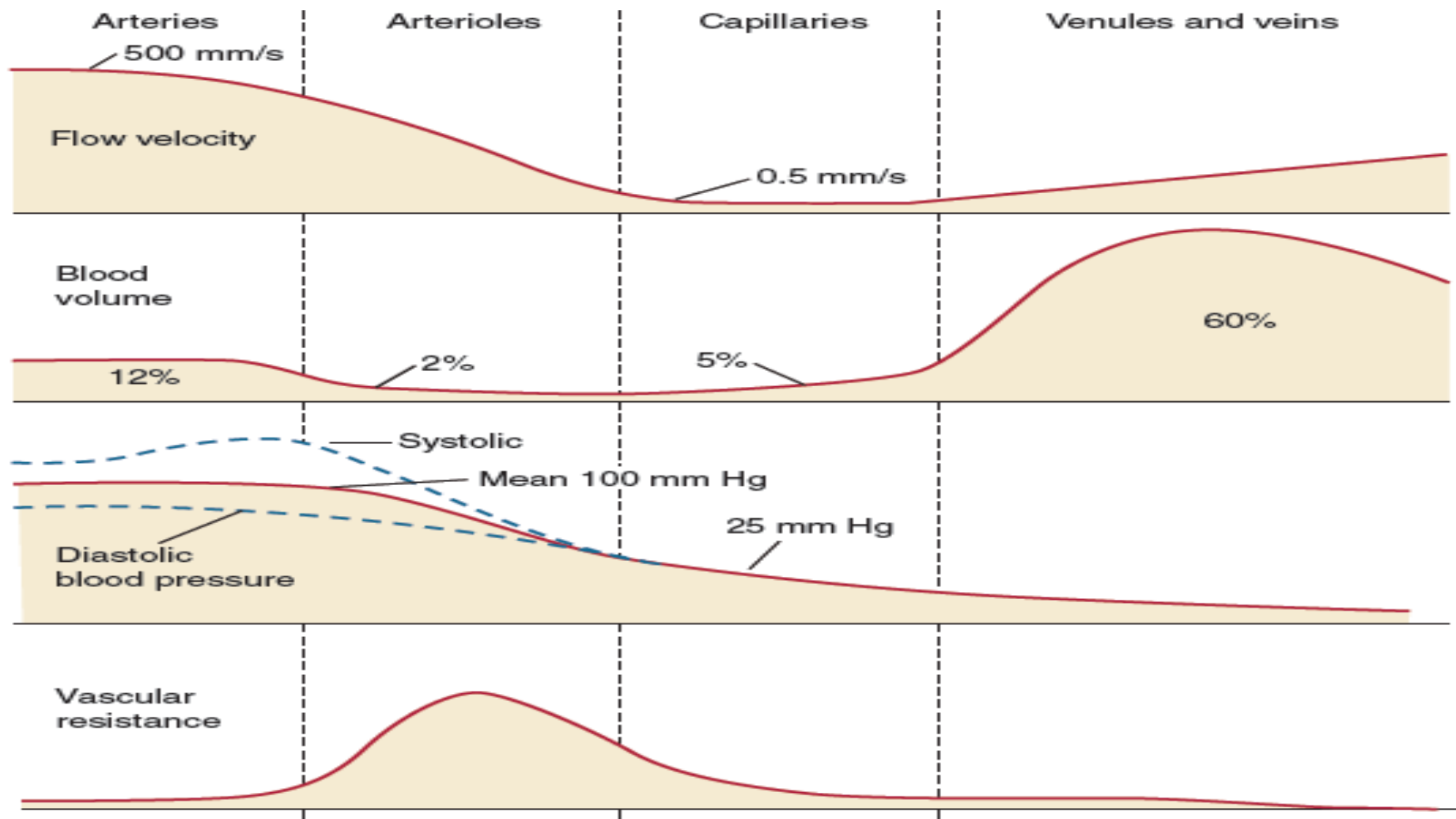
Blood Vessels

- Contain blood and allow pressure to be maintained
- Direct blood flow to specific organs
- Ensure adequate nutrition and removal of waste
- Regulate capillary blood pressure
- Permit direction of blood flow to be altered
- Together with the heart, permit regulation of blood pressure

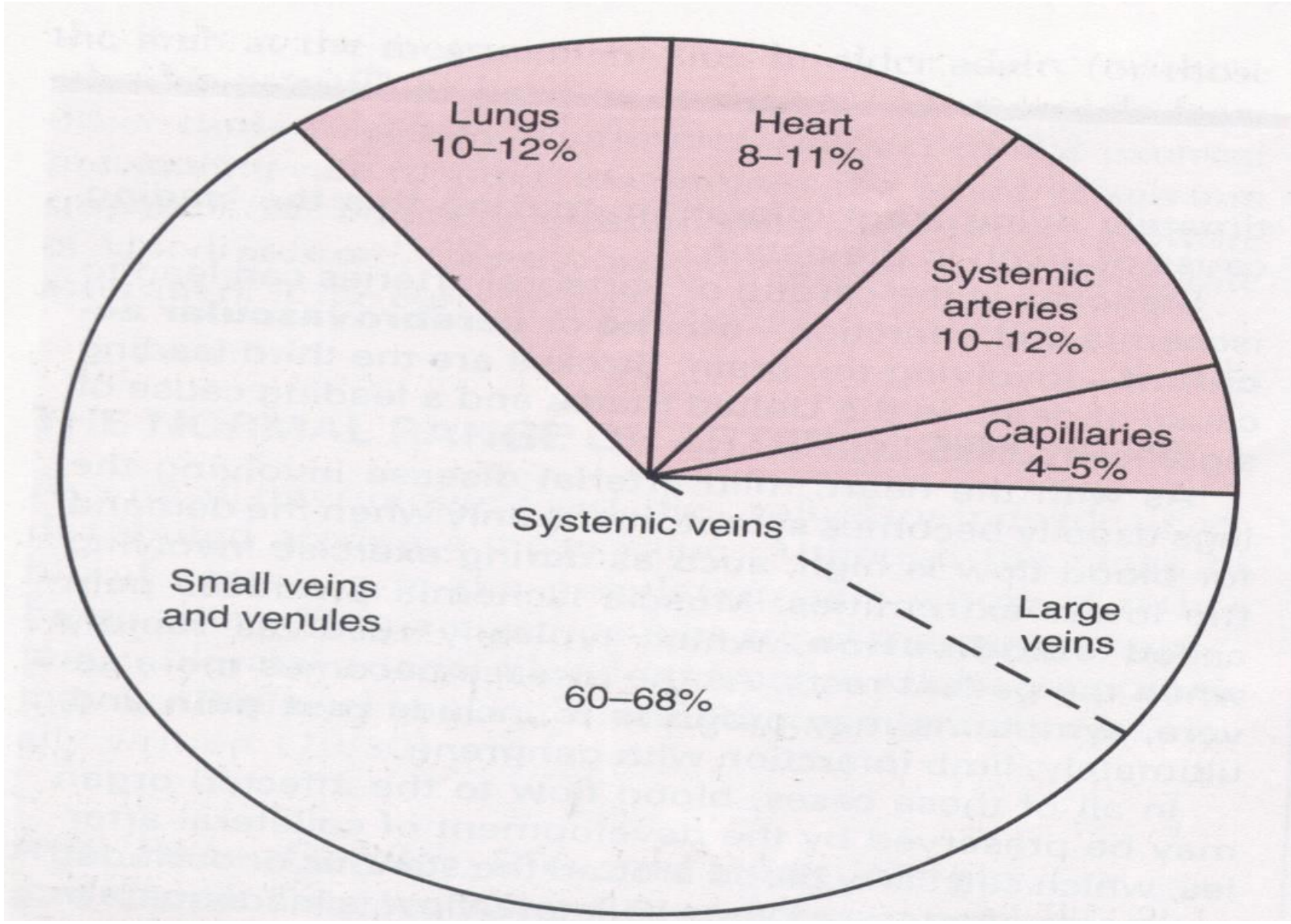


Blood Vessels

- **Specializations**
 - Aorta and elastic arteries → conductance vessels
 - Muscular arteries → distribution vessels
 - Arterioles → resistance vessels (pressure/flow regulation)
 - Capillaries → Exchange vessels
 - Venules and veins → Capacitance vessels (blood volume)
 - Shunt vessels → arteriovenous anastomoses (in fingers, toes, ears)
→ heat transfer

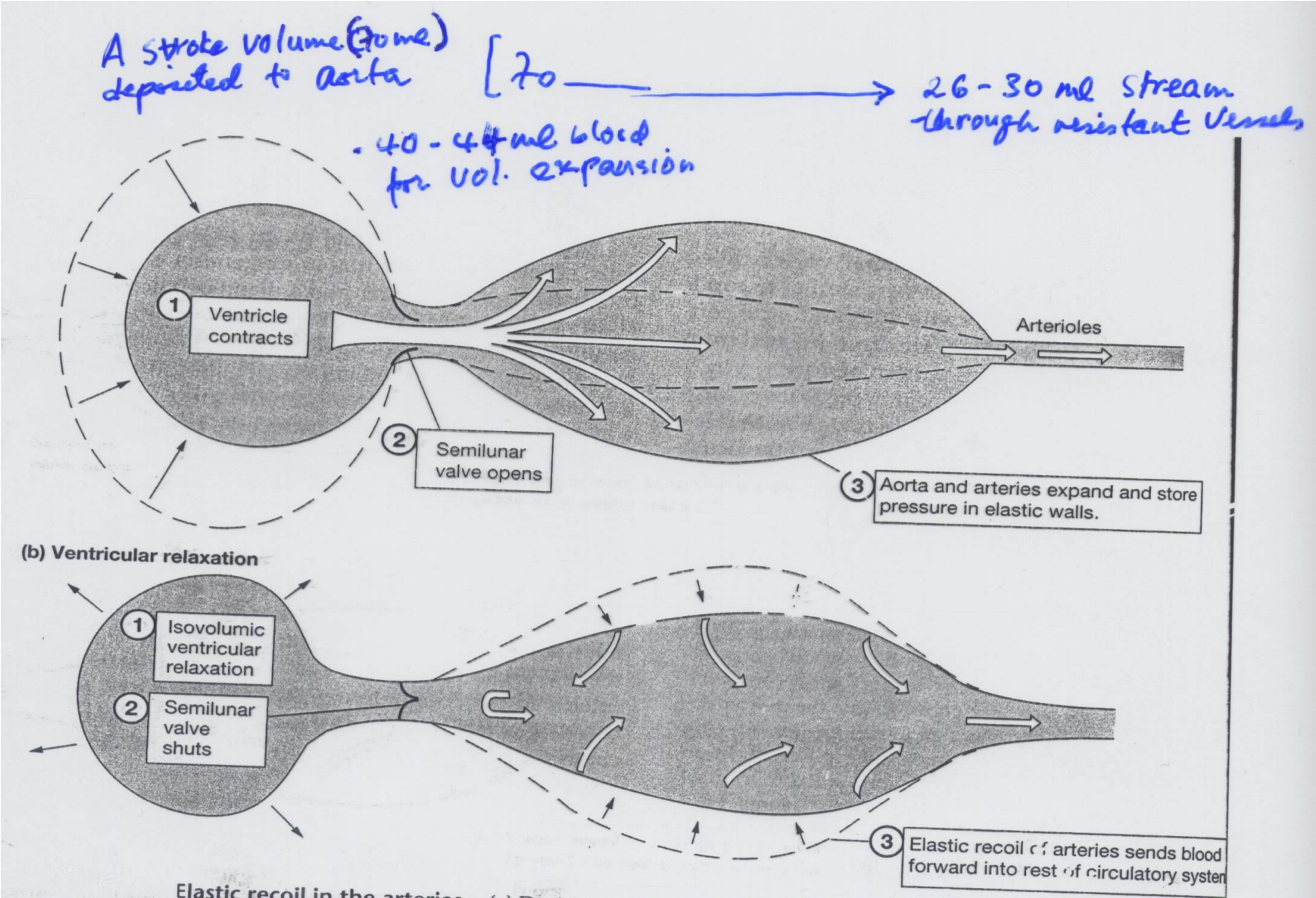


- Rapid flow → aorta (smallest cross-sectional area)
- Slowest flow → capillaries (largest cross-sectional area)



Arterial System

- Allow continues blood flow to the periphery
- LV ejection of blood to aorta → distention of aortic wall (peak at systole, decline at diastolic)
 - Part of the energy of ventricular contraction → forward flow during systole
 - Remainder → distend arterial tree (and stored as potential energy)
 - Release of this potential energy (by recoil of arterial wall) propel blood distally (*provide diastolic “run off”*)
- Alteration of systolic and diastolic propulsion provide continues (rather than pulsatile) blood flow to peripheral tissues



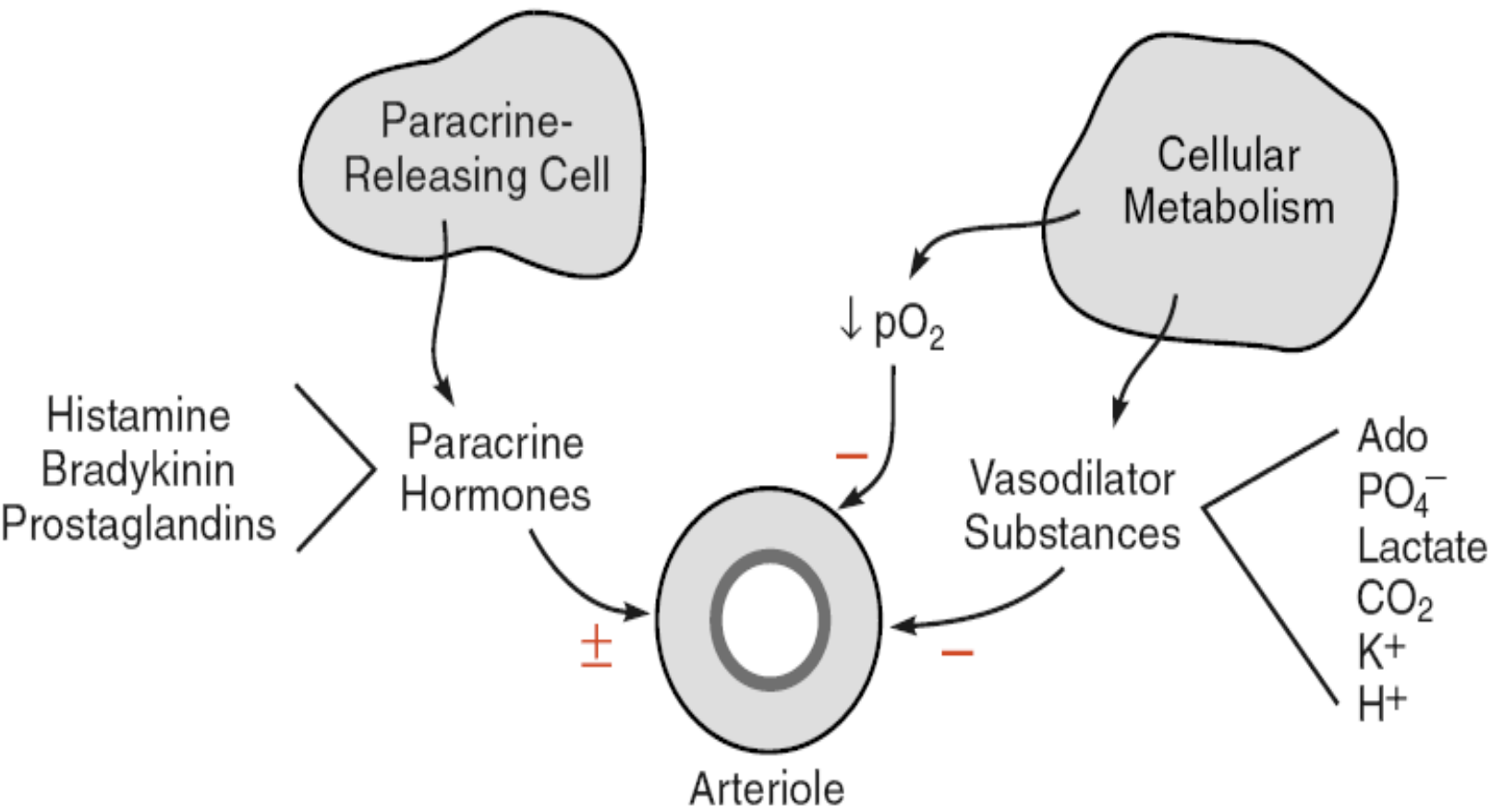
Arterioles

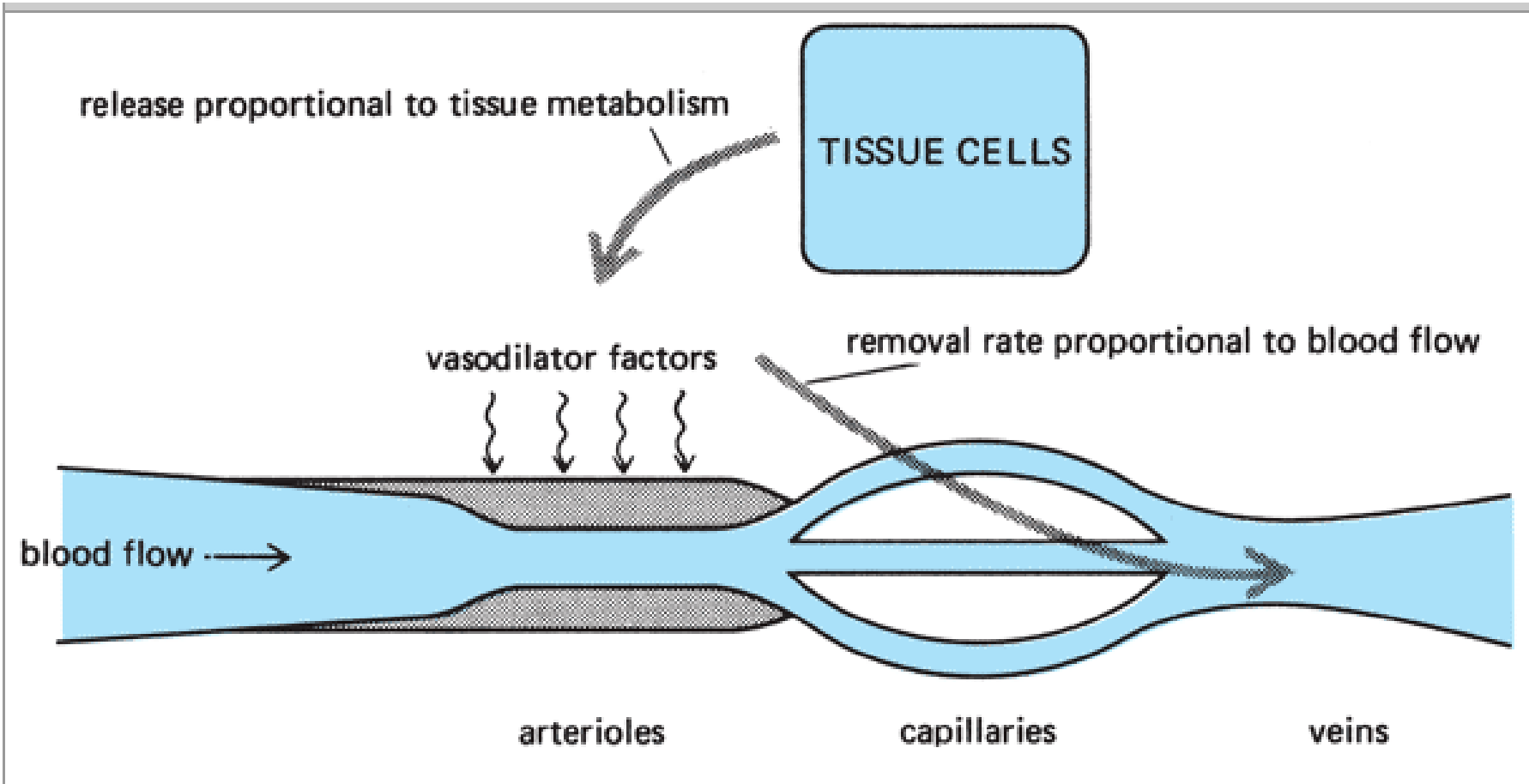
- Determine blood distribution to different organs
- Resistant vessels
- Few elastic fibers
- Have smooth muscles
- Constrict or dilate (radii subject to precise physiological control)
- **Control mechanisms**
 1. Local control
 2. Central control

Arterioles

1. Local control

- Individual organs possess inherent mechanism for altering resistance
- Metabolic autoregulation (Active hyperemia): exercise increased blood flow (dilation)
- Chemicals:
 - $\downarrow\text{PO}_2$, $\uparrow\text{PCO}_2$, H^+ , bradykinin and other metabolites cause vasodilation
 - Adenosine is also a potent vasodilator (coronary vessels)





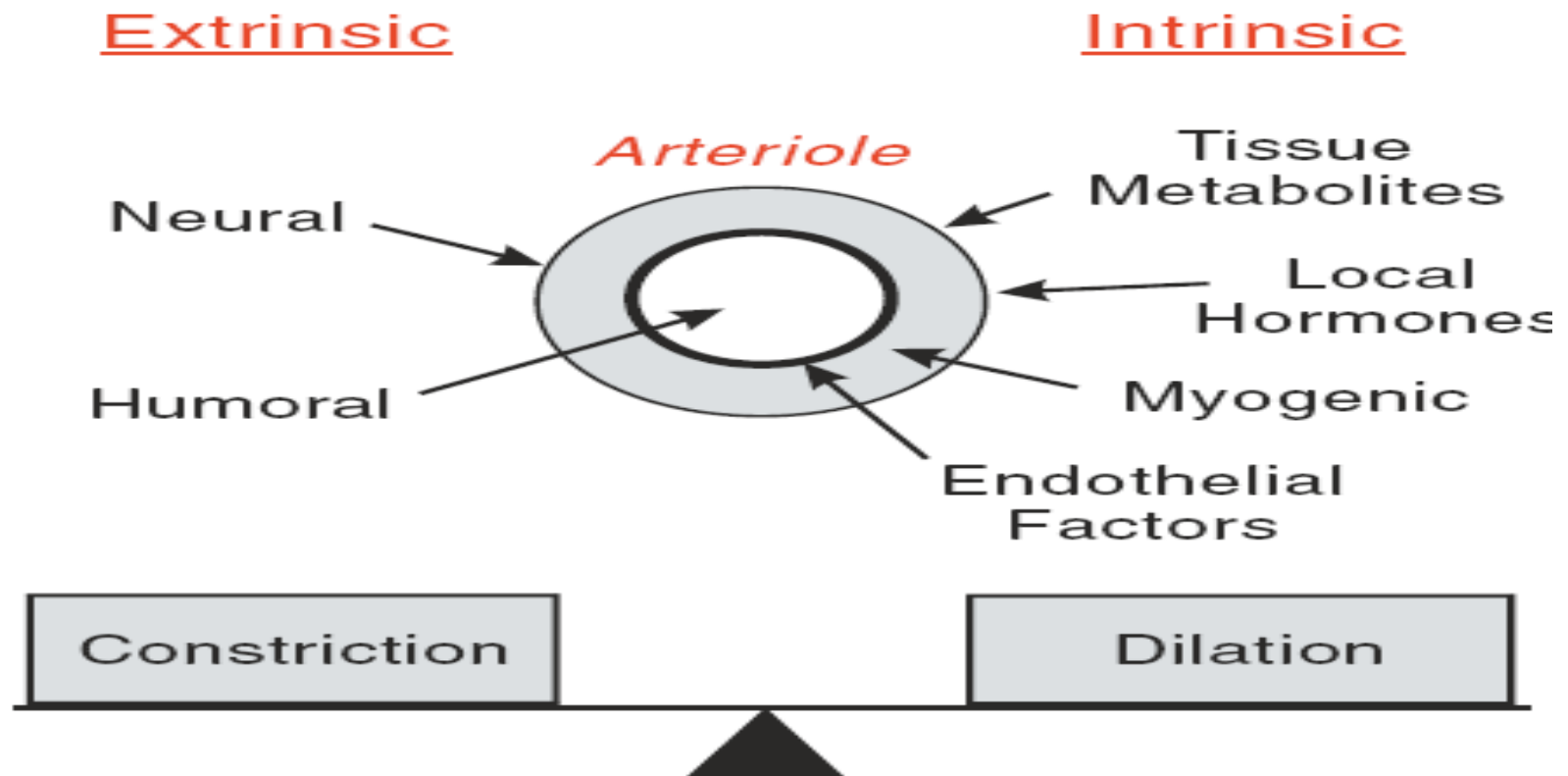
Arterioles

2. Central control

- Concerned with maintaining MAP (mean arterial pressure)
- Parasympathetic supply → Generally has no effect
- Sympathetic supply (NE):
 - On α receptors → arteriolar constriction
 - Reduces flow and increases MAP
 - On β_2 receptor → arteriolar dilatation (skeletal muscle + coronary vessels)

Arterioles

- Vascular tone** → state of vessel tone determined between constrictor and dilator influences



Capillaries

- Characteristics:
 - Have thin endothelial layer
 - No smooth muscle, no change in inner diameter ($\theta \approx 5\text{-}10\mu\text{m}$, $L \approx 1\text{mm}$, velocity of blood flow $\approx 1\text{mm/s}$)
 - Small capillary diameter & thin endothelial wall minimizes diffusion path
 - Narrow lumen, thus RBCs change shape to pass through
 - Large surface area $\rightarrow$ ideal for exchange (nutritional flow)

Capillaries

- Capillary density:
 - High in cardiac and skeletal muscles
 - Low in subcutis and cartilage
- Intracapillary pressure varies from arterioles to venules:
 - At rest = 25 and 12mmHg
 - During arteriolar vasoconstriction = 12 and 8mmHg
 - During arteriolar vasodilatation = 40 and 25mmHg

Capillaries

- Types:

1. Continuous capillaries – tight junctions

- Distance between endothelial cells is $\approx 5\text{-}30\text{nm}$
- In brain are low permeable to ions and most hydrophilic molecules (blood -brain barrier)

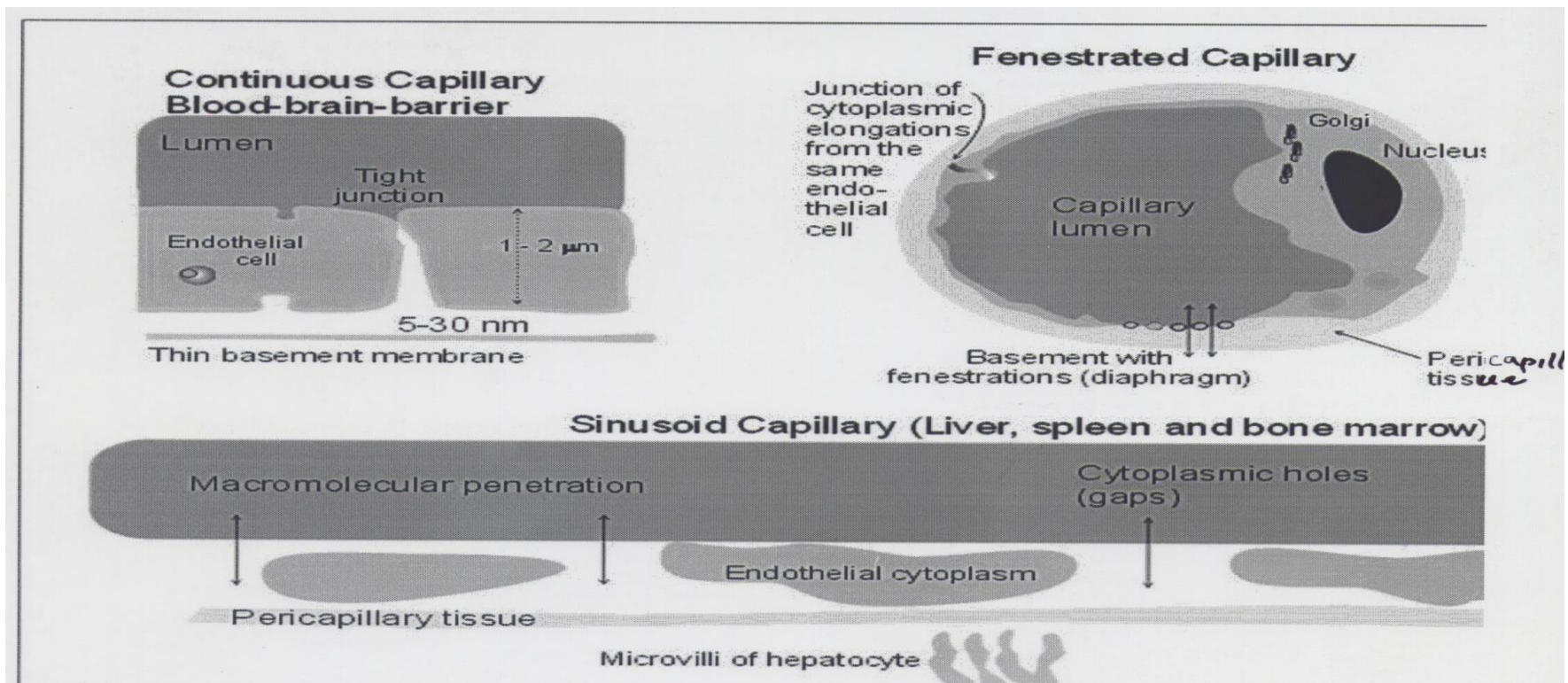
2. Fenestrated capillaries

- Found in organs that transport lots of water
 - Bowels, glomerular capillaries, pancreas and salivary glands
- Fenestrae keep plasma proteins back

Capillaries

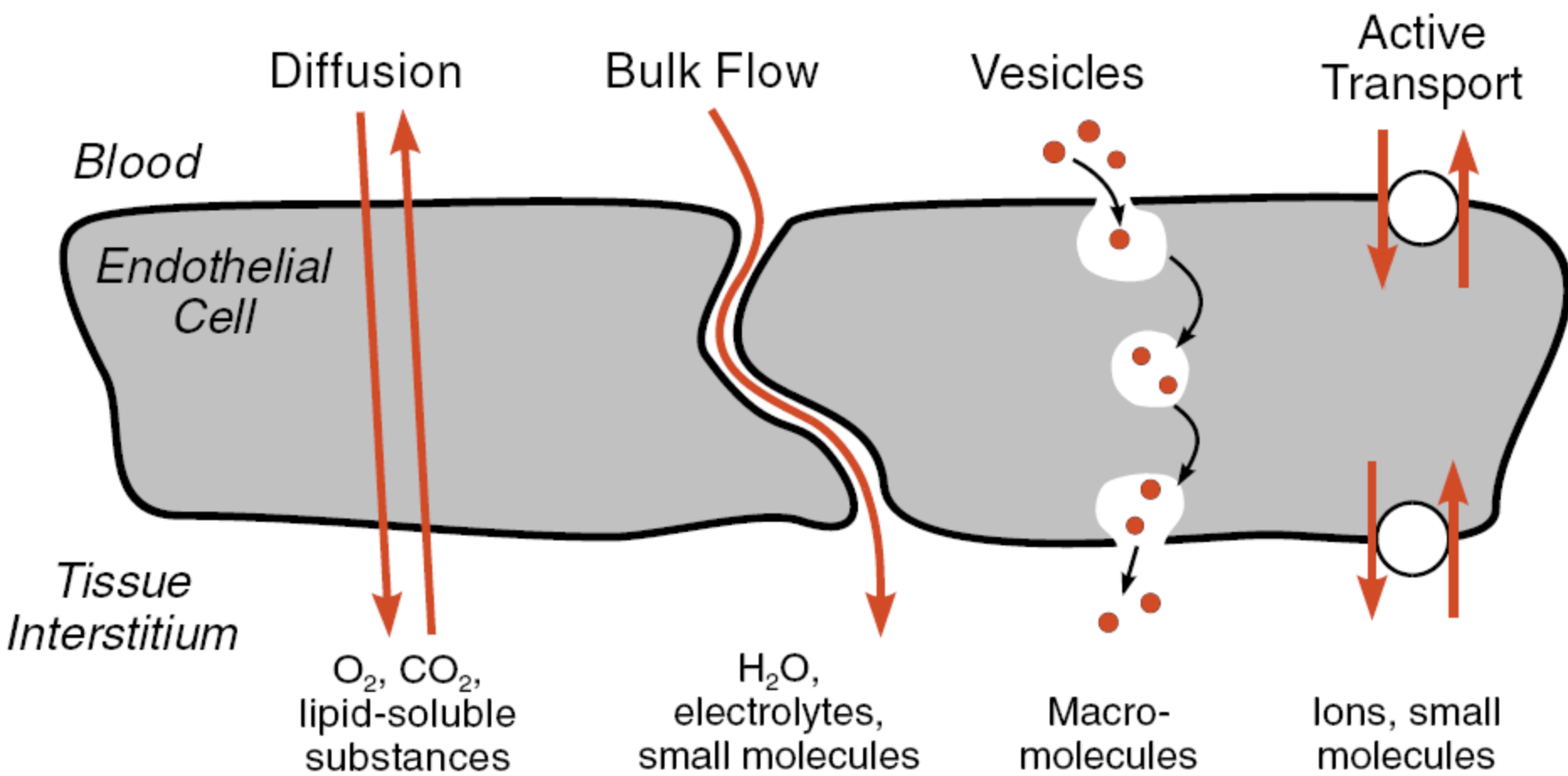
3. Sinusoid capillaries

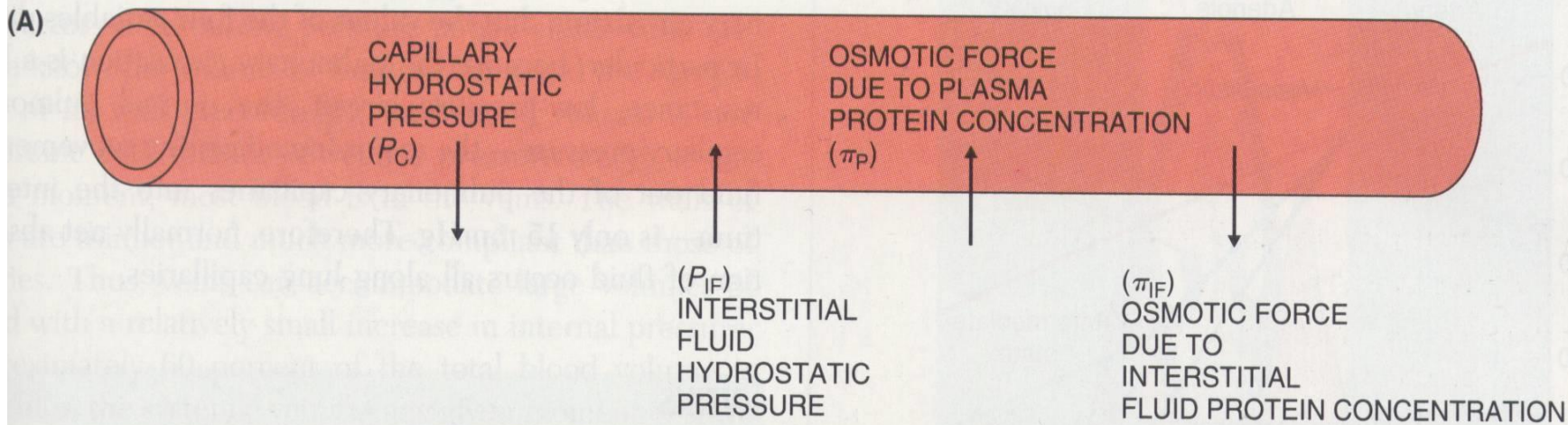
- Have broad openings between endothelial cells
- Found in tissues bathed in plasma (liver, spleen, and bone marrow)



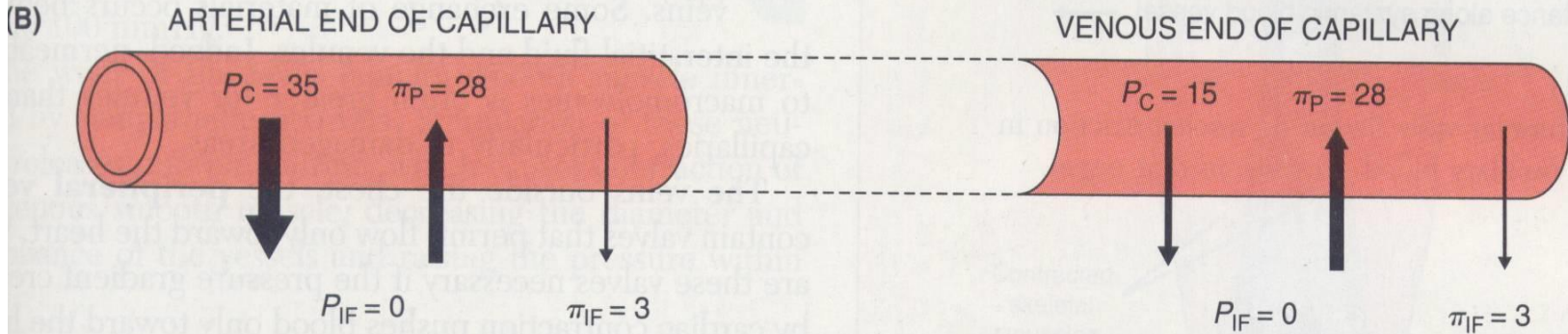
Capillary Exchange

- Starling's Law:
- Movement across capillary is influenced by:
 - Nature of substance (size, shape, lipid solubility)
 - Balance between hydrostatic and colloid osmotic pressure across the membrane
 - Capillary surface area available for exchange
 - Physical characteristics of the capillary membrane (continuous, fenestrated, sinusoid)





NET FILTRATION PRESSURE = $(P_C - P_{IF}) - (\pi_P - \pi_{IF})$



NET FILTRATION PRESSURE = $(35 - 0) - (28 - 3) = 10 \text{ mmHg}$

NET FILTRATION PRESSURE = $(15 - 0) - (28 - 3) = -10 \text{ mmHg}$
10 mmHg FAVORING ABSORPTION

Venous System

- **Veins**

- Thin walled, Low resistant, Low pressure and Highly distensible (compliant) vessels
- Small amount of elastic tissues and smooth muscle veins of dependent parts of the body have valves
- **Valves** prevent backflow of venous blood
- **Pressure**
 - ~ 10mmHg at capillary end
 - ~ 0 mmHg at entrance of vena cava to RA

Venous System

- **Function:**
 - Pressure regulation (ΔP) → Constrict and enlarge
 - Store large quantity of blood (>60%) (making blood available when required)
 - Propel blood forward by means of “**venous pump**”
 - Regulate CO

Venous System

- Venous return is improved by:
 - **Skeletal muscle pump** of lower extremities: assist venous return during quite standing, walking or running
 - **Venous valves**: (present only in extremities) are important adjuncts to skeletal muscle pump
 - **Respiratory pump**: helps to pump blood back to the heart during inspiration (decreased intrathoracic pressure).

Venous System

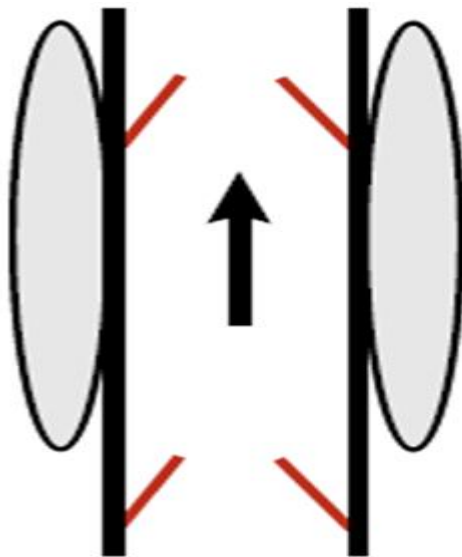
- Skeletal muscle pump
 - a. Standing at rest
 - Valves are open → Blood flow upward towards the heart
 - b. Contraction of the muscle
 - Compresses the veins so that the increased pressure in the veins drives blood upwards and closes the valves at lower region just below point of muscle contraction

Venous System

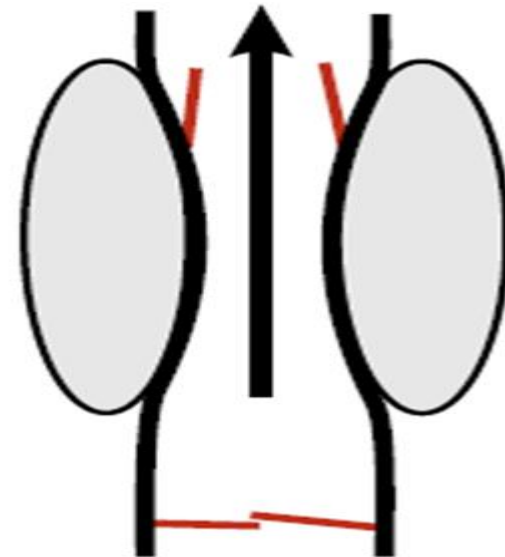
c. Immediately after muscle relaxation

- The pressure in the previously compressed venous segment falls and the reverse pressure gradient causes the upper valve to close (to prevent back flow)

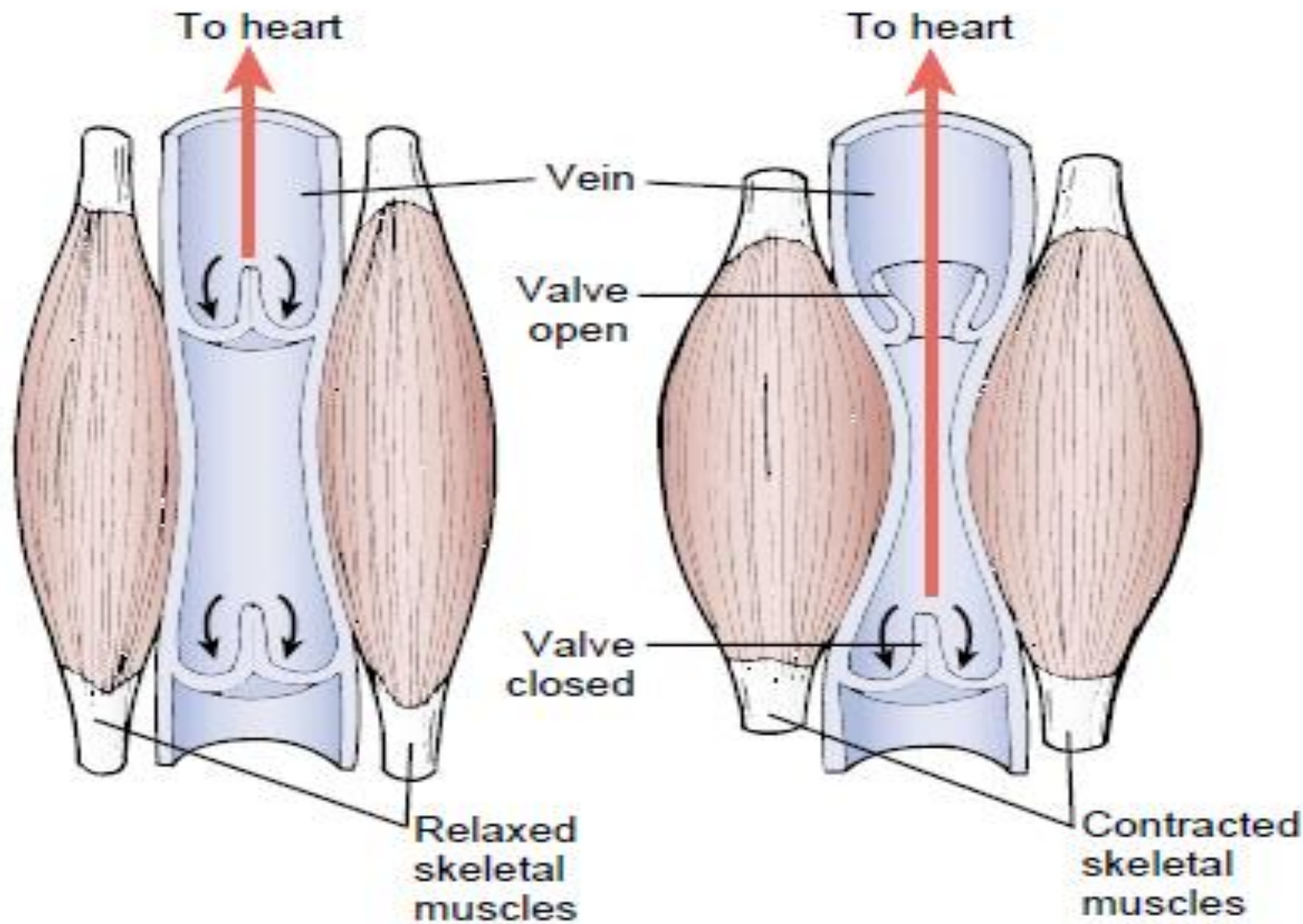
Skeletal muscle pump



Relaxed



Contracted

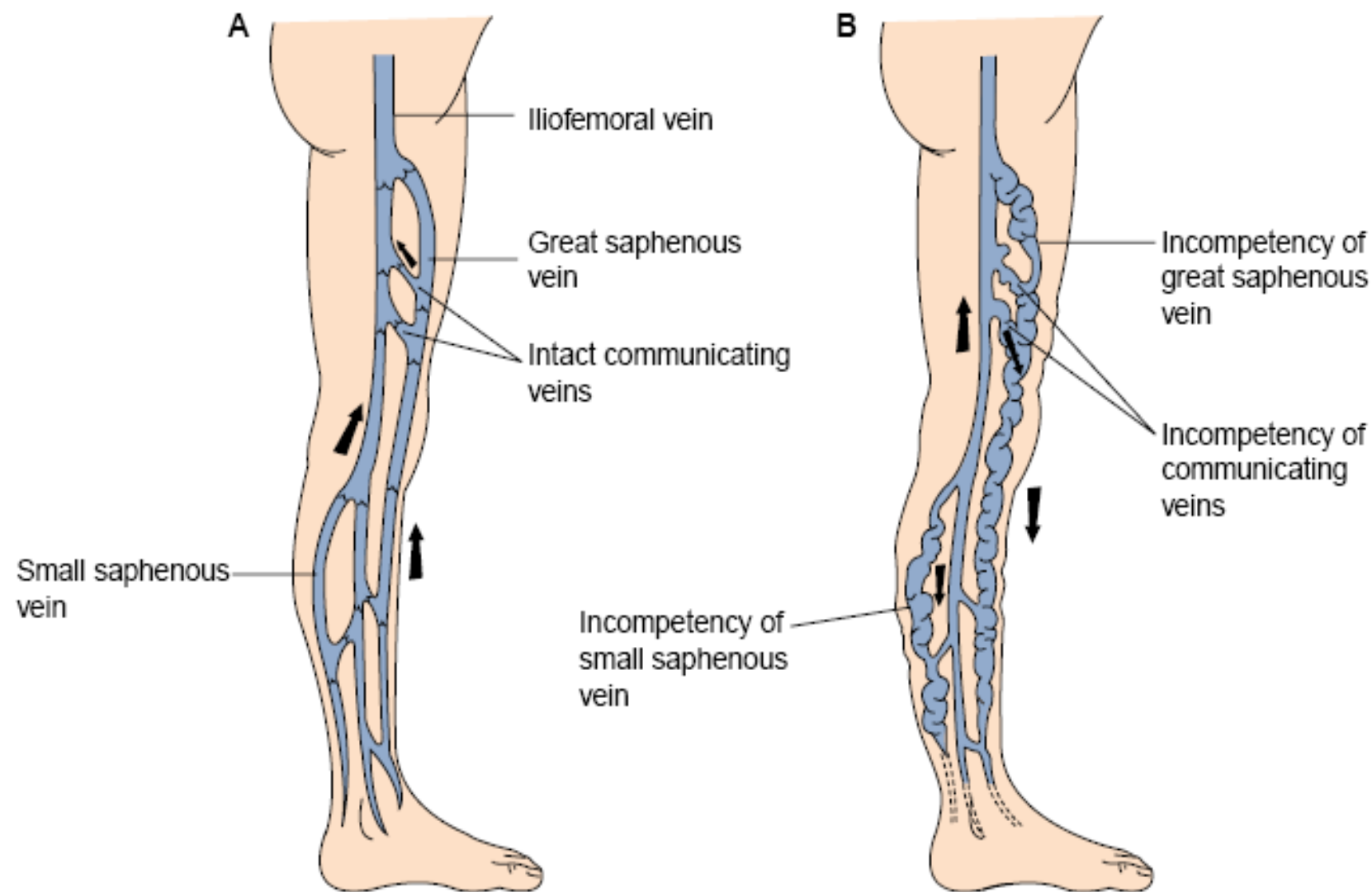


Venous System

- **Varicose Veins**

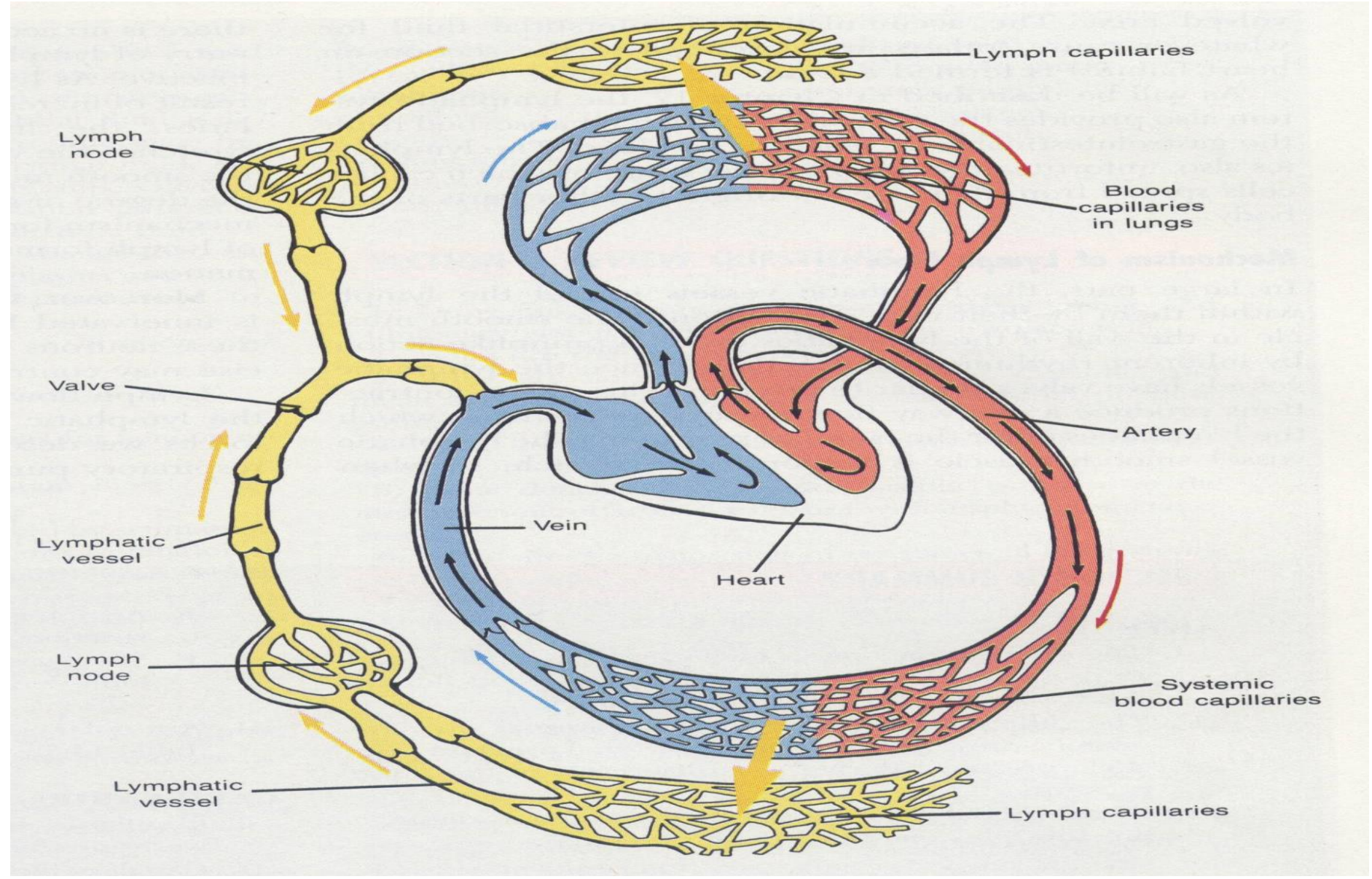
- Dilation and valvular insufficiency of dependent veins, especially of the legs → (valves become knotted and swollen)
- **Causes:** inheritance, inflammation, thrombosis, occupational predisposition/injury, prolonged obstruction as in pregnancy

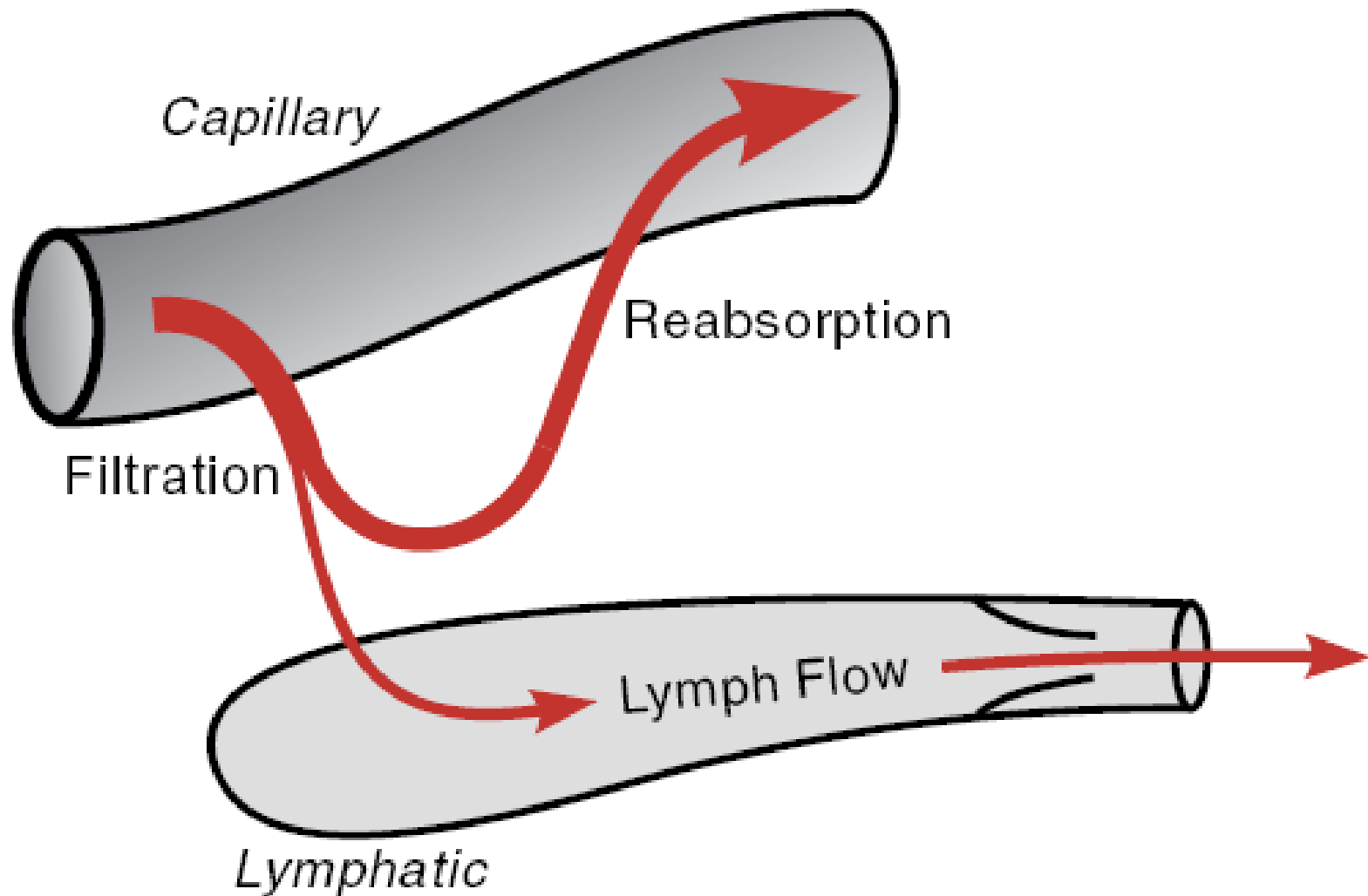
Chapter 10: Alterations in Blood Flow



Lymphatic System

- Lymphatic's
 - Porous, thin walled highly permeable (dying blood cells, bacteria)
 - All vessels empty into thoracic duct (left) and right lymphatic ducts
 - Finally drain into subclavian and internal jugular vein on their respective sides
 - Lymph flow – sluggish, 2-4L/day
 - Flow increased by: muscular activity, massage, increased volume, pressure, inflammation





$$\text{Filtration} = \text{Reabsorption} + \text{Lymph Flow}$$

Lymphatic System

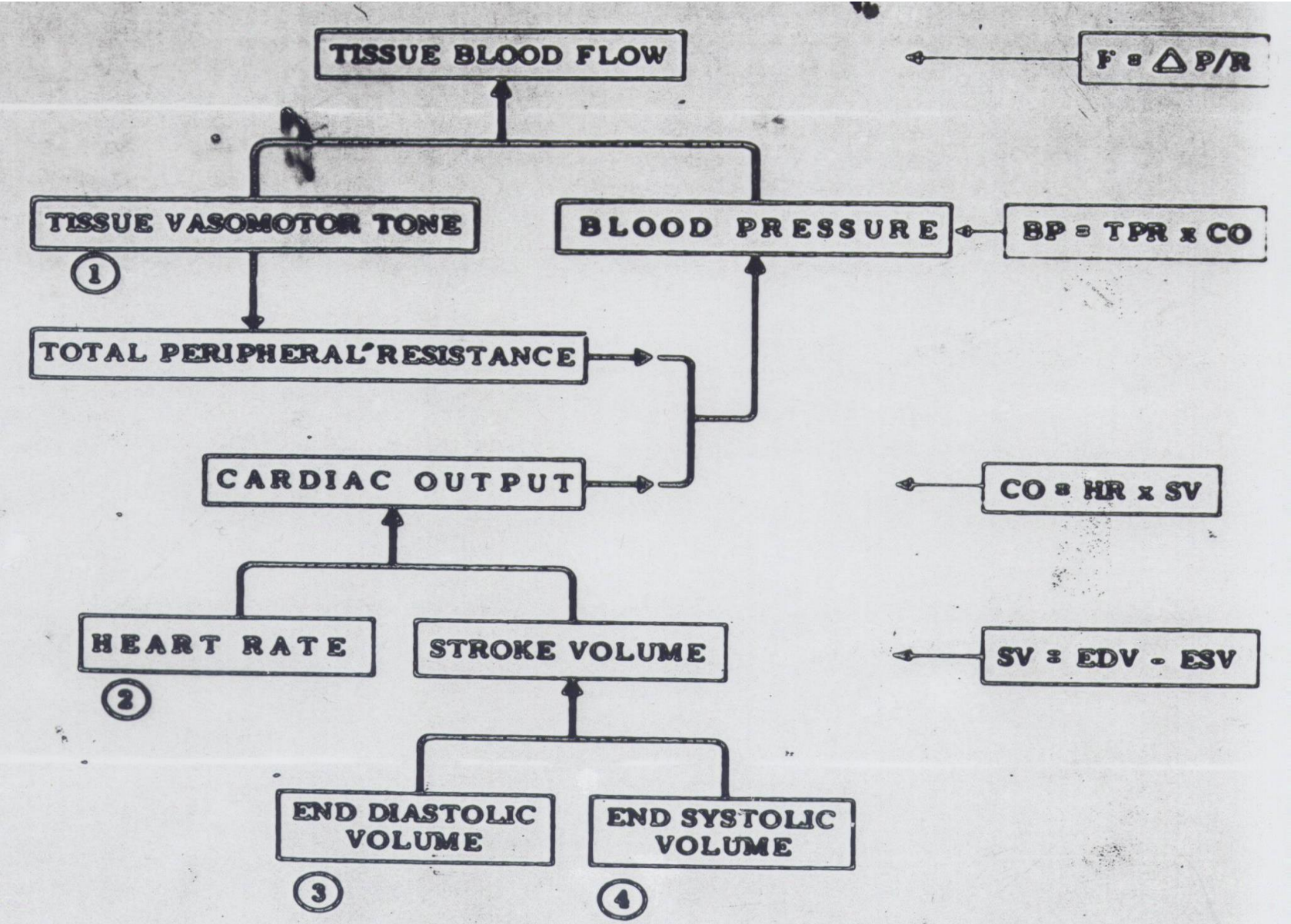
- **Function**
 - Clear excess fluid, proteins, lipids and foreign metabolites of interstitial space
 - Secondary interstitial drainage system, thus supplement venous drainage

Regulation of Peripheral Circulation

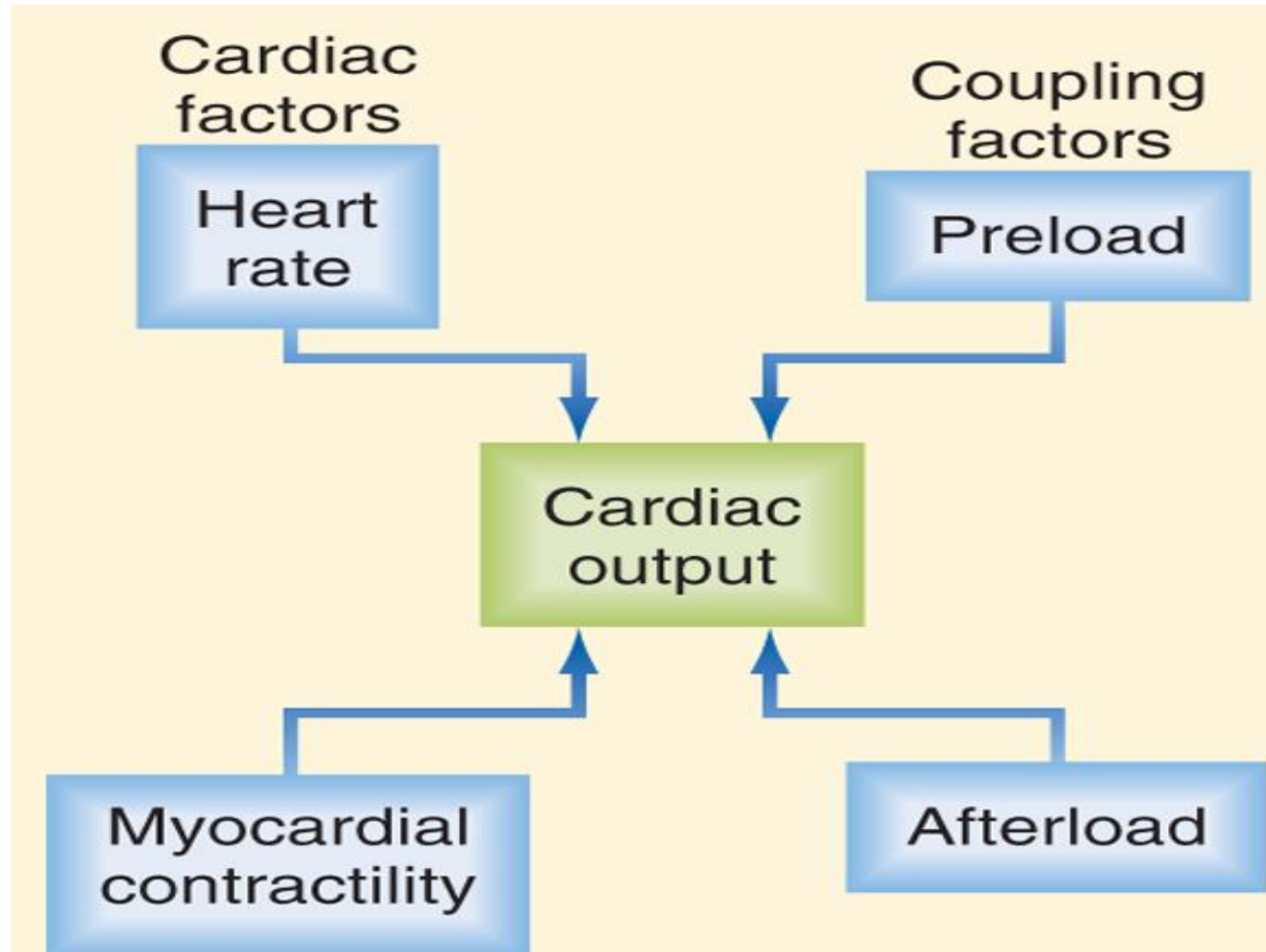
- The most important parameters controlled are:
 - Cardiac output
 - Distribution of blood to the different organs and tissues
 - Arterial blood pressure
 - Blood volume

Regulation of Peripheral Circulation

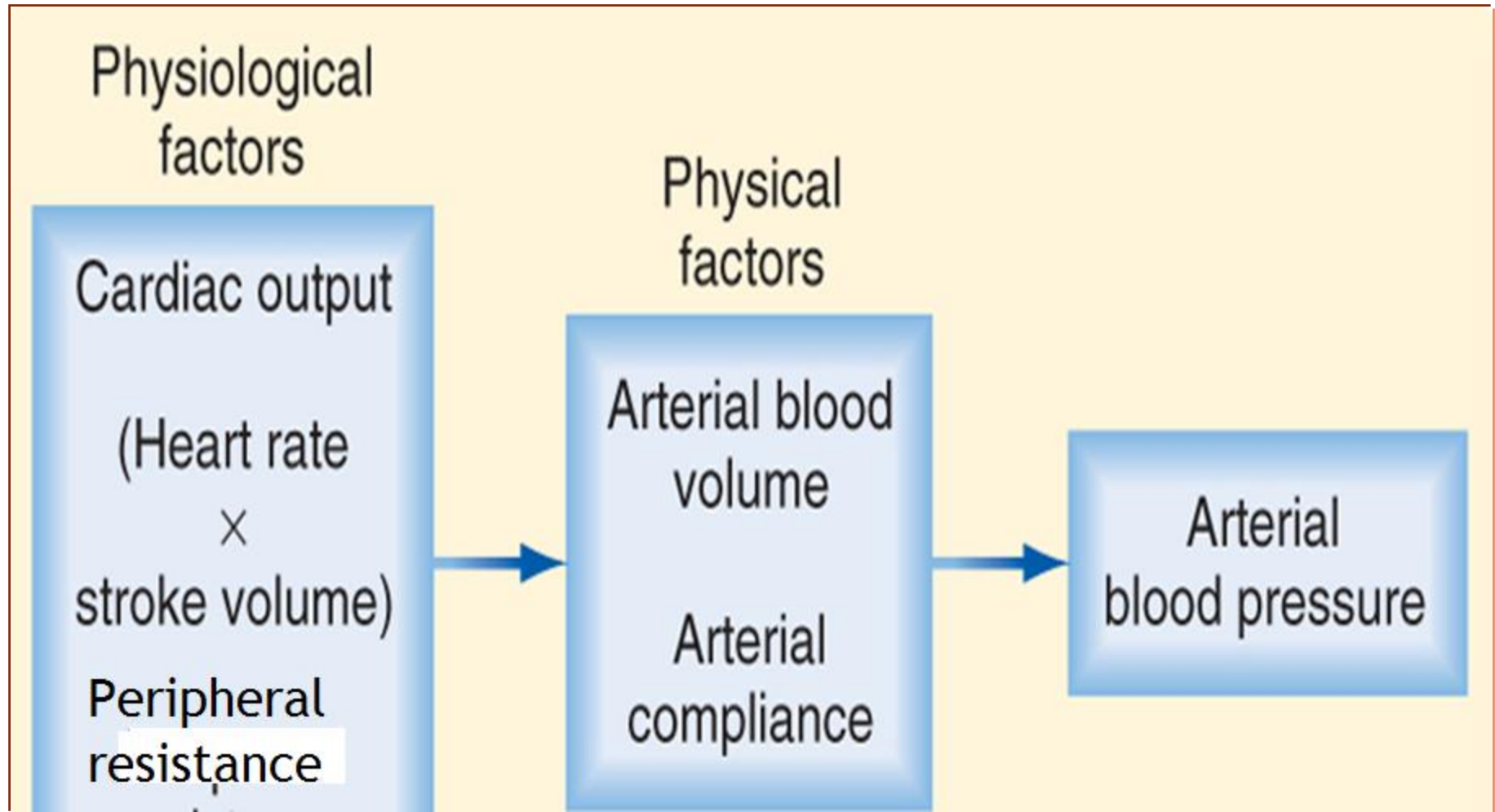
- Regulation of overall circulation:
 - Regional and supra-regional (higher level – heart and vasomotor center in the Medulla oblongata)
 - Effectors → heart and blood vessels
- Crucial mechanism:
 - Adjustment of **TPR** and **CO** to maintain pressure gradient for flow through the vascular system
 - Adjust vessel capacity and blood volume



Factors influencing cardiac output



Factors influencing BP



Edema

- Accumulation of excess fluid in interstitial space.
- Develops in dependent areas (eg. Feet)
- More evident in expansion limited tissues (eg. Around the eye)
- Causes of Edema:

1. General increase in venous pressure

- Increase venous pressure → increase venular pressure → increased pressure in arterial end of capillary
- **Result** = increased filtration and interstitial fluid accumulation
- Common in: Pregnancy, prolonged standing, CHF

Edema

2. Hypoproteinemia

- Decreased plasma protein → decreased plasma oncotic pressure
→ more filtration → fluid accumulation
- Causes of Hypoproteinemia:
 - Nephrosis (albumin lose in urine)
 - Inadequate protein in diet (nutritional edema)
 - Severe burns (capillary damage)

Edema

3. Lymphatic obstruction

- Lymph vessels return the small amount of protein “leaked” through capillaries
- If not returned, accumulate in interstitial space
- Thus prevent return of fluid into venular end of capillaries →

Edema

- Elephantiasis (Filariasis) → massive edema in leg → (lymph vessel obstruction by Tse tse fly)

Edema

4. Local edema

- Extensive venous obstruction with elevation of venous and capillary pressure
- Eg. Wide spread venous thrombosis

5. Left ventricular failure or mitral valve stenosis

- Pulmonary capillary hydrostatic pressure $>$ oncotic pressure = pulmonary edema
- Pulmonary edema causes interference with gas exchange in the lungs

Edema

6. Capillary injury

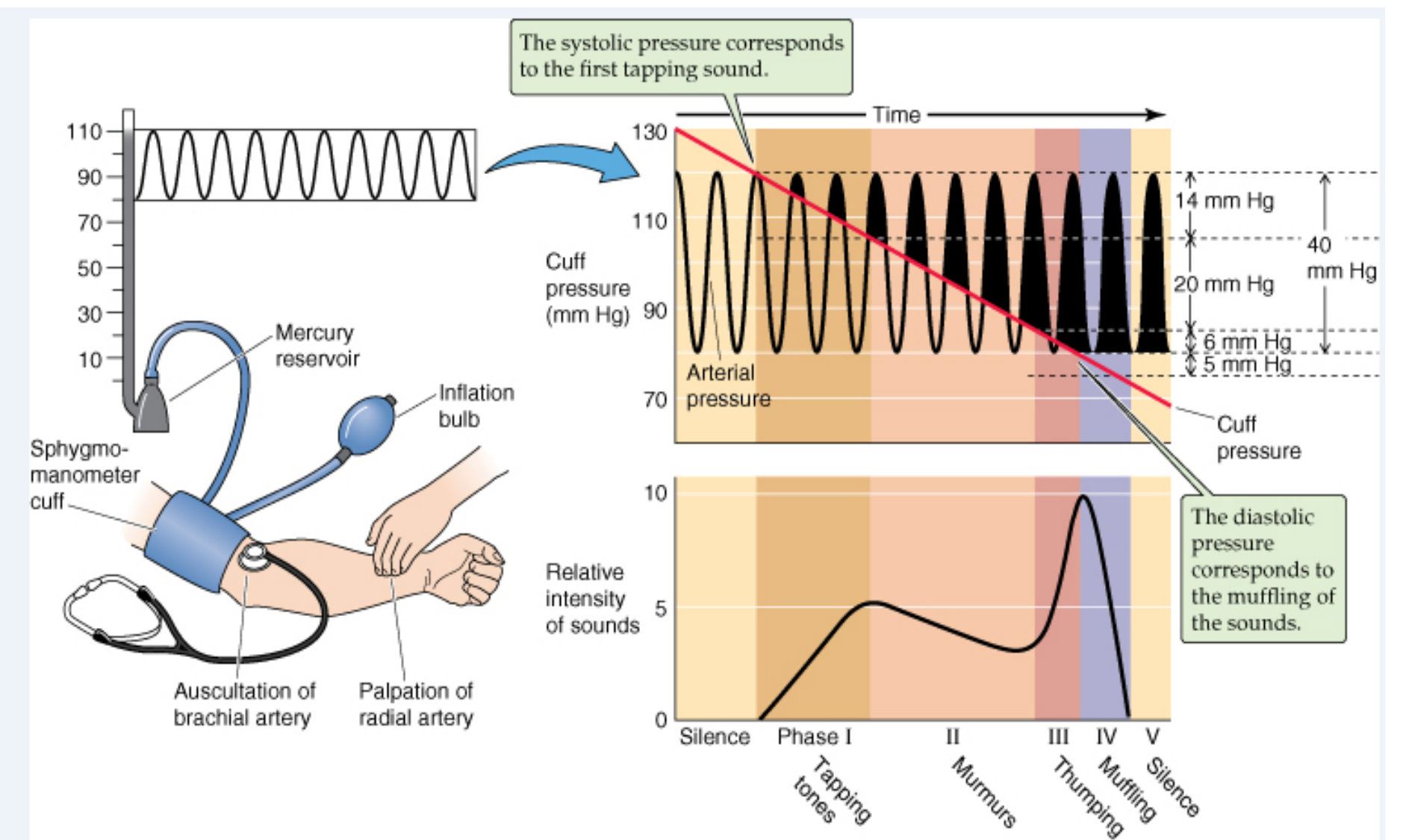
- Toxins, severe burns etc. increase capillary permeability greatly
- Fluid and protein leakage into interstitial space → edema
- Treatment of burns is replacement of lost fluid and plasma proteins
- .
- .
-

Blood Pressure Regulation

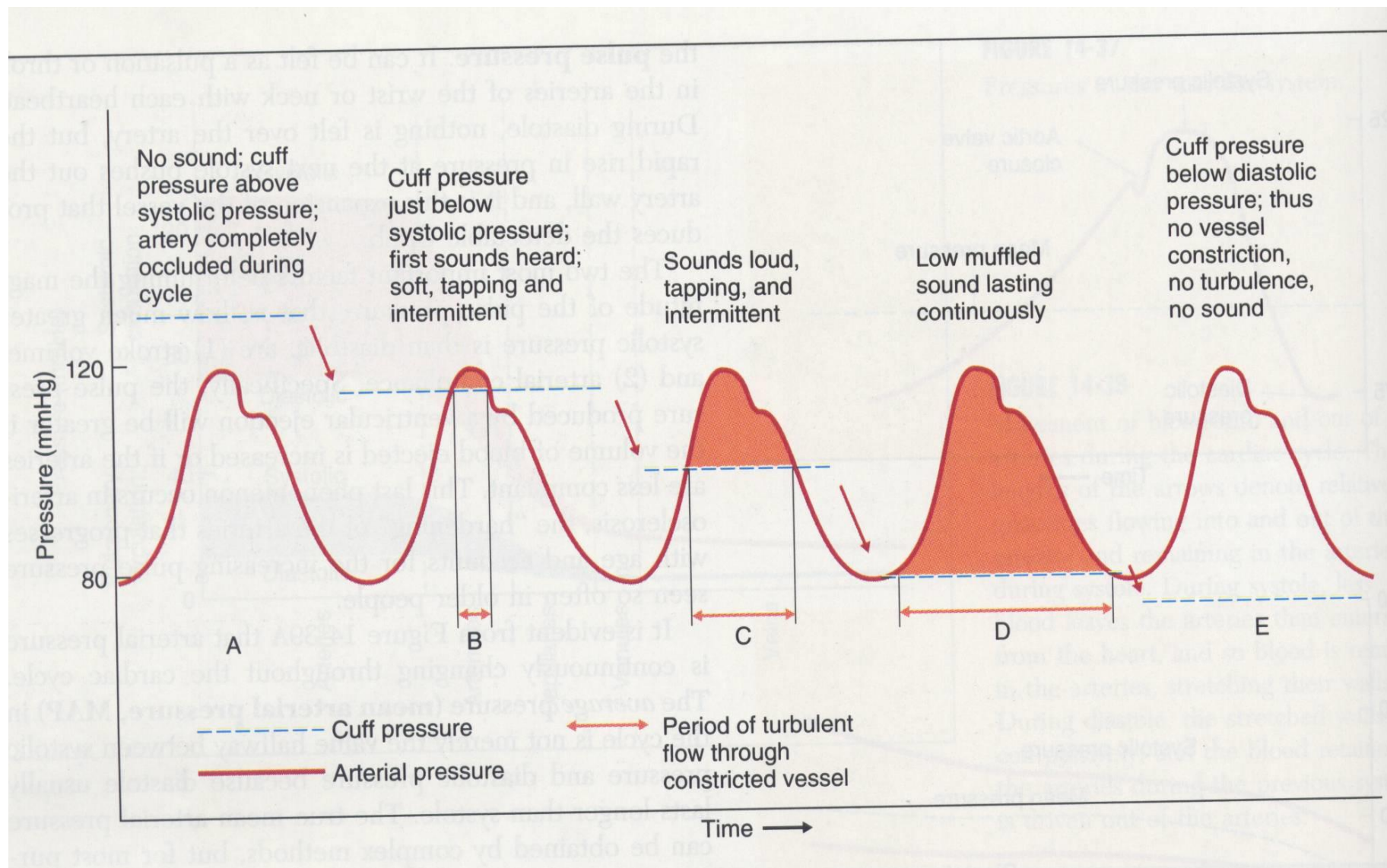
Blood Pressure

- Pressure exerted by circulating blood upon the walls of blood vessels
- BP varies between maximum (systole) and minimum (diastole) during each beat
- Other influences on BP: gravity, valves in veins, pumping from contraction of skeletal muscles...
- **Normal Blood Pressure**
 - Average blood pressure: age dependent
 - 95/65 mmHg → 1year
 - 100/65 mmHg → 6-9 years
 - 110/65-140/90 mmHg → Adults

Blood Pressure Measurement

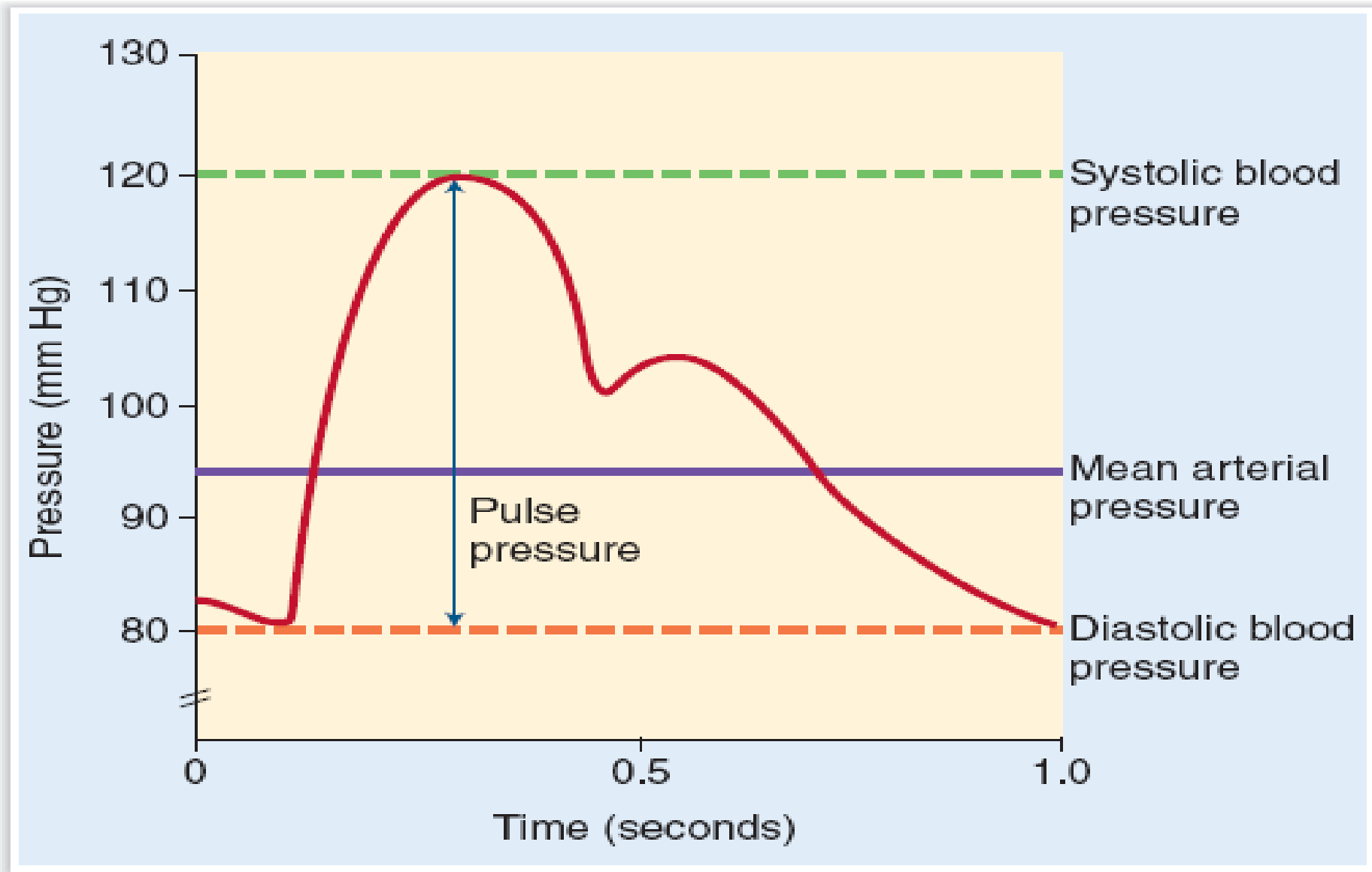


Blood Pressure Measurement

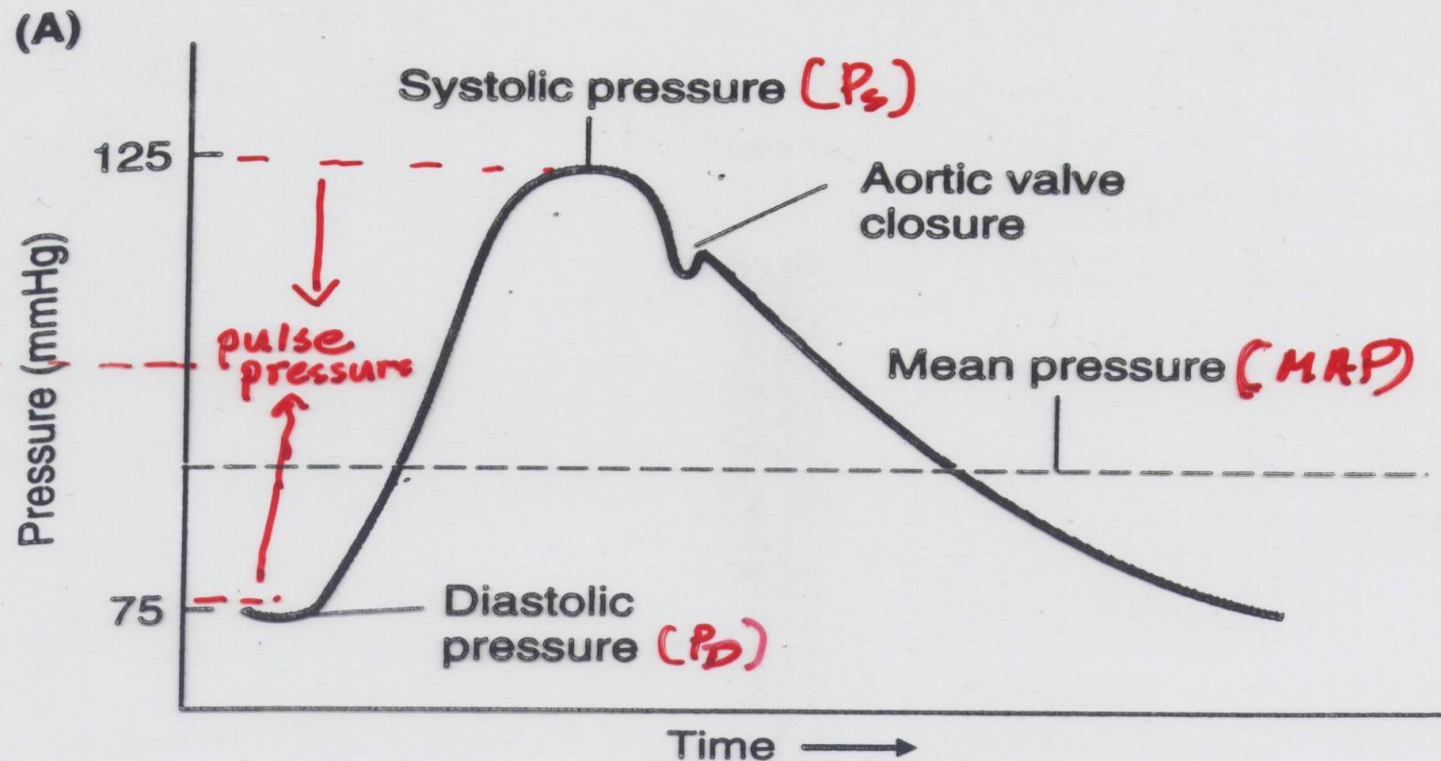


Blood Pressure

- Mean Arterial Pressure (MAP)
 - Average BP over a cardiac cycle
 - Determined by CO, systemic vascular resistance (SVR) and CVP
 - $MAP = (CO \times SVR) + CVP = CO \times TPVR$
 - $MAP \sim P_d + 1/3(P_s - P_d)$
 - Pulse Pressure ($P_s - P_d$) → up & down fluctuation of arterial pressure



Arterial blood pressure



• change in
arterial
pressure

$$MAP = P_D + \frac{1}{3} (P_s - P_D)$$

Blood Pressure

- Mean Arterial Pressure (MAP)
 - CO is determined by:
 - Blood volume + Venous tone
 - Stroke volume + Heart rate
 - TPVR is determined by:
 - Neuronal (sympathetic tone)
 - Hormonal factors (EP, Ang-II...)
- Mean BP (due to pumping and resistance) decreases as blood moves away from the heart

Blood Pressure

- Physical factors influencing arterial blood pressure
 - Cardiac output (HR X SV)
 - Volume of blood:
 - High salt intake, increased volume and pressure
 - Also dependent on ANS and RAAS
 - Resistance (R) – depends on:
 - Radius, length, viscosity and smoothness of blood vessels
 - Lumen of blood vessels: constriction, dilation
 - Viscosity (thickness of fluid) – Anemia, Polycythemia... change viscosity

Blood Pressure Regulation

- Maintenance of blood pressure depends upon:
- **Intrinsic control (central)**
 - Frank-Starling mechanism → Preload + Afterload
- **Intrinsic control (peripheral)**
 - Via change in peripheral vascular resistance
 - Metabolic control → inadequate supply of oxygen or other metabolites promote vasodilatation
- **Auto regulation**
 - Assures constant blood flow under changing pressure or resistance

Blood Pressure Regulation

- **Extrinsic control:**
 - Reflex + Hormonal + Renal
 - Micro-vascular control systems
- NB: normal range of BP is controlled by integration of all mechanisms
- BP regulation is classified into 3 groups according to time of action:
 - A. Short term control mechanisms
 - B. Intermediate term control mechanisms
 - C. Long term control mechanisms

Blood Pressure Regulation

A. Short term control mechanisms

- Predominantly vasomotor adjustment and neuronal control
- Includes:
 1. Baroreceptor (Stretch) reflexes
 2. Chemoreceptor reflexes
 3. CNS Ischemic reflexes

Blood Pressure Regulation

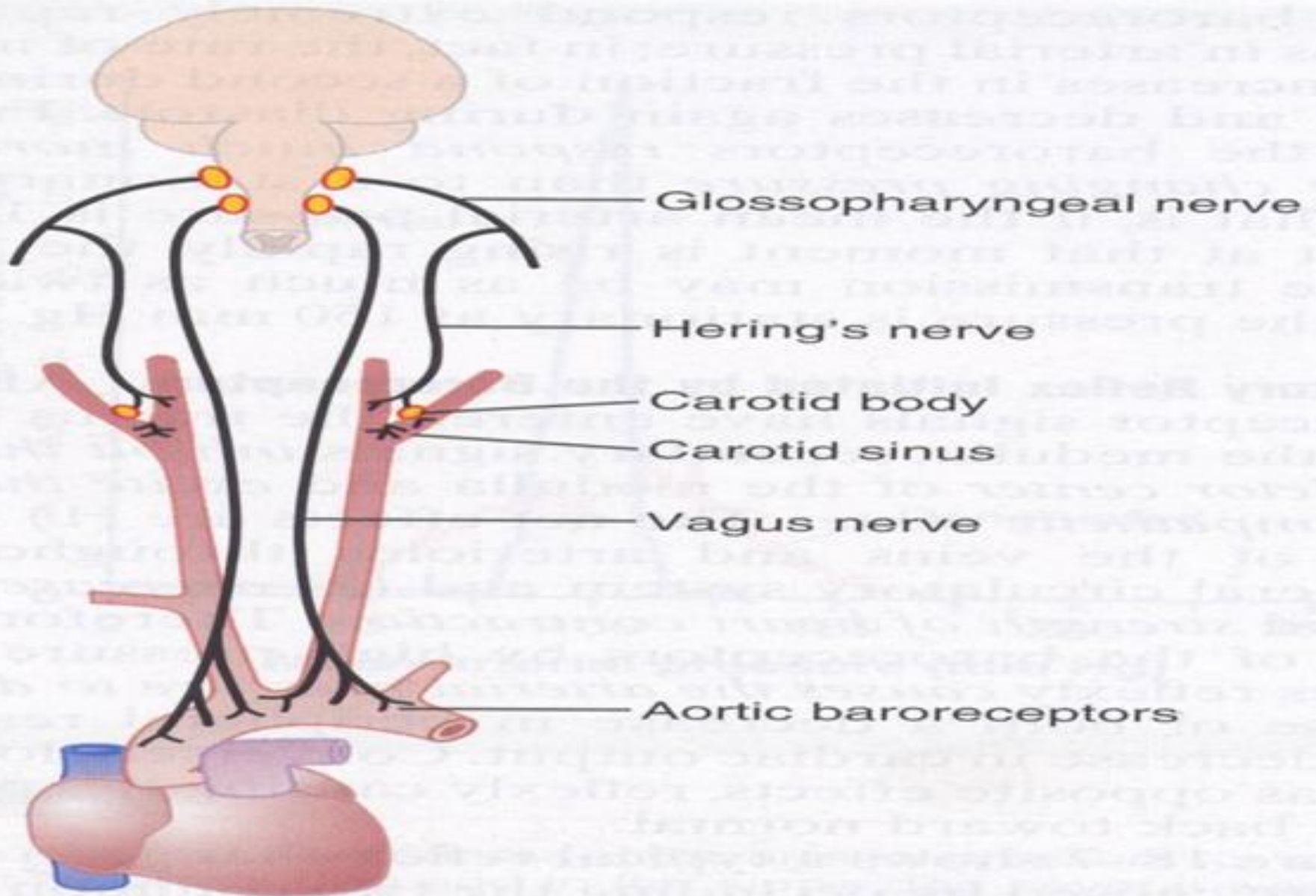
A. Short term control mechanisms

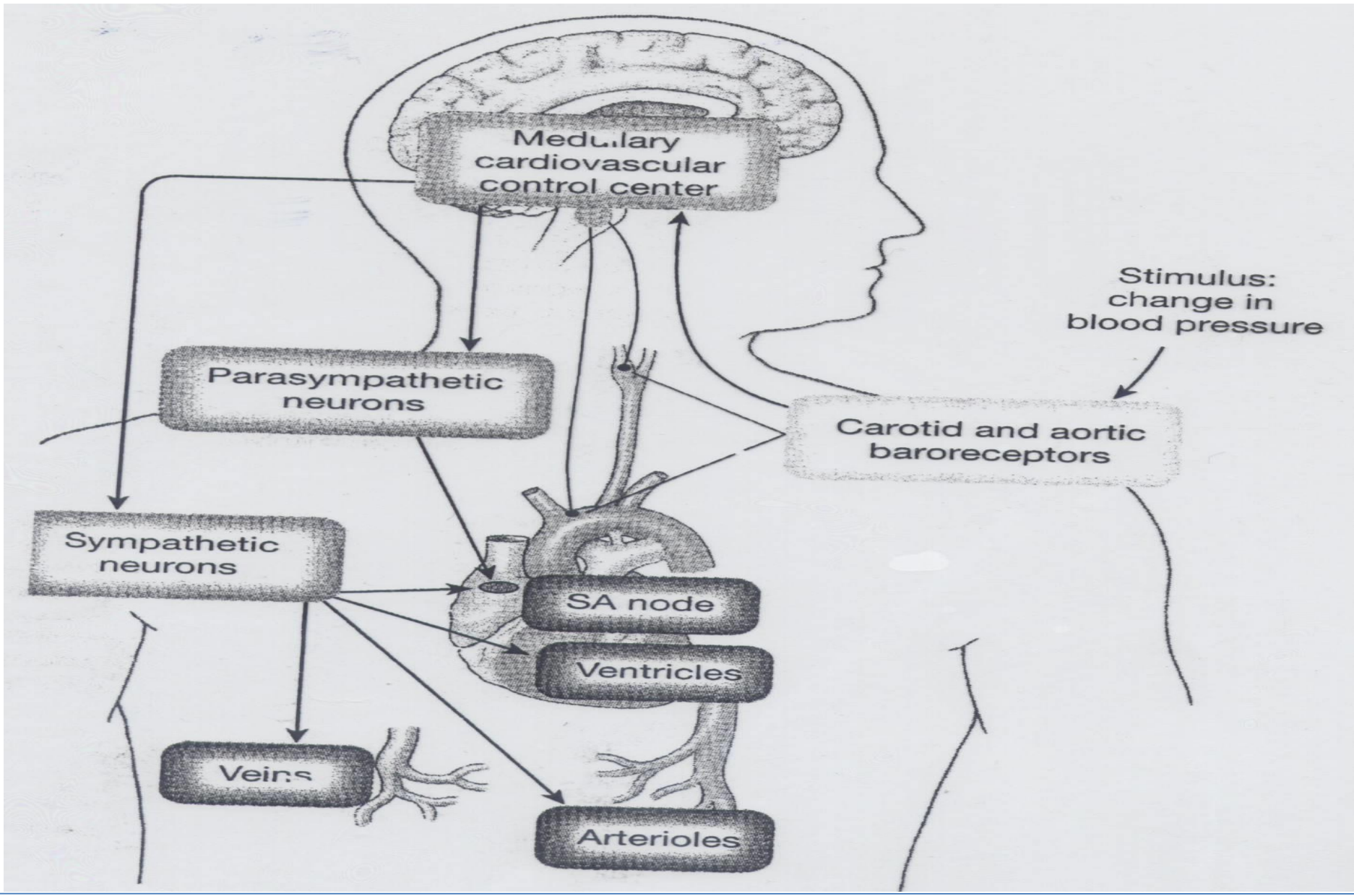
- Common characteristics:
 - Rapid onset of action – within few seconds
 - Response vigorous, but if activated continuously, within a few days, it either dies out completely (baroreceptors) or attenuated -(chemoreceptors, CNS Ischemic response)
 - Vasomotor responses are supplemented by hormonal mechanisms (EP), and with delayed action; vasopressin

Blood Pressure Regulation

1. Baroreceptor reflexes

- Location of Baroreceptors:
 - Aortic arch → greater functional importance
 - Carotid sinus
 - Wall of large thoracic arteries
 - Wall of cervical arteries

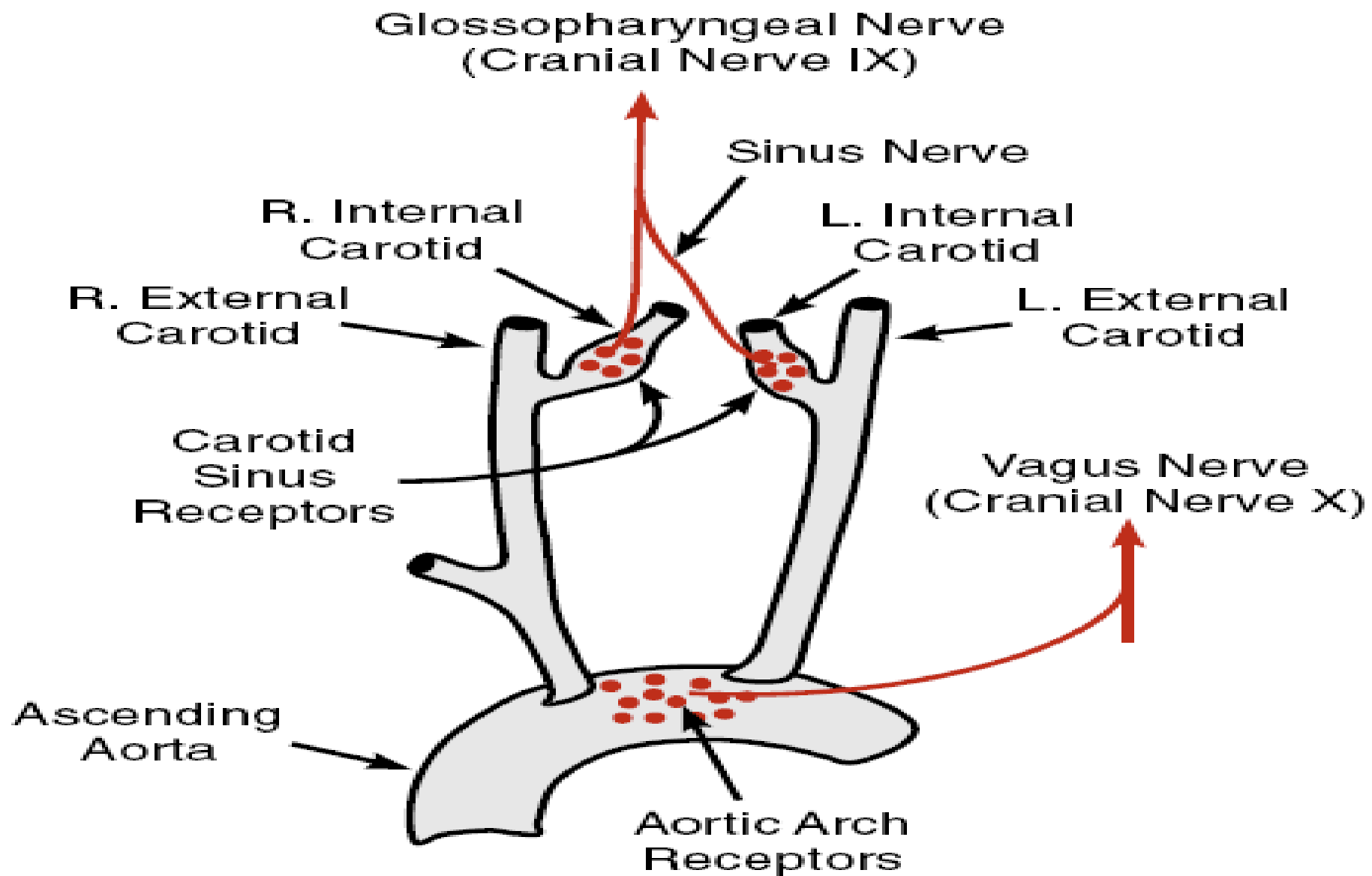




Blood Pressure Regulation

1. Baroreceptor reflexes

- Baroreceptors convey information about:
 - Mean arterial pressure (**MAP**)
 - Amplitude of pressure fluctuation
 - Steepness of pressure rise – rate of pressure change
 - More sensitive to decrease than increase pressure ($\Delta P/\Delta t$)
 - More sensitive to sudden change
- Stimulation of baroreceptors causes stretch → changes set point



Glossopharyngeal nerve

Vagus
nerve

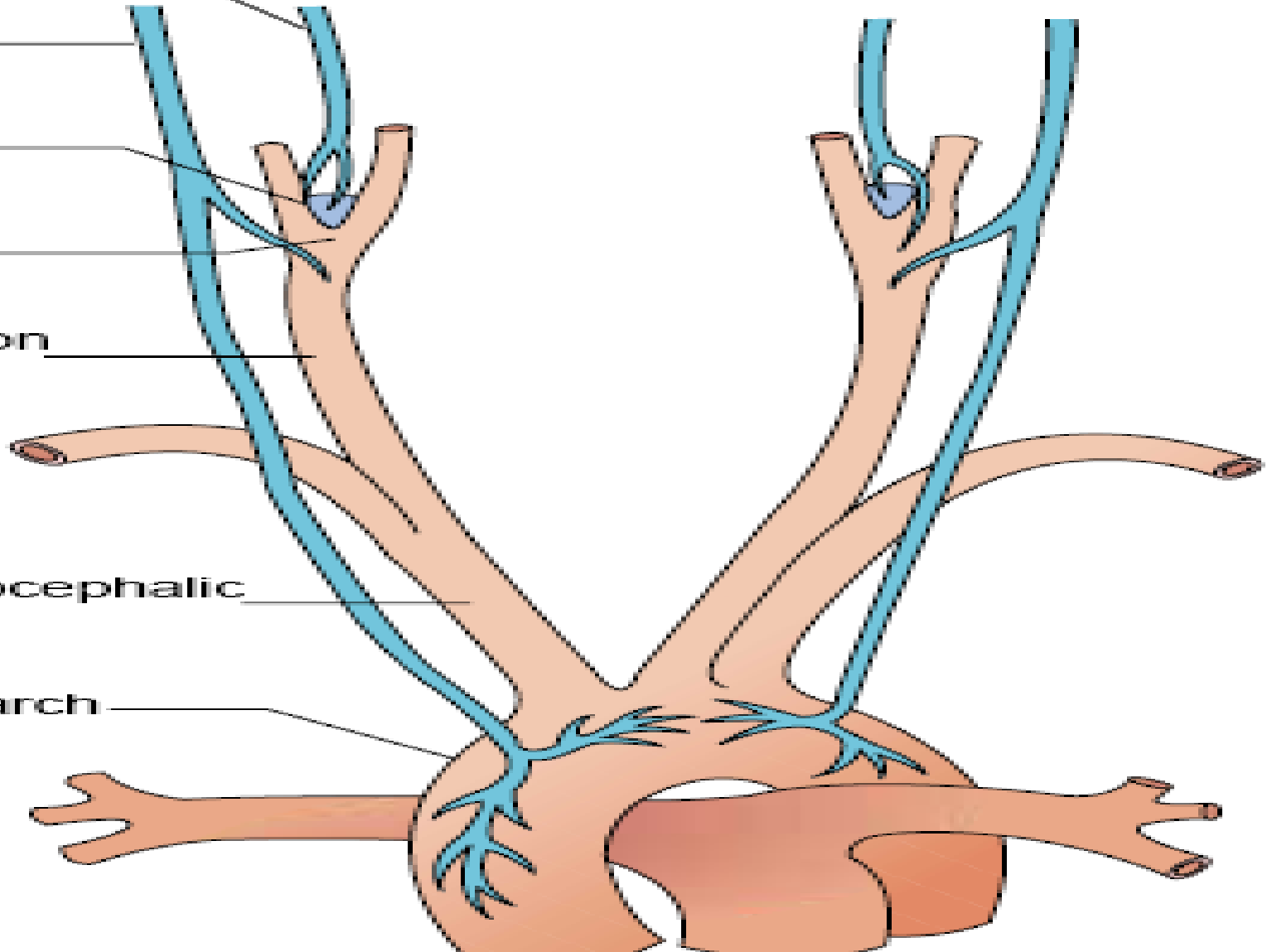
Carotid
body

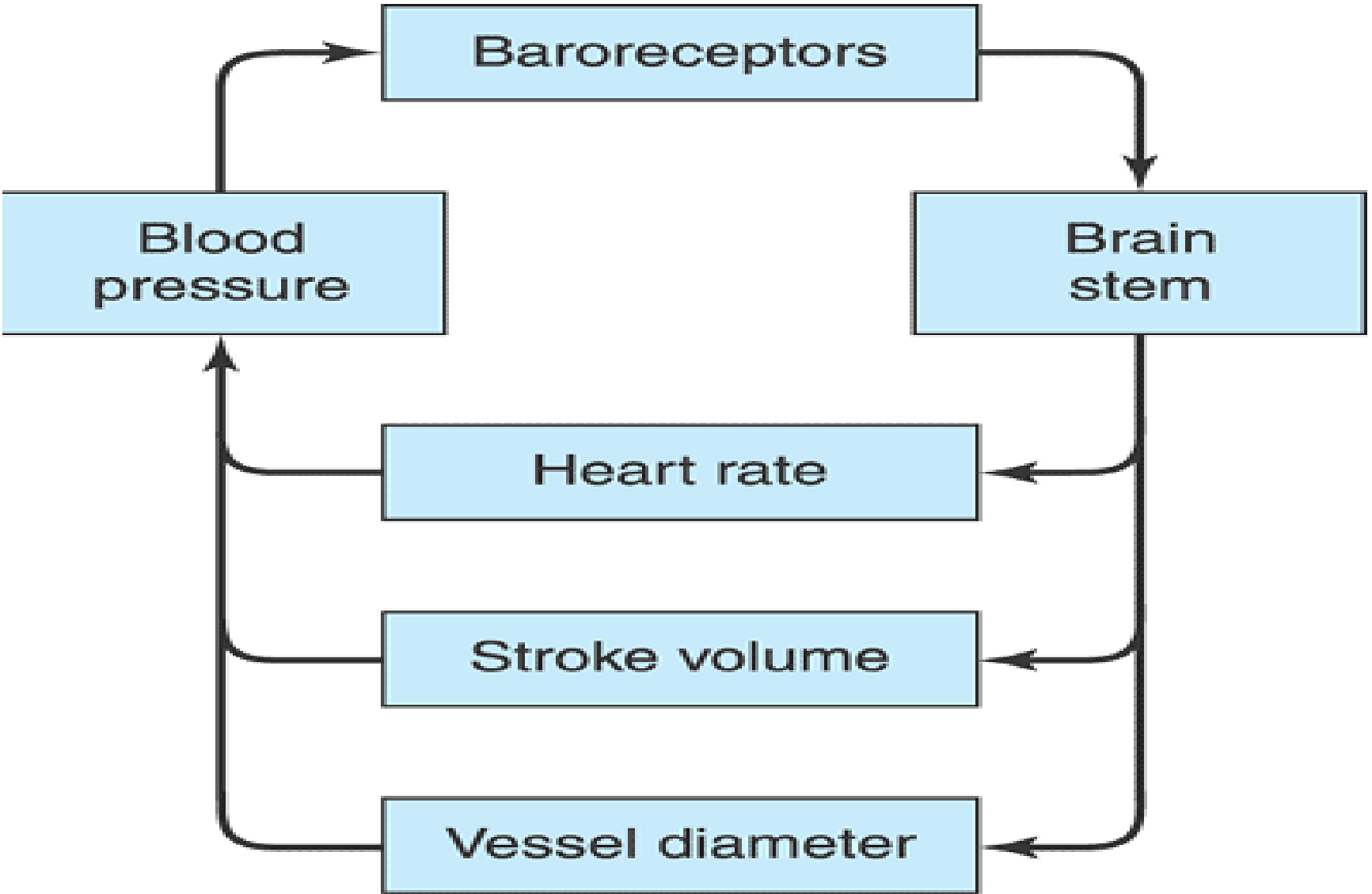
Carotid
sinus

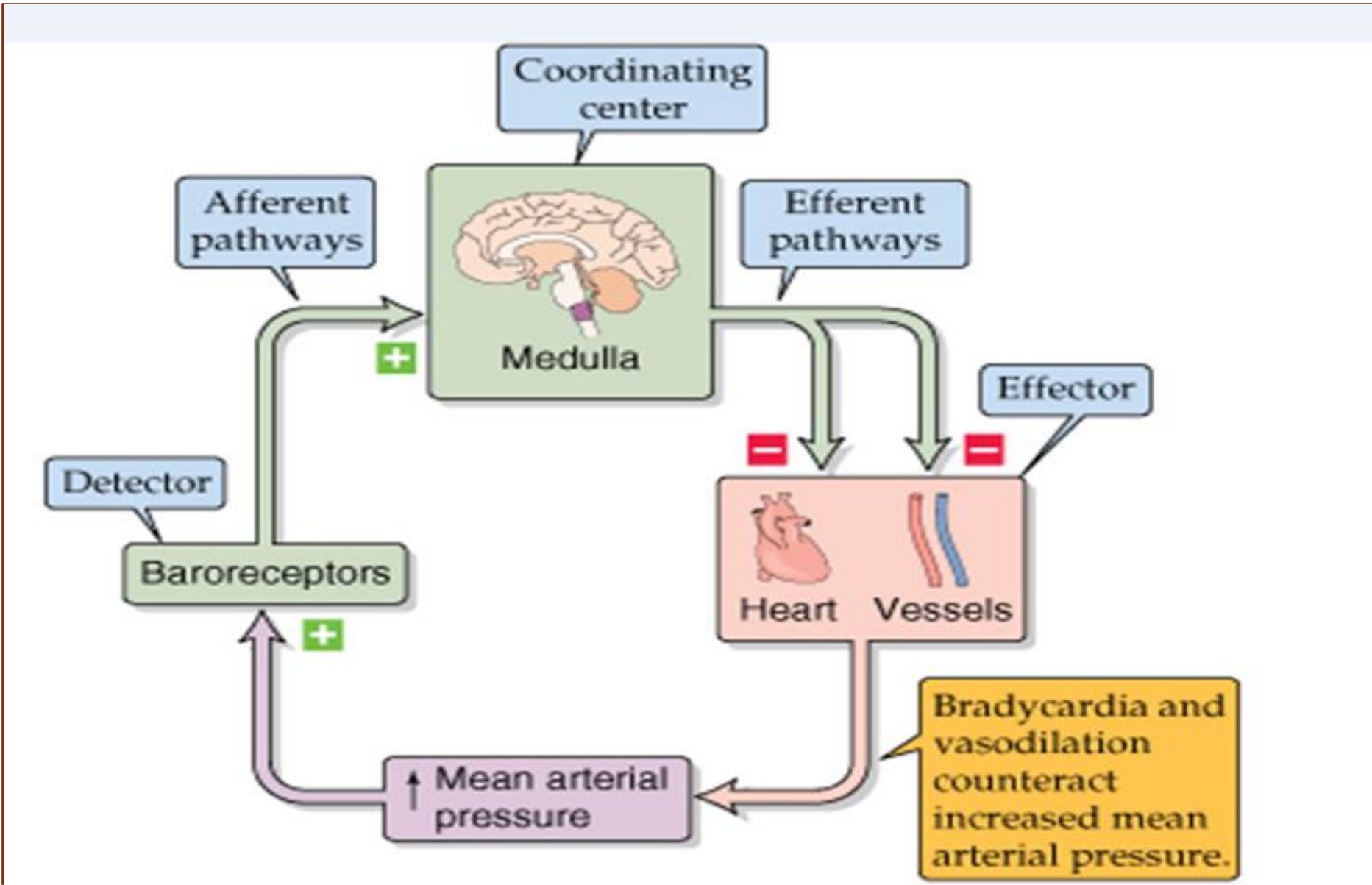
Common
carotid
artery

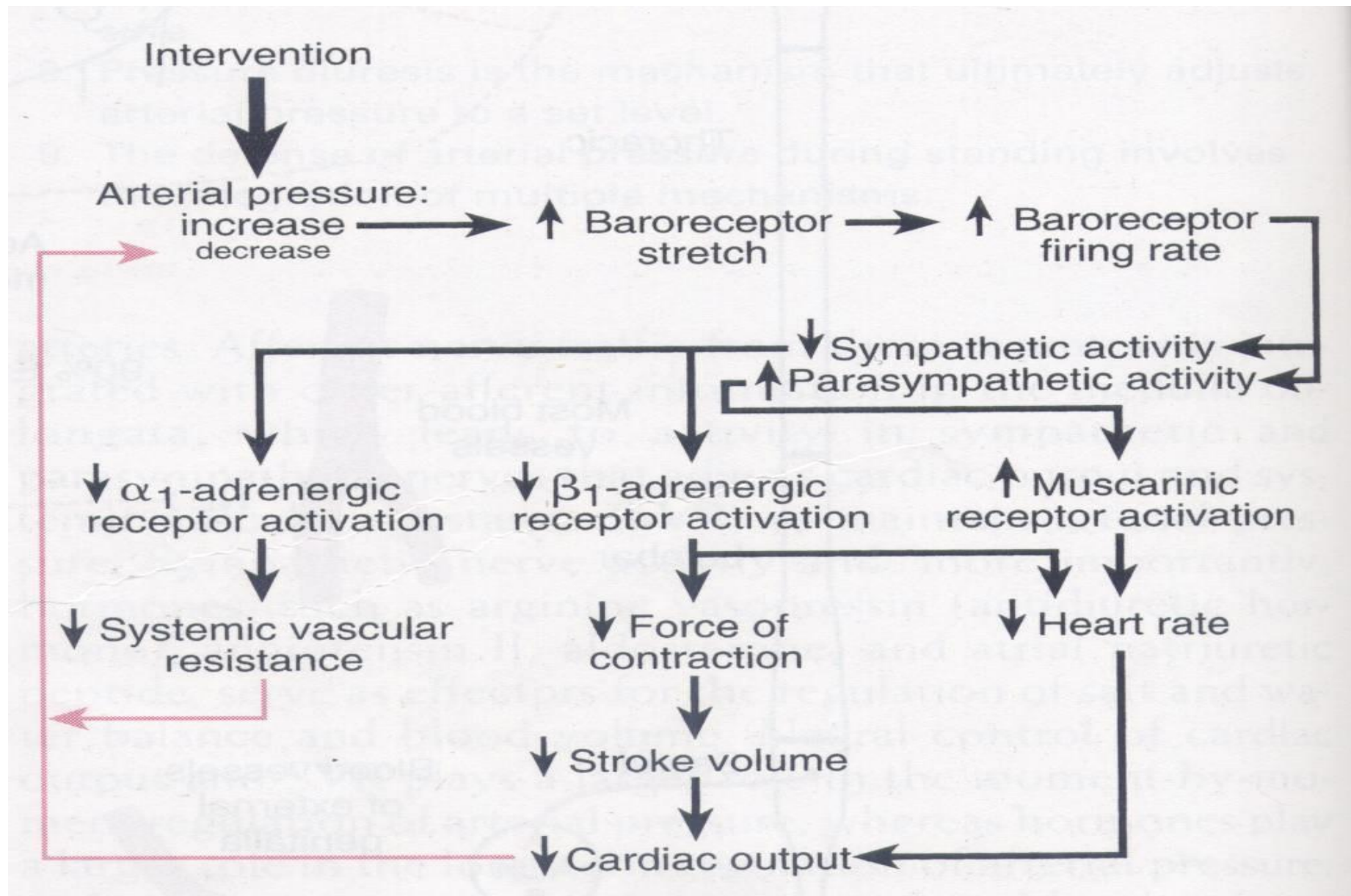
Brachiocephalic
artery

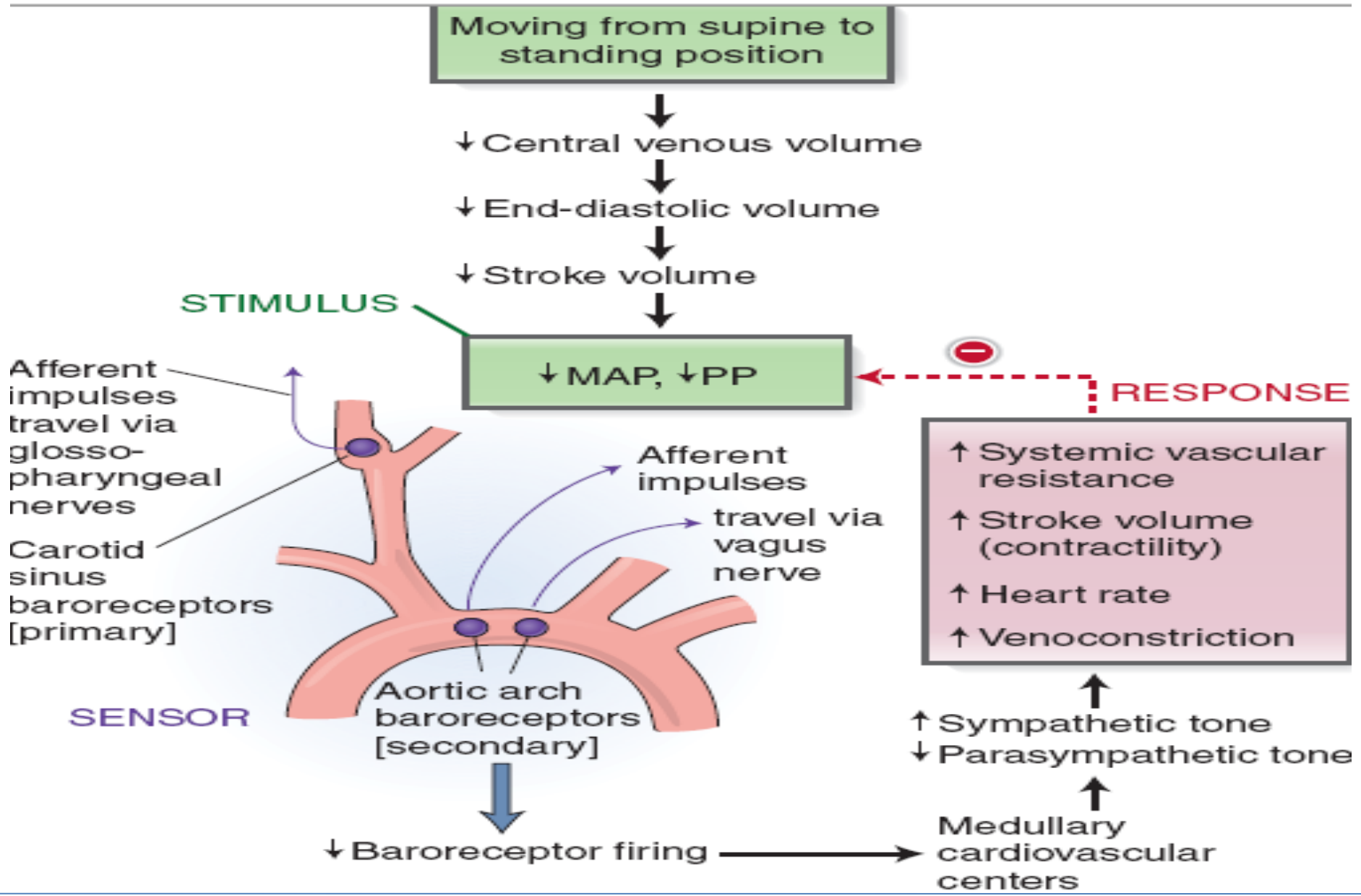
Aortic arch





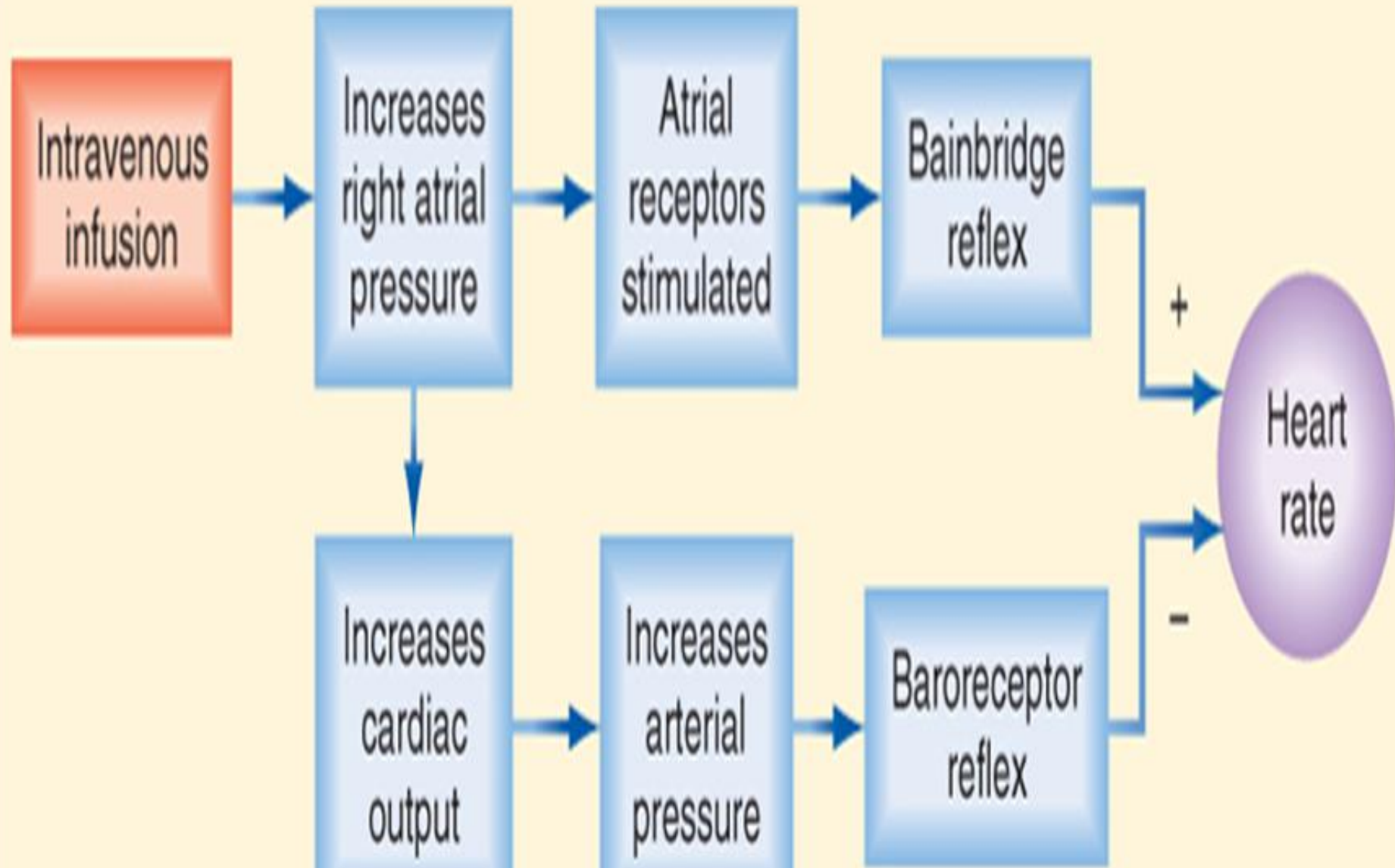


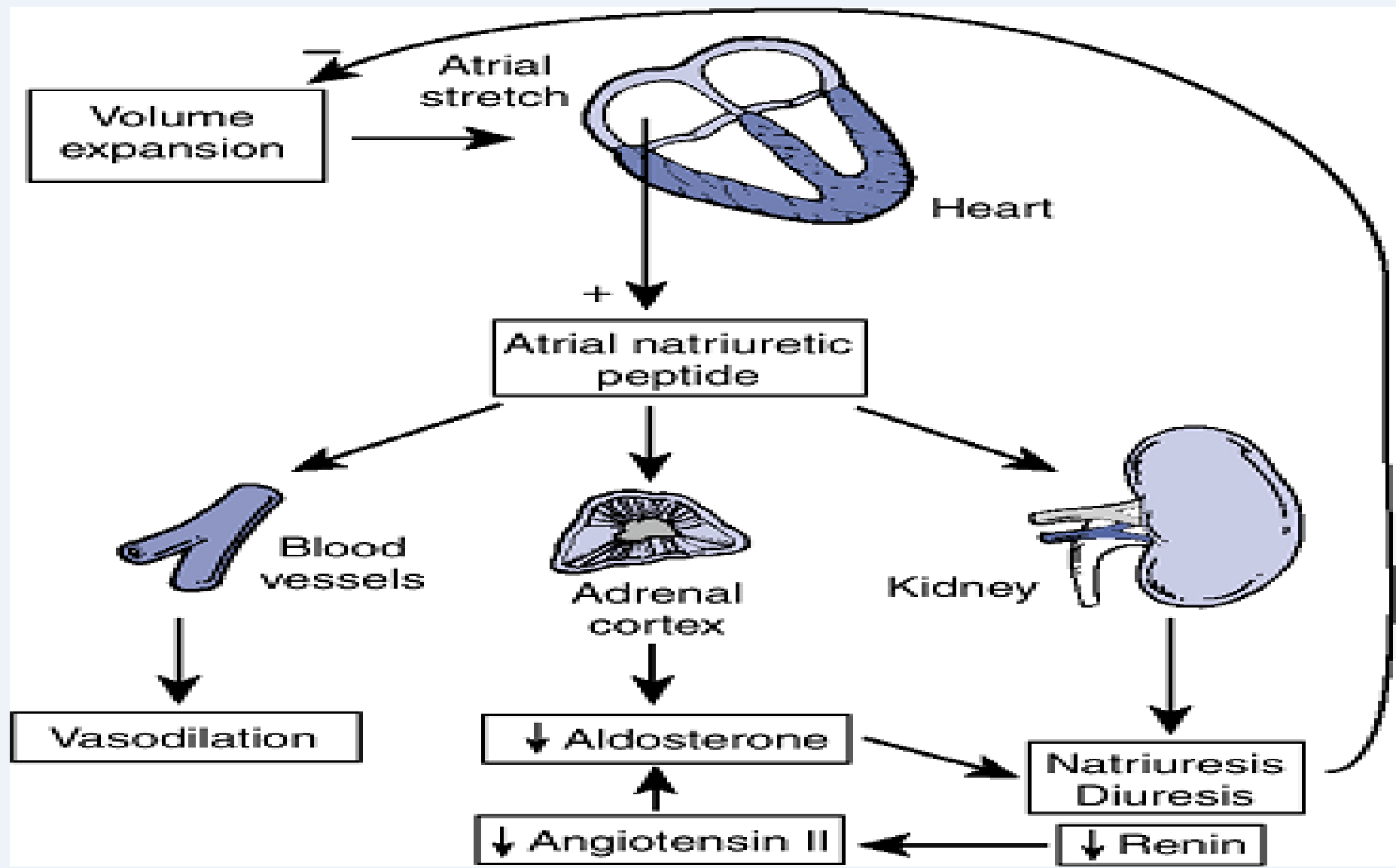


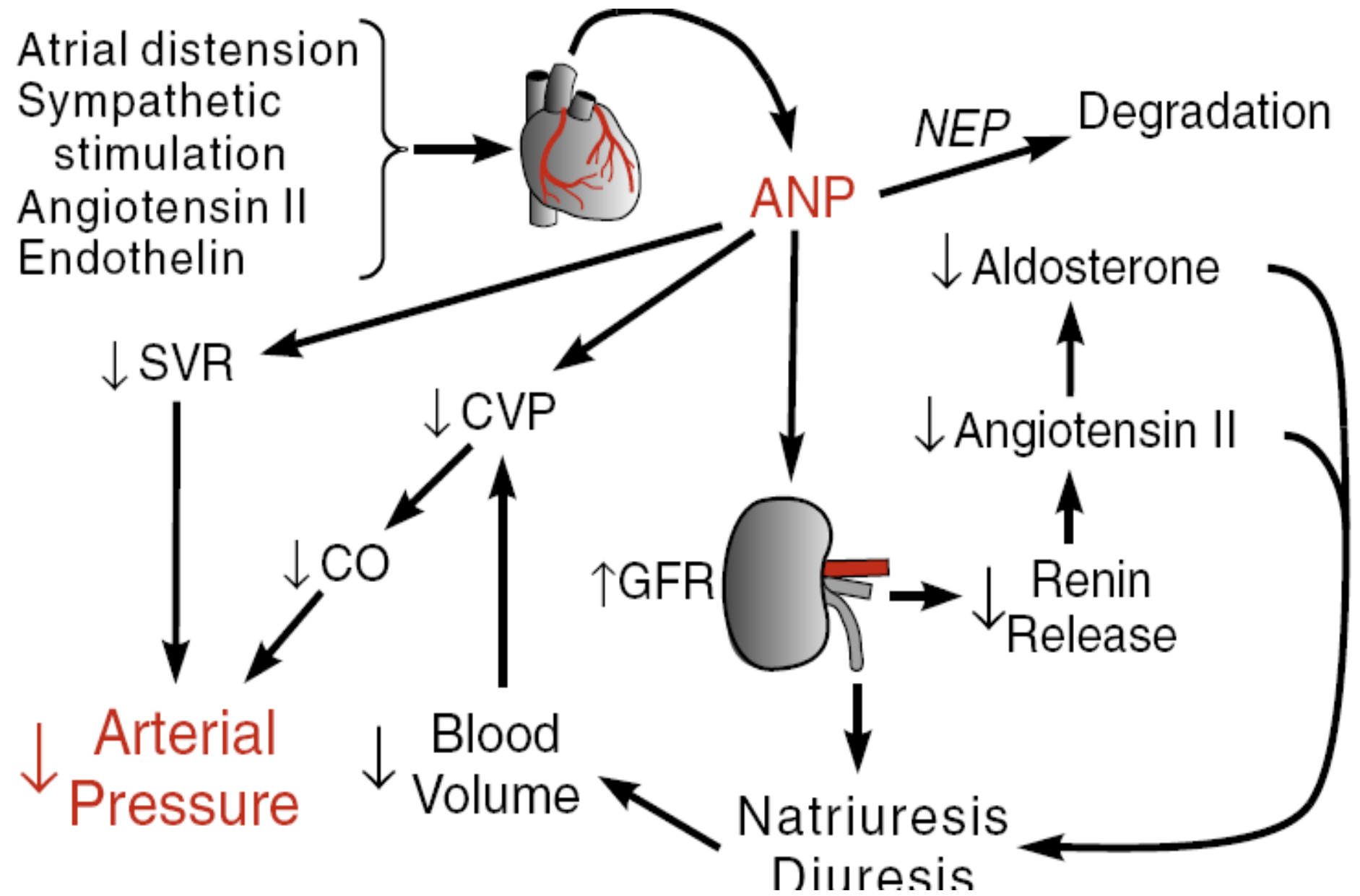


Blood Pressure Regulation

- Cardiac stretch receptor reflex
 - Atrial natriuretic peptide (**ANP**) – released from atria in response to increased atrial pressure (stretch)
 - Increase in venous return → stretches RA
 - Atrial stretch elicits 2 reflexes to increase CO:
 - **Stretch of SA node** → increase HR → helps to pump extra blood returning to the heart
 - **Bainbridge reflex** → extra stretch in RA → impulse to vasomotor center → Sympathetic stimulation and Vagus inhibition → Increase Heart rate → helps to pump out the excess venous return



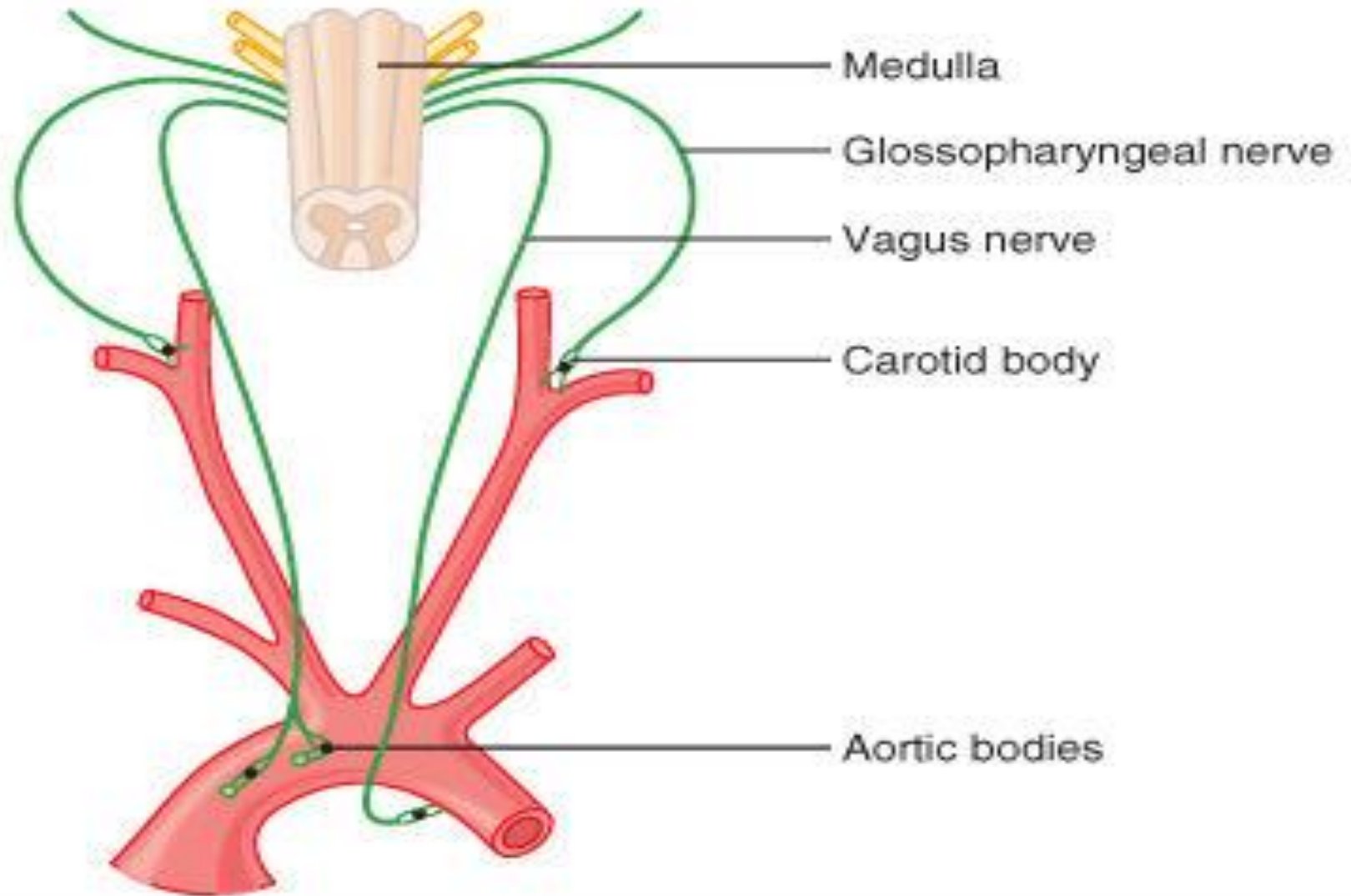




Blood Pressure Regulation

2. Chemoreceptor Reflexes

- Chemoreceptors
 - Location
 - Carotid and aortic bodies (near aortic sinus and carotid sinus)
 - Stimulus
 - Decreased PO_2 , increased PCO_2 , and increased H^+
 - Response
 - Hyperventilation (increased minute volume)

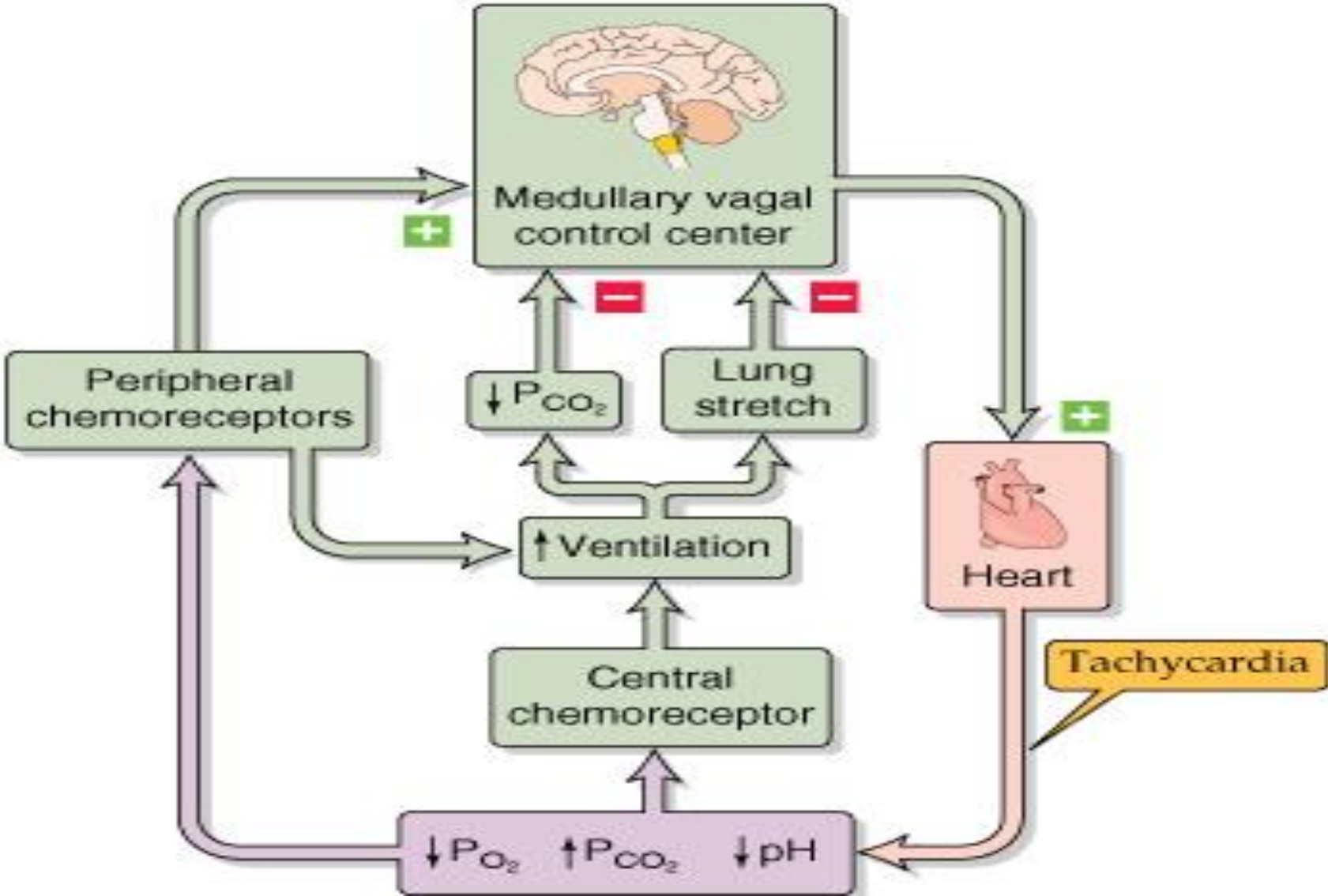


Blood Pressure Regulation

2. Chemoreceptor Reflexes

- Secondary effect: increased O_2 delivery to tissues (heart, brain)
 - General vasoconstriction
 - Increased arterial pressure
- Chemoreceptor reflex is protective in:
 - Hypoxia due to altitude (low arterial PO_2)
 - Circulatory shock (peripheral ischemia)

B INTEGRATED RESPONSE



Blood Pressure Regulation

3. CNS Ischemic Response

- Raises arterial pressure in response to diminished blood flow in vasomotor center of the brain
- Cerebral ischemia increases $\text{PCO}_2 \rightarrow$ which stimulates sympathetic area of medulla oblongata $\rightarrow$ causes increase in arterial pressure (as high as heart can pump)
- Thus prevent further decline in arterial pressure when blood flow to brain decreases dangerously

Blood Pressure Regulation

3. CNS Ischemic Response

- This control system is the “last ditch” mechanism for blood pressure control
- Emergency arterial pressure control system → acts rapidly and very powerfully

Blood Pressure Regulation

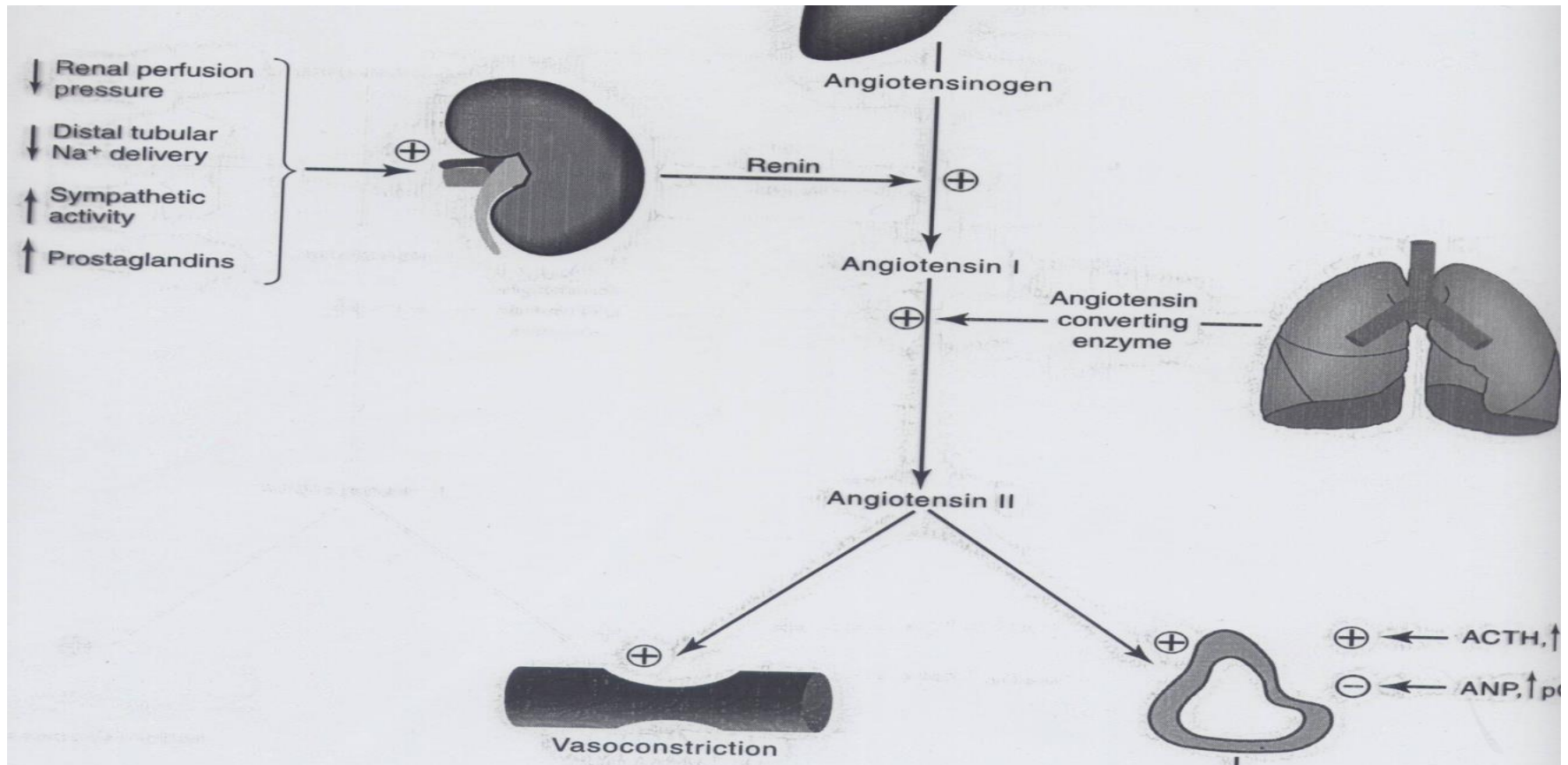
B. Intermediate term control mechanism

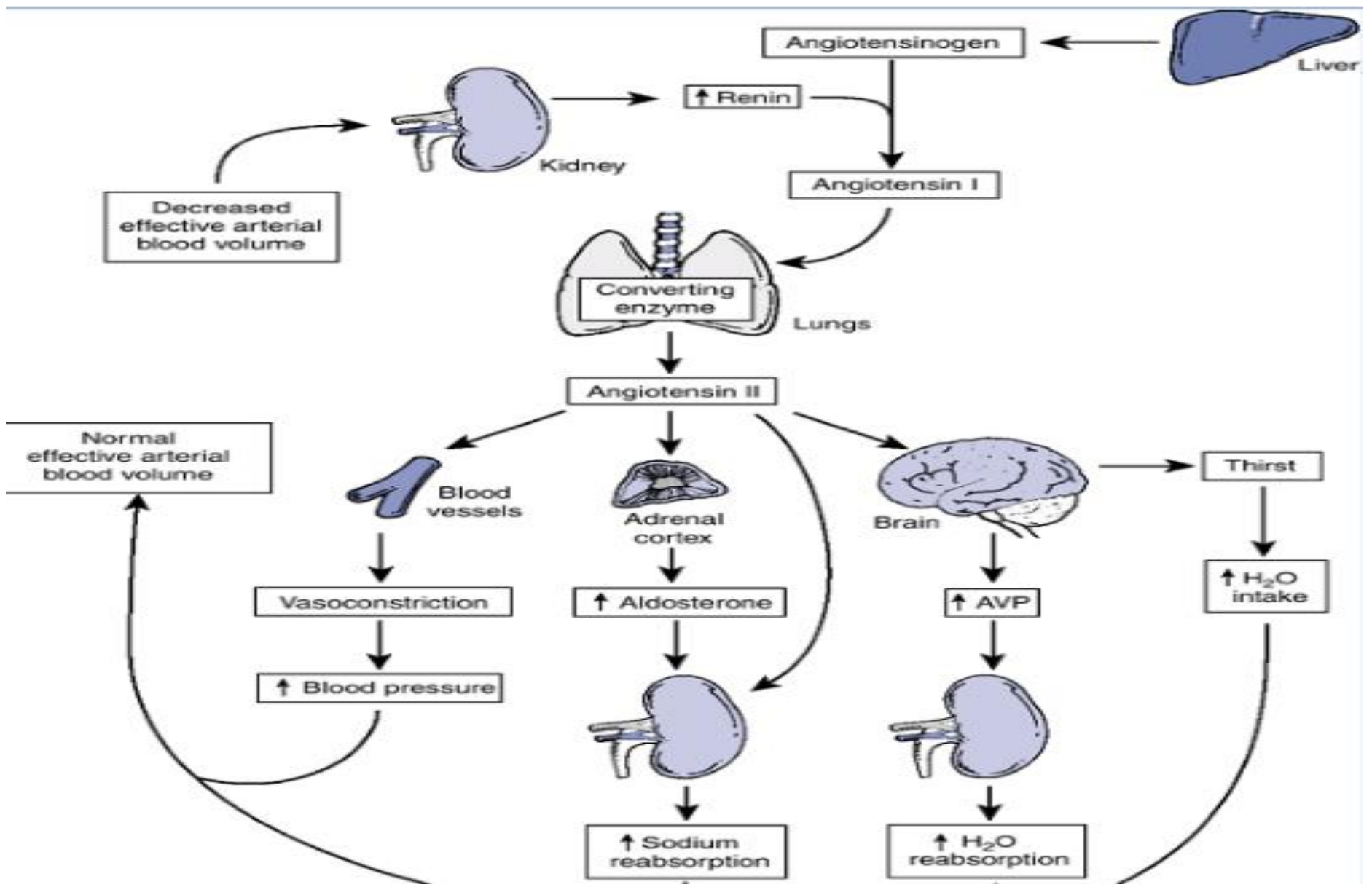
- Trans-capillary volume shifts in response to changes in capillary pressure
- $\uparrow$ Pressure $\rightarrow$ $\uparrow$ stress on veins $\rightarrow$ veins relax & $\downarrow$ pressure
(stress relaxation)
- $\downarrow$ intravascular volume $\rightarrow$ opposite effect
- Beginning of arteriolar vasoconstriction by Ang-II

Blood Pressure Regulation

C. Long term control mechanism

- Renin–Angiotensin–Aldosterone system (RAAS)



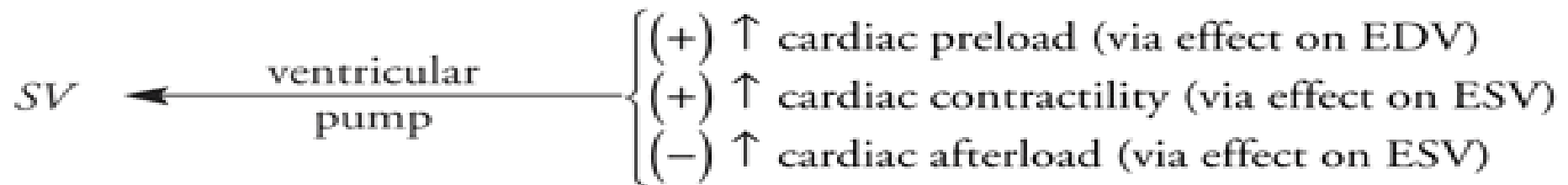


$$\bar{P}_A = CO \times TPR$$

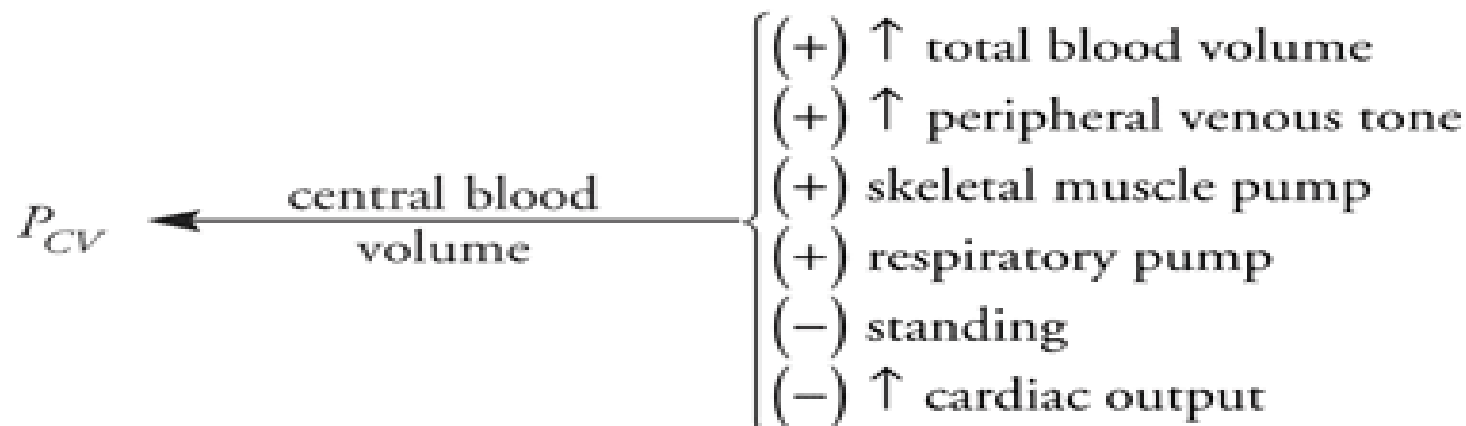
$$CO = SV \times HR$$

$$SV = EDV - ESV$$

$$ejection\ fraction = SV / EDV$$



$$cardiac\ preload \propto P_{CV} \quad (\text{"}\propto\text{" means "is proportional to"})$$



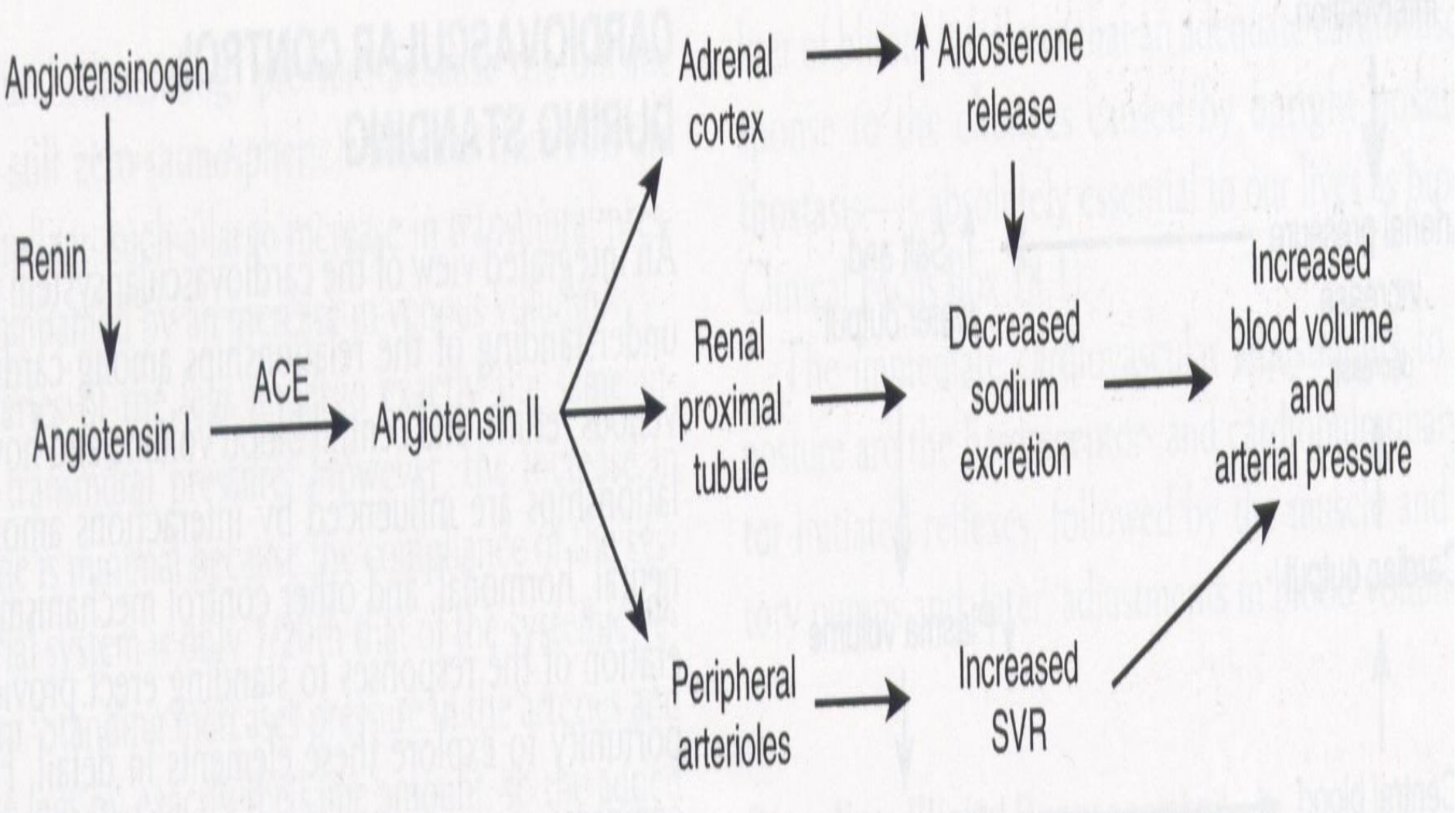
venous tone ← ————— { $(+)$ ↑ sympathetic activity (via NE, α -receptors)

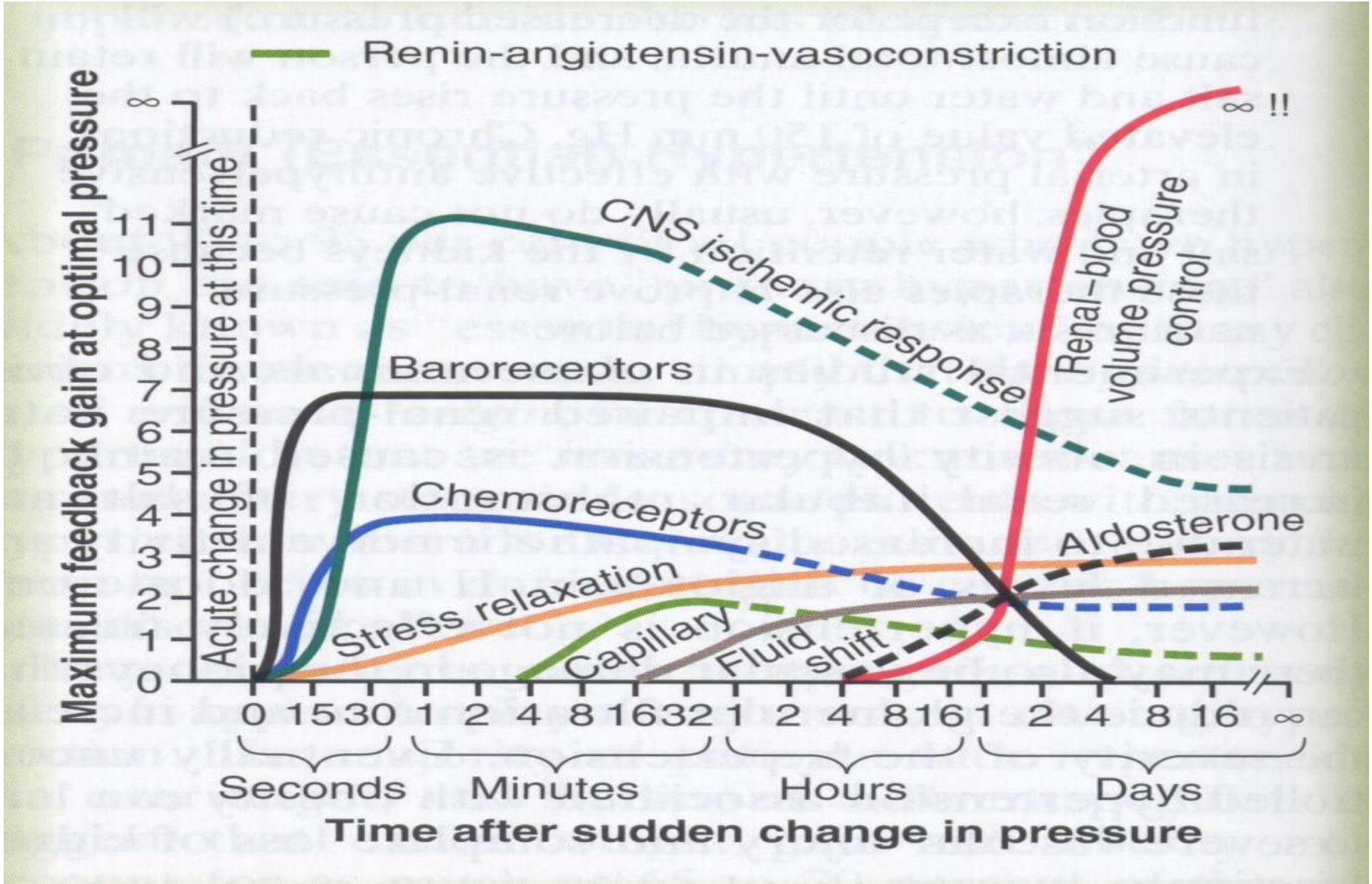
contractility ← $\xrightarrow{\text{ventricular cells}}$ { $(+)$ ↑ sympathetic activity (via NE, β -receptors)

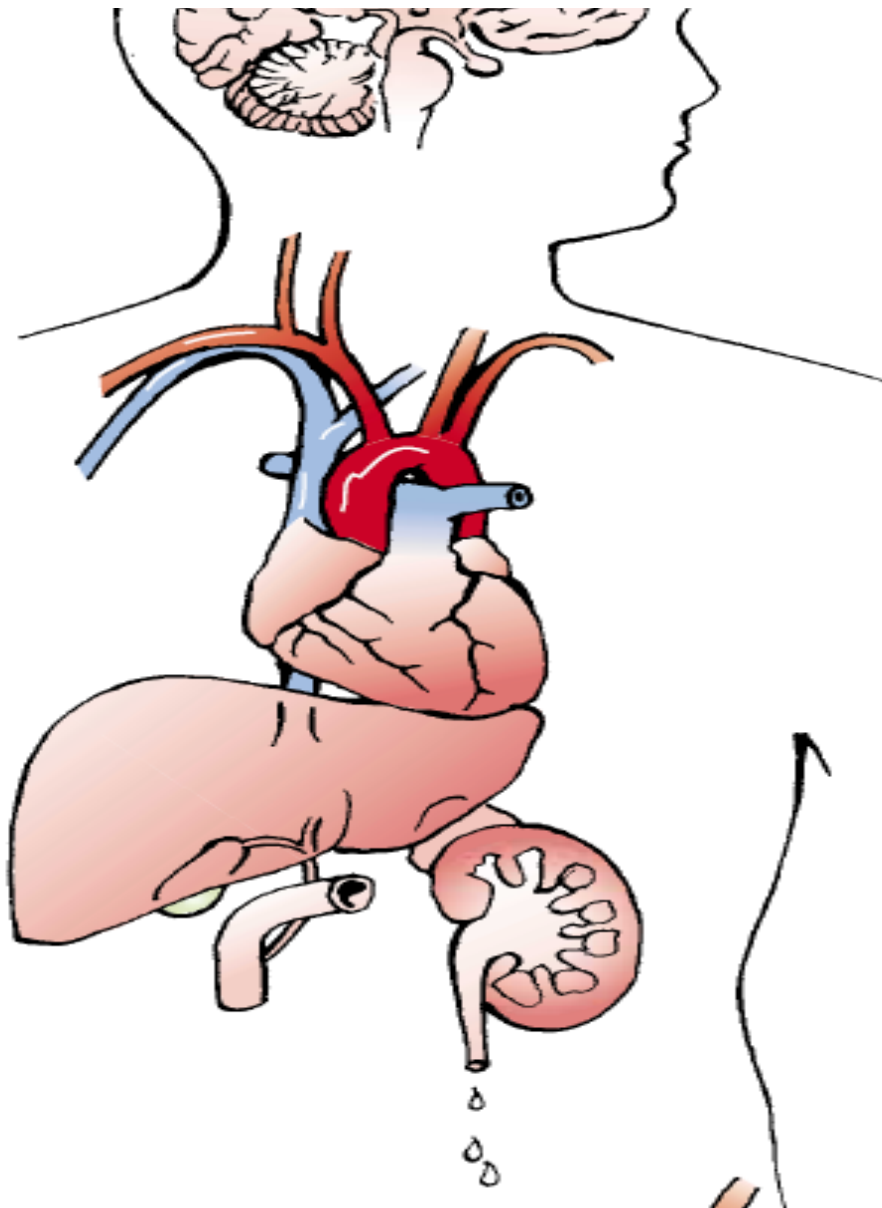
cardiac afterload $\propto \bar{P}_A$

HR ← $\xrightarrow[\text{firing rate}]{\text{SA node cell}}$ { $(+)$ ↑ sympathetic activity (via NE, β -receptors)
 $(-)$ ↑ parasympathetic activity (via ACh)

TPR ← $\xrightarrow{\text{arteriolar tone}}$ { $(+)$ ↑ sympathetic activity (via NE, α -receptors)
 $(-)$ ↑ local metabolites (↑ local metabolic rate)





**Brain**

- Thirst
- Increased production and release ADH

Cardiovascular system

- Increased heart rate
- Increased force of cardiac contraction
- Increased systemic vascular resistance
- Decreased blood flow to the
kidney
gastrointestinal tract
skin
skeletal muscles
- Constriction of the veins

Adrenal gland

- Increased production and release of aldosterone by the adrenal cortex
- Increased production and release of the catecholamines (epinephrine and norepinephrine) by the adrenal medulla

Liver

- Constriction of veins and sinusoids with mobilization of blood stored in the liver

Kidney

- Increased retention of sodium and water
- Decreased urine output

Capillary bed

- Increased reabsorption of water into the capillary from the interstitial spaces due to constriction of the arterioles with a resultant decrease in capillary pressure

Blood Pressure Regulation

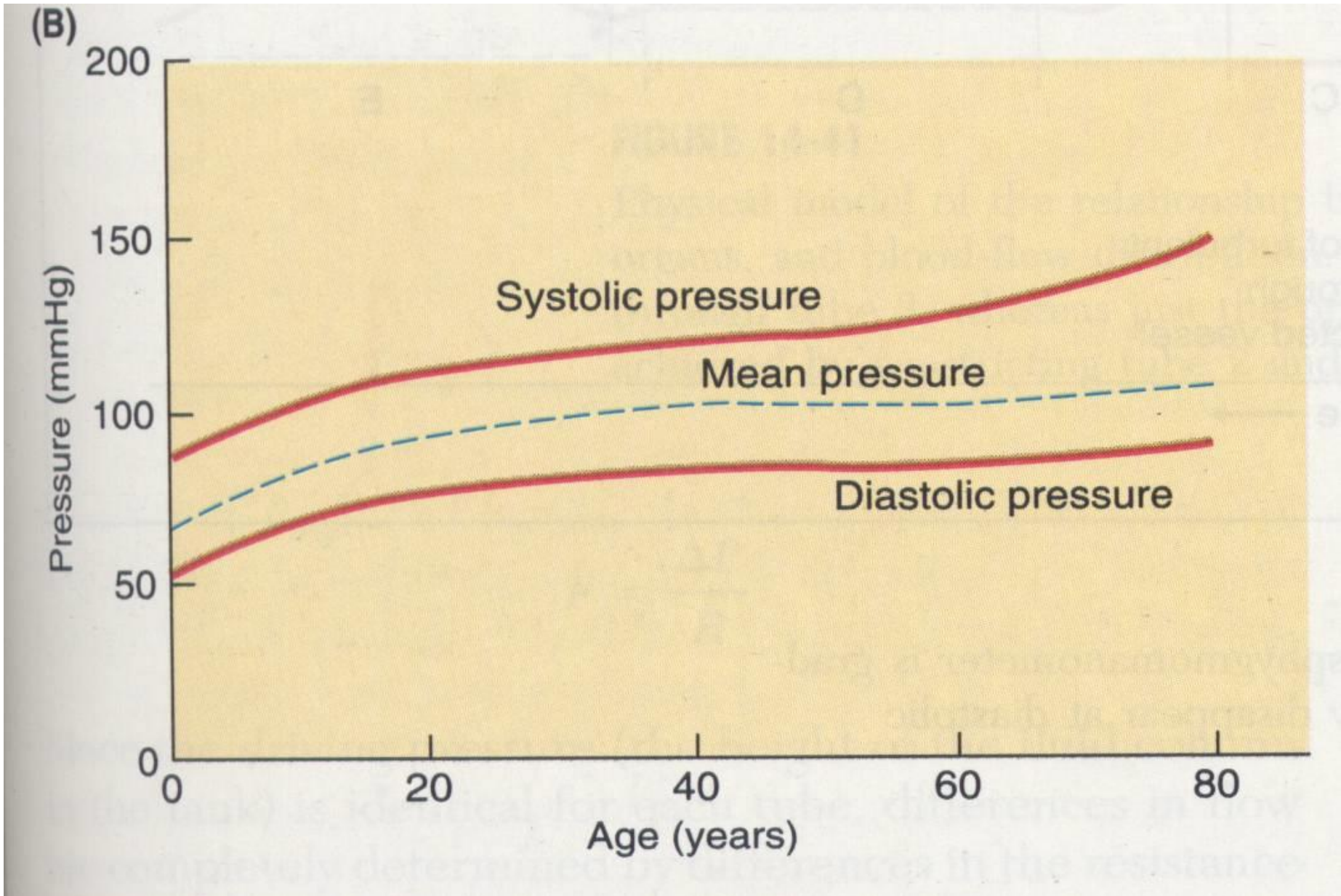
- Central control of blood pressure
 - Concerned with maintaining MAP
- Neuronal:
 - Parasympathetic supply – Generally has no effect
 - Sympathetic supply
 - Releases NE → acts on α receptors
 - Causes arteriolar constriction
 - Reduces flow and increases MAP

Blood Pressure Regulation

- Central control of blood pressure
- Hormonal:
 - Circulating epinephrine
 - Acts on α receptors → Causes arteriolar constriction
 - Acts on β_2 receptors → Causes arteriolar dilation
 - Response depends on which receptors tissue expresses

Blood Pressure Regulation

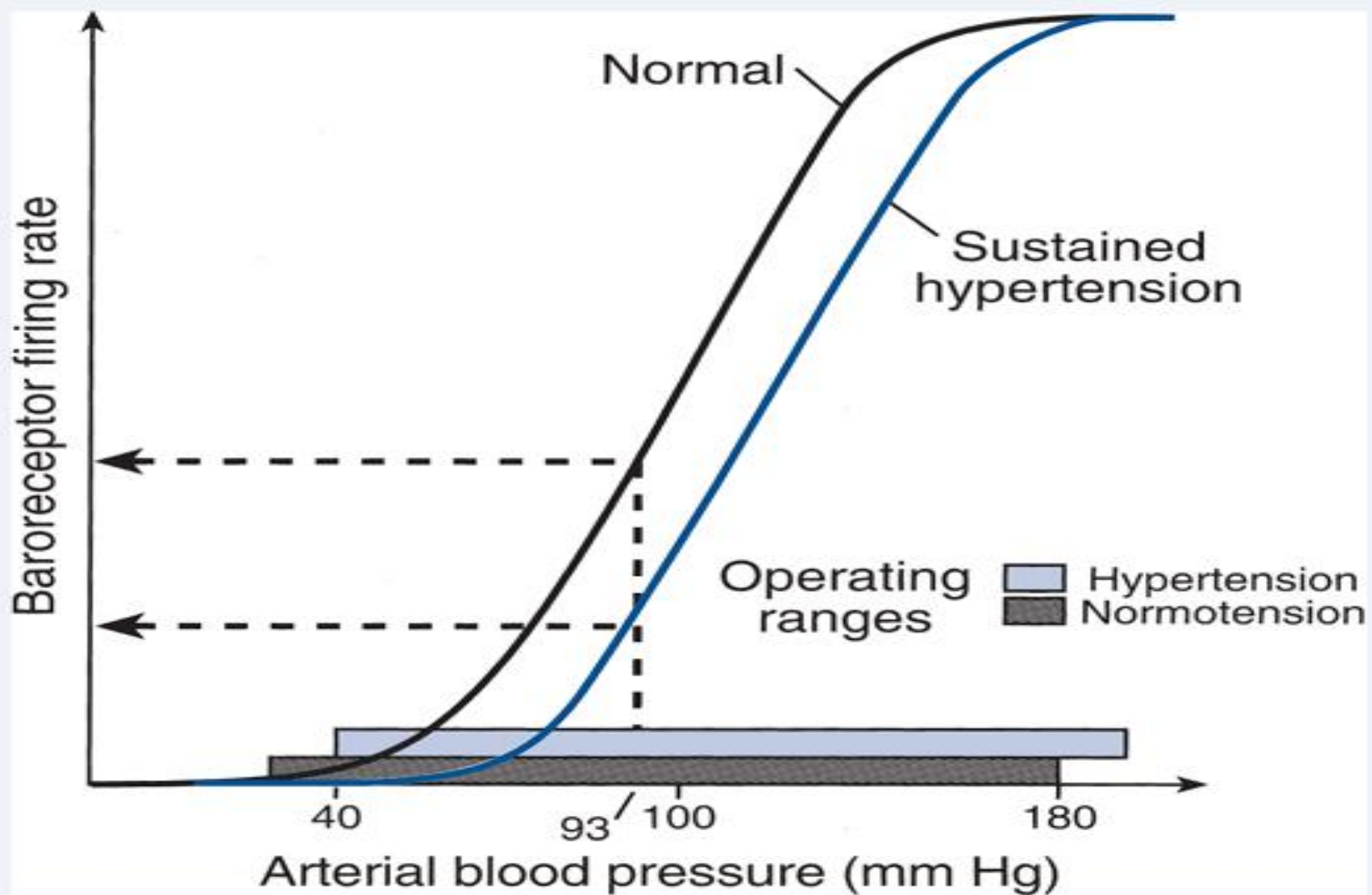
- Age related change in blood pressure
 - Two common changes in CVS with age:
 - Increased peripheral vascular resistance → morphological changes to the heart
 - Disproportionate increase in systolic pressure mainly due to arterial stiffness



Clinical Correlates

Pathophysiology of Blood Pressure

- Hypertension
 - $BP \geq 140/90 = 18-49$ yrs
 - $160/95 = \text{over } 50$ yrs
- More common in male than female
- Severe hypertension 4 X more common in black men than white
 - Possible reason = higher renal vascular resistance



Hypertension

- Types:
 - **Systolic hypertension** → Systolic rise more prominent
 - Arterial vessel wall thickened (hypertrophy of vascular smooth muscle)
 - Minimized compliance
 - Elderly → ↓ dispensability of arterial wall
 - **Diastolic hypertension** ⇒ → TPR

Hypertension

- Types:
 - Primary/essential hypertension
 - Cause of hypertension can't be determined (95% of cases)
 - Secondary hypertension
 - Hypertension due to identifiable cause (~5%)
 - Causes:
 - Renal disorder + endocrine (hyperaldosteronism) + cardiovascular diseases are the main causes
 - Pregnancy or drugs are also related

Secondary Hypertension

1. Renal disorder

- Accounts 80% of all cases
- Chronic glomerulonephritis, pyelonephritis and other permanent damage of kidneys where water and salt are retained
- Hyperparathyroidism and Calcium overload leads to renal failure and hypertension
- Renal artery stenosis (eg. Atherosclerosis), chronic renal inflammation (glomerulonephritis) all lead to hypertension

Secondary Hypertension

1. Renal disorder

- Renal artery stenosis → reduces glomerular pressure → causes release of Renin (juxtaglomerular apparatus) → release aldosterone → water and salt retention → **Hypertension**

Secondary Hypertension

2. Hyperaldosteronism

A. Primary Hyperaldosteronism – Conn's Syndrome

- Isolated rise in serum aldosterone due to hyper function of **Zona- glomerulosa** of the adrenal cortex (not the renin release)

B. Secondary Hyperaldosteronism

- Abnormally high stimulation of the adrenal zona glomerulosa
- Serum concentration of whole renin-angiotensin-aldosterone cascade are increased

Secondary Hypertension

3. Cushing's syndrome

- Increased glucocorticoid concentration in the blood
- Classical Cushing's disease is caused by excess liberation of ACTH from Adenohypophysis

4. Pregnancy hypertension

- In the last 3 months pregnancy, some develop hypertension, edema and proteinuria (pre-eclampsia or toxemia of pregnancy)
- If this condition develops into severe hypertension with fits and lung edema it is called eclampsia

Hypertension

- Etiologic risk factors:
 - **Positive family history** of hypertension
 - **Sex** → younger men and older women more hypertensive
 - **Race** → black more than white
 - **Body mass** → increased body mass increased risk
- Others that can be controlled through behavioral modification :
 - Body composition + Diet
 - Stress + Neuronal + Hormonal
 - Renal regulation

TABLE 9-3 CAUSES OF HYPERTENSION**Essential hypertension (90% to 95%)**

- Unknown causes
- Involves:
 - increased blood volume
 - increased systemic vascular resistance (vascular disease)
- Associated with:
 - heredity
 - abnormal response to stress
 - diabetes and obesity
 - age, race, and socioeconomic status

Secondary hypertension (5% to 10%)

- Renal artery stenosis
- Renal disease
- Hyperaldosteronism (primary)
- Pheochromocytoma (catecholamine-secreting tumor)
- Aortic coarctation
- Pregnancy (preeclampsia)
- Hyperthyroidism
- Cushing's syndrome (excessive glucocorticoid secretion)

Hypertension

- Hypertension shortens life expectancy in at least 3 ways:
 1. Excessive workload on heart and elevated arterial pressure
 - ⇒ Coronary artery disease
heart attack
 - ⇒ Congestive heart disease
 2. Rupture of major vessels in the brain
 - Hypertrophy and eventual obstruction of cerebral blood vessels
 - Both leads to cerebral ischemia and death of major brain portion
→ Stroke
 3. Damage to kidneys → leading to kidney failure

Hypertension: R_x

- Therapeutic Principles:
 - Untreated hypertension shortens life by ~20 yrs
 - Target organs damaged → heart + aorta + brain + eyes + kidneys...
- Systemic hypertension can be treated through one of the following strategies:
 - Reduction of total blood volume (and thus stroke volume) with diuretics results in reduction of the driving pressure
 - Reduction of cardiac frequency reduce cardiac output and thus the driving pressure
 - Reduction of TPVR with vasodilators reduce driving force

Hypertension: R_x

1. β -adrenergic receptor blockers

- Reduce cardiac function and reduce BP
 - Eg. Propranolol $\rightarrow$ antagonist at both β_1 and β_2 receptors
- Propranolol acts on heart and reduces chronotropic (reduced HR) and inotropic effect (reduced force and cardiac output)

2. Angiotensin Converting Enzyme (ACE) inhibitors:

- Captopril + Enalapril + ...
- Reversibly inhibit ACE and thus act as vasodilators of both resistance and capacitance vessels

Hypertension: R_x

3. Calcium channel blockers:

- Nifedepine + Verapamil + ...
- Bind to Ca²⁺ channels → Inhibit Ca²⁺ entry into the cells
- Acts as effective vasodilators → they relax smooth muscles of arterioles + inhibit cardiac contractile force
- Overall effect → vasodilation → diminishes TPVR + reduced afterload
- Thus, CO is improved despite cardiac contractile depression

Shock

- General circulatory insufficiency (fall in arterial pressure) associated with rapid HR, RR, With pale, moist and grey skin
- Causes:
 - Disorders in the heart (Cardiac insufficiency + ...)
 - Cardiogenic shock can be caused by restricted filling (AV stenosis...)
 - Myocardial disorders (Angina + MI...)
 - Restricted ventricular ejection (semilunar insufficiency or shunts)
 - Disorders in vessels (vascular insufficiency developing into vascular shock)

Shock

- Types:

1. Hypovolemic shock

- Hemorrhage (rapid loss of blood)
- Trauma + Burns (plasma loss)
- Fluid loss in vomiting and diarrhea
- Hypovolemic shock can be:
 - A. Absolute hypovolemia
 - B. Relative hypovolemia

Shock

A. Absolute hypovolemia

- Blood loss + plasma loss
- Severe diarrhea + vomiting + excessive sweating
- Intestinal obstruction with luminal fluid accumulation
- Urinary loss of proteins/salt/water
- Excessive use of diuretics
- Hypoaldosteronism

Shock

B. Relative hypovolemia → due to Vasodilation

- Septic shock from viral or bacterial infection
- Anaphylactic shock
- Neurogenic shock:
 - Vasodilatation by severe pain + stress
 - Anesthetics
 - Brain stem lesions close to vasoconstrictor center

Shock

2. Cardiogenic shock

- Myocardial infraction
- Pulmonary embolism
- Cardiac tumor

3. Obstructive shock (Obstruction of blood flow)

- Pulmonary embolism
- Cardiac tumor
- Tension Pneumothorax

Shock

4. Distributive shock (vasogenic or low resistant shock)

A. Neurogenic shock

- Loss of vasomotor tone through out the body → thus increasing vascular capacity

B. Anaphylactic shock

- Allergic reaction → histamine induced
- Increase capillary permeability
- Increase vascular resistance + Arteriolar dilation

C. Septic shock

- Dissemination of bacterial toxin → increase capillary permeability

